UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

Master's Thesis

Design and Sizing of an Energy Storage System for a Hybrid Tugboat

Thesis as a requirement to obtain the degree of
Magister en Ciencias de la Ingeniería Electrónica

Thesis presented by
Leonardo Solis Zamora

Supervisor
Dr. Marcelo Pérez Leiva

Evaluation Commitee
Dr. Christian Rojas, UTFSM, Valparaíso, Chile
Dr. Jon Are Suul, NTNU, Trondheim, Norway

January 2025 , Valparaíso, Chile

UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA

CONSTANCIA DE VALIDACIÓN Y CONFIDENCIALIDAD DE MONOGRAFÍA A REPOSITORIO ACADÉMICO

1.- IDENTIFICACIÓN DEL TRABAJO ACADÉMICO

Tipo de monografía (marcar una opción): ☐ Memoria o trabajo de título ☑ Tesis de Postgrado
Título del trabajo: Design and Sizing of an Energy Storage System for a Hybrid Tugboat
Nombre del candidato(a): Leonardo Solis Zamora
Carrera / Grado: Magíster en Ciencias de la Ingeniería Electrónica
Campus: Casa Central Valparaíso **Departamento:** Electrónica

2.- VALIDACIÓN DEL PROFESOR GUÍA/DIRECTOR DE TESIS

Yo, Marcelo Pérez Leiva, en mi calidad de profesor(a) guía/director(a) del trabajo académico mencionado anteriormente **DEJO CONSTANCIA** que:

- He revisado esta versión del documento y corresponde a la versión final aprobada del trabajo.
- El trabajo cumple con los requisitos académicos y de formato establecidos por la institución.

3.- EVALUACIÓN DE CONFIDENCIALIDAD POR PROPIEDAD INDUSTRIAL (marcar una opción)

☐ El trabajo **NO** contiene información que amerite confidencialidad y puede ser publicado de inmediato en repositorio con acceso abierto.
☑ El trabajo **CONTIENE** información con potenciales implicancias de propiedad industrial o intelectual y requiere un periodo de confidencialidad (embargo) por (marcar una opción):

☐ 6 meses ☑ 12 meses ☐ 2 años ☐ 3 años ☐ 5 años ☐ 10 años

Fundamentación de la necesidad de confidencialidad (obligatorio si se solicita embargo):

Uso de información en AC3E para proyectos con empresas del rubro marítimo.

4.- FIRMAS

Profesor(a) guía o director(a) de memoria o tesis:

Fecha: 23-01-2026 Firma: **Marcelo Pérez Leiva**

Estudiante o Candidato(a):

Fecha: 23-01-2026 Firma: **Leonardo Solis Zamora**

Este formulario debe ser insertado como página 2 de la memoria o tesis, completado y firmado por estudiante y profesor(a) antes de la entrega en portal PRISMA de Biblioteca USM.

Abstract

The global maritime industry is increasingly prioritizing sustainability, promoting the adoption of various hybrid and fully electric propulsion systems for different vessel types. This document presents a design and sizing methodology of an energy storage system for a hybrid tugboat, customized to meet the operational demands while minimizing environmental impact. The proposed framework integrates analysis of load profile, energy storage sizing, battery technologies selection, and propulsion system optimization to achieve an optimal balance between performance, and emissions reduction. The results demonstrate that a well-designed energy storage system for a hybrid tugboat can achieve reductions in fuel consumption and greenhouse gas emissions without compromising the availability and robustness required for towing and transit operations.

Resumen

La industria marítima mundial está dando cada vez más prioridad a la sostenibilidad, promoviendo la adopción de diversos sistemas de propulsión híbridos y totalmente eléctricos para diferentes tipos de buques. En este documento se presenta una metodología de diseño y dimensionamiento de un sistema de almacenamiento de energía para un remolcador híbrido, personalizado para satisfacer las exigencias operativas y minimizar el impacto medioambiental. El marco propuesto integra el análisis del perfil de carga, el dimensionamiento del almacenamiento de energía, la selección de tecnologías de baterías y la optimización del sistema de propulsión para lograr un equilibrio óptimo entre el rendimiento y la reducción de emisiones. Los resultados demuestran que un sistema de almacenamiento de energía bien diseñado para un remolcador híbrido puede lograr reducciones en el consumo de combustible y las emisiones de gases de efecto invernadero sin comprometer la disponibilidad y la robustez necesarias para las operaciones de remolque y tránsito.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Motivation and Background

The growing global concern over climate change has significantly pressured industries to adopt sustainable practices. The maritime sector, responsible for approximately 2.5% of energy sector CO_2 emmisions as shown in figure 1.1, has continuosly increased in emissions [1] [2].

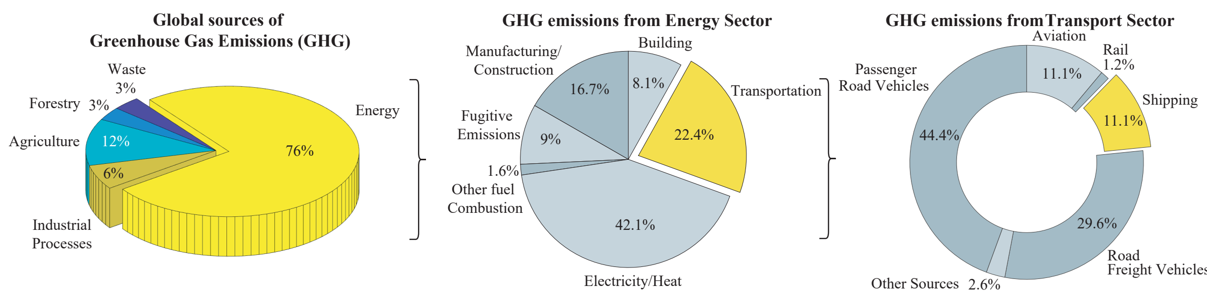
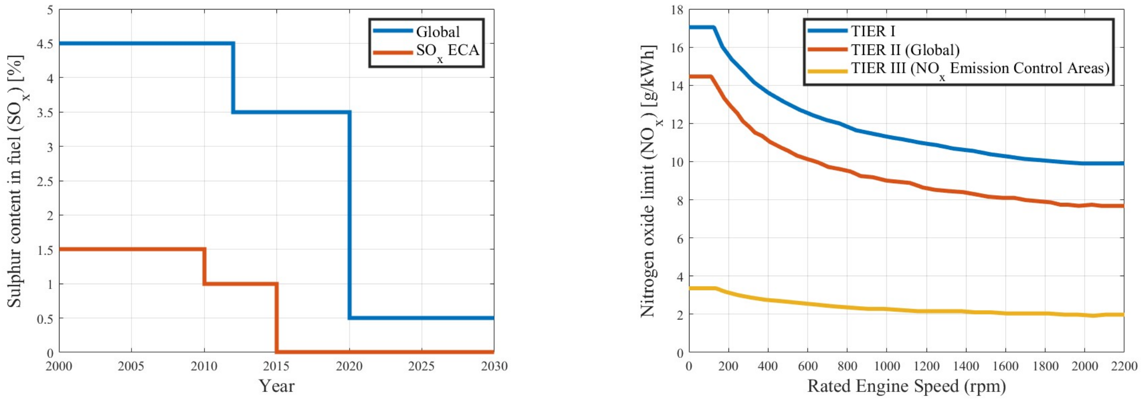


Figure 1.1: Global sources of greenhouse gas emissions and breakdown of CO_2 emissions from the energy sector and the transport subsector [1].

The International Maritime Organization (IMO) is implementing strategies to reduce the emissions from international shipping by establishing standards, imposing restrictions and provinding incentives. The initial strategy avoids a reduction in total GHG emissions from international shipping by at least 50% by 2030 compared to 2008, consistent with the Paris Agreement. IMO's current global standard for SO_x emissions is 0.5% mass by mass (m/m) by 2030 per the **International Convention for the Prevention of Pollution by Ships (MARPOL 73/78) Annex VI**, as shown in figure 1.2a.

Similarly, the emission limit for NO_x from engines is dependent on the type of the engine and the nominal engine speed, as shown in figure 1.2b. The global limit has been set to 14.4 g/kWh for engines with speeds less than 130 rev/min and 7.7 g/kWh for engines with speed greater than 2000 rev/min [3].



(a) Changes in SO_x limits over time. (b) NO_x limits in terms of engine speed.

Figure 1.2: The MARPOL Annex VI emission limits for global and Emission Controlled Areas (ECAs) perspective [3].

With the **Net-Zero Famework (NZF)** approved bye the IMO in April 2025, the course is being set for the global maritime fuel transition [4]. The NZF aims to meet the emission reduction targets set out in the 2023 IMO GHG Stratey. Pending adoption in October 2025, a GHG fuel intensity (GFI) metric, in combination with a two-tier GHG pricing mechanism, will requiere operators to reduce their ship's GHG emissions intensity by 21% by 2030, with financial penalties for those who fail to do so. Beyond 2030, the requirements become rapidly stricter.

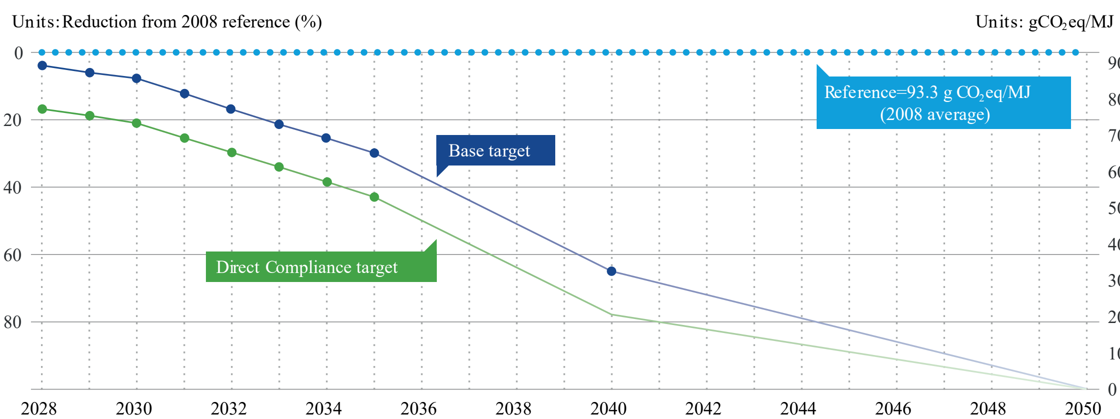


Figure 1.3: GFI reduction factors and reference value in NZF [4].

The NZF promotes low-GHG fuels and technologies to comply with the IMO's 2023 Strategy: reducing emissions by 20% by 2030 and 70% by 2040 (compared to 2008), reaching net zero by 2050., as is shown in fig 1.3. The NZF regulates the GFI metric (gCO_2eq/MJ) under a well-to-wake approach, tightening limits from 2028 onwards. This system of incentives and penalties requires shipowners to evaluate solutions that minimize consumption and reduce the carbon intensity of their vessels.

Many vessels chartered in the coming years could still be in operation in 2050, so it is necessary to take into account the strict requirements that lie ahead in order to maintain their commercial appeal, asset value, and profitability in the coming decades. Furthermore, beyond 2030, ships in operation may need to consider modernization options that allow for the use of fuels with low greenhouse gas emissions.

Incentives to reduce emissions and improve equipment efficiency are motivating ship manufacturers to adopt zero-emission technologies. Fuel cells and energy storage systems such as batteries (main source of electrical energy) and supercapacitors (high power peaks and improved battery system life) reduce greenhouse gas emissions and are key factors in achieving low-emission operation of marine vessels.

1.1.1 Propulsion Architectures

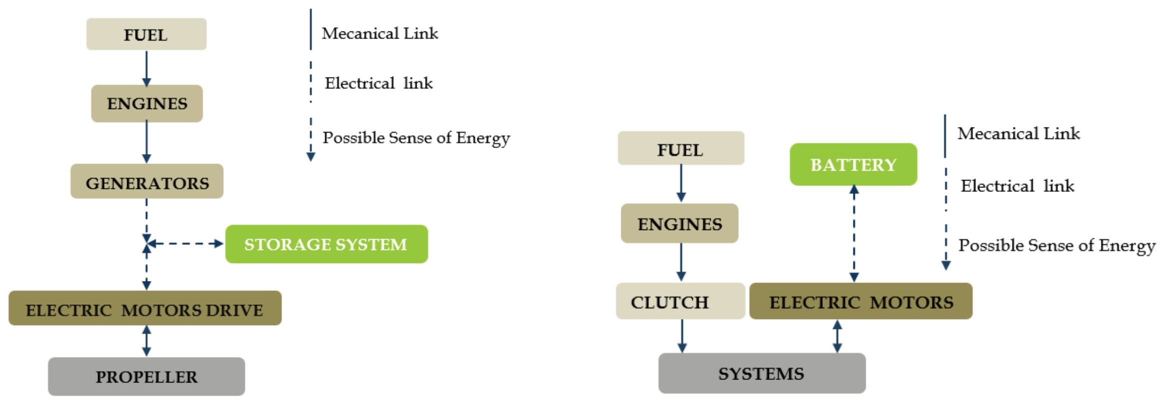
While conventional maritime transport is characterized by restricting the amount of fuel produced by diesel engines in propulsion systems through Tier standards, which translates into a continuous decrease in fossil fuel consumption, fully electric and hybrid propulsion architectures, made possible by innovative high-performance energy storage systems (batteries, fuel cells, supercapacitors, etc.), are a promising way to significantly reduce fossil fuel consumption, pollutants, and greenhouse gas emissions [6].

For full-electric propulsion, the main advantages include better dynamic performance (start-up, arrest, changes in speed, and regenerative braking), reduced fuel consumption due to the more efficient and flexible use of generators that work within their minimum specific fuel consumption region, and the possibility of having shorter shaft lines because the electric motor must be installed close to the propeller [7].

The hybrid propulsion system uses a combination of diesel engines or gas turbines with electric motors to provide a wide range of propeller speeds [7]. One of the main advantages of this configuration concerns an increase in efficiency at low loads. Hence, less fuel consumption, fewer emissions, and a small footprint are achieved. Hybrid propulsion systems (HPS) are interesting when there is a large variation in power demand during operations. The design rules of this type of systems can differ significicatly for different types of vessel, mainly due to the different types of mission profiles and operation requirements [6].

Hybridization of the propulsion chains are based on the separation or simultaneous use of several different energy sources. The choice of different power sources, and sources combinations is designed to adapt to the ship power and energy requirements which include the needs of low and high power, the needs of flexibility in changes of rhythm and the needs of redundancy and reliability.

As can be seen in the following subsections, there are three main architectures for hybrid propulsion chains [6]:



(a) Series Hybrid architecture. (b) Parallel Hybrid architecture.

Figure 1.4: Hybridization schemes [6].

a) Series Hybrid Architectures

In this type of HPS architecture, two kind of energy system (combustion engines and electric systems) are used in series. The propulsion is done by a variable speed electrical drive (propulsion motor and power converter), as can be seen in the figure 1.4a. The electrical power of the propulsion motor is generally supplied by a set of generators driven by combustion engines. Another option is to use fuel cells to provide electric power or a combination of fuel cells and generators driven by combustion engine. Energy storage systems (ESS) (battery, supercapacitors, etc.) can be connected to the electrical bus to provide propulsion power. This ESS can be used separately or with the generators. In this configuration, electric power is used as an energy vector between the combustion engine and the propulsion motor, so there will be no direct mechanical connection between the engine and the propeller. This configuration allows dissociating the operating points of the combustion engine and of the propeller. This is why it allows increasing the global efficiency of the system by optimizing the operating point of combustion engine and propeller in efficiency point of view. Of course, this advantage would be more significant if several generators and ESS are used, because it allows providing the required propulsion power by choosing an optimal combination of sources in order to keep the running combustion engines in their optimal operating area. However, for small ships, the possibility of multiplying the sources is very limited by volume and mass constraints.

b) Parallel Hybrid Architectures

Parallel hybrid architecture combines two kinds of propulsion motors: electric motors and combustion engines which are mechanically coupled by the same shafts through gearboxes and clutches, as can be seen in the figure 1.4b. The electrical motors and drive are supplied by electrical ESS and/or independent power sources generators driven by combustion engines and/or Fuel Cell for example. The two propulsion systems can be used together or separately. In most cases one of the propulsion systems is devoted to be used for low speed operations (classically the electric system) and the other one (classically the combustion engine) is used for high speeds. The first system can also be used as a boost system to provide a supplementary power during transient (starting, acceleration, etc.). This architecture reduces the number of elements compared to hybridization series and allows optimizing the sizing of each of the energy sources.

c) Series-Parallel Hybrid Architectures

The series-parallel architecture combines both architectures defined previosly. This association allows the passage of a parallel-type operation to a serial type operation.

Hybrid Propulsion systems are based on the combination of several power sources and energy storage systems.

1.1.2 Power Generation Systems

There are different power generation systems associated with propulsion systems for boats. The main ones are listed below:

a) Steam Turbine

A steam turbine is a device that extracts thermal energy from pressurized steam and uses it to do mechanical work on a rotating output shaft [12] [13]. Steam turbines work by expanding steam through a series of blades mounted on a shaft, which turns a propeller to move the ship. They are particularly favoured for their smooth operation, high power output, and efficiency at high speeds, making them suitable for large vessels like liquefied natural gas (LNG) carriers and naval ships.

Steam turbines revolutionized ship propulsion with their introduction in the late 19th century, offering advantages such as small size, reduced maintenance, and low vibration. Despite the need for precise reduction gears or turbo-electric transmission to match the high RPM of turbines to the effective propeller speeds, the benefits included light weight and compact engine rooms.

b) Gas Turbine

Gas turbine engines are defined as aerospace engines that convert the chemical energy in fuel into thermal energy through combustion with air, resulting in the rapid expansion of gas that can perform work in the turbine and exhaust nozzle [14].

The gas turbine engine is recognized for its relative mechanical simplicity and its capacity to use a broad spectrum of fuels. The excellent power–weight ratio of gas turbine engines can be effectively exploited. The performance of gas turbine engines is affected by compressor efficiency, turbine efficiency, pressure ratio, peak cycle temperature, and combustion efficiency. Compressor and turbine efficiencies are continuously increasing through better design and manufacturing techniques, but little room is left for improvement.

The first time a gas turbine was mounted on a watercraft occurred in 1947 in the United Kingdom, when a diesel free power turbine was mounted on a modified gunboat as a proof of concept [15]. This craft endured sea trials for 4 years, until the technology was proven to be a viable concept for the future of naval propulsion. The HMS Grey Goose became the first ship powered entirely by gas turbines, launched in the United Kingdom in 1953, fulfilling its mission to demonstrate the viability of gas turbine propulsion at sea. Since the inception of this technology, gas turbine engines or combinations of them with other engine technologies have been utilized in a wide range of watercraft, ranging from hovercraft all the way up to full-size aircraft carriers, with power outputs starting at single digits of MW all the way to hundreds of MW.

c) Diesel Engines

A diesel engine is a prime mover with the function to convert chemical energy stored in fuel to mechanical energy [6] [8]. Compared to other prime movers such as a gas turbine or steam turbine, the diesel engine has a high efficiency. Its low fuel consumption is the main reason it is widely used in marine applications.

This power source is characterized by a high level of autonomy and by a relatively low cost compared to other sources. Diesel engines are characterized by a high level of greenhouse gases and pollutants emission particularly if they are used at a lower level of power than their rated power.

d) Fuel Cells

Fuel Cells (FCs) work as power sources without greenhouse gas emissions [30] [30]. Instead of burning fuels, FCs have various techniques to transfer chemical energy to electric energy, which include:

- Alkaline Fuel Cell (AFC)
- Proton Exchange Membrane FC (PEMFC)
- Direct Methanol FC
- Solid Oxide FC (SOFC)

The fuel in FC can be methanol, diesel, hydrogen, LNG, and so on. The dc output of FCs is converted to the required voltage through dc/dc converters to fit the dc bus.

FCs have the advantage of a long lifetime, which is attractive in marine applications. However, the slow dynamic response is the main barrier of using FCs in shipboard microgrids due to their slow internal electrochemical and thermodynamic responses [30].

1.1.3 Energy Storage Systems

An intermediate energy storage system can be used to better manage energy and reduce the size of power generation system. The integration of energy storage systems offers interesting solutions for energy efficiency and optimal energy use.

a) Ultra-capacitors

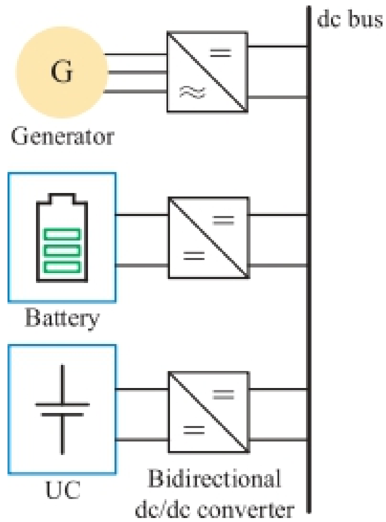


Figure 1.5: Block diagram of dc bus with UC-ESS [30].

Ultra-Capacitors Energy Storage Systems (UCSS), also known like Supercapacitors, can be interesting intermediate energy storage systems with short power transients [6]. They are characterized by a high value of charge and discharge peak power and can be used to rapid dynamic storage with very low specific energy. They are usually associated with other systems, such as batteries, generators and fuel cells, to manage high power short transients, as is shown in figure 1.5. Consequently, combining UCs and batteries not only can meet the long-term load demand but also reduce peak power stresses on the battery packs [30].

b) Flywheels

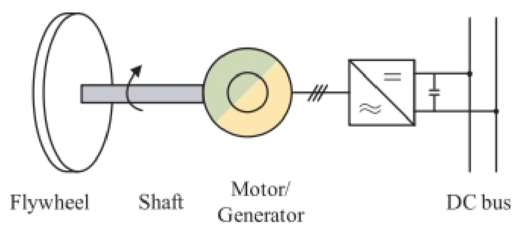


Figure 1.6: Structure of the flywheel ESS (FESS) [30].

Flywheels Energy Storage Systems (FESS) can be used in some particular cases. Like ultra-capacitors, they are characterized by a high power density but a relatively low energy density. Currently, FESS is becoming widely applied in various applications including the marine industry.

Typically, FESS is a kind of electric supply device that stores energy in the form of kinetic energy in a rotating flywheel connected to the shaft of an electric machine [30]. The FESS structure is depicted in Fig. 1.6, containing rotor (flywheel), shaft, motor/generator, and power electronic interface. The amount of stored energy depends on the form, mass, and rotation speed of the flywheel. In charging mode, the flywheel speeds up and store the kinetic energy in the high-speed rotational disk; while in discharging mode, the stored kinetic energy will be released by slowing down the rotating flywheel.

However using flywheels leads to implement safety systems to protect crew and passengers from possible flywheel failure. So their use in small ship context seems very limited.

c) Batteries

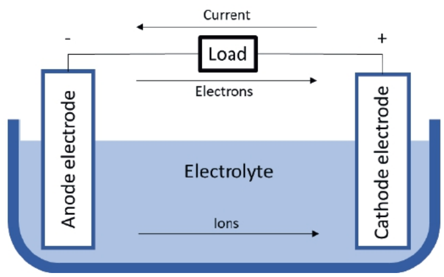


Figure 1.7: Components of a battery [20].

Table 1.1: Energy Storage Systems Comparison [22].

Technology	Battery	Supercapacitor	Flywheel
Energy Density (Wh/kg)	50-250	0.05-5	5-100
Power Density (W/Kg)	50-2000	100k	1000
Lifecycle (cycle)	400-9000	50k	20k-100k
Lifetime (years)	5-10	5-8	15-20
Overall Efficiency (%)	85-99	60-65	15-20

The oldest and most classical method for storing electrical energy is using batteries. In general, as can be seen in fig. 1.7, a battery is comprised of two different poles – a positive electrode called the cathode and a negative electrode called the anode. Then some material is used inside the battery, called electrolyte, that enables ions, or charge carriers, to be transferred back and forth between these poles by electrochemical reactions [20]. A separator can be placed between the cathode and anode, preventing them from touching each other. Hence, when the poles are connected by an electrical conducting material, electrons will flow through the external electrical circuit, and ions to flow through the electrolyte. This process allows energy to be stored or produced in the battery. The chosen energy carrier material, electrode and electrolyte composition, and the shape of the electrodes determine the properties of the batteries.

Battery technology has matured significantly during the past two decades as not only the world but also energy has become increasingly mobile [21].

As shown in the table 1.1, the battery provides the highest power density but lacks overall efficiency compared to energy storage systems. The flywheel, on the other hand, is the most durable system, offering a cycle life of between 20,000 and 100,000 cycles and a service life of between 15 and 20 years. However, due to its low power density, it may not provide enough power for a vessel.

The comparation seen in table 1.1 helps to understand why batteries are usually chosen over other types of energy storage systems: because of their reliability and high energy density.

1.1.4 Battery Systems in Maritime Applications

Battery application onboard ships can have multiple functional roles. While batteries can fully power a vessel for short distance or duration, improving performance and energy efficiency of the overall vessel is often the key purpose.
In 2018, the European Maritime Safety Agency (EMSA) commissioned DNV GL to perform a study on the use of electrical storage systems in shipping, with the objective of providing an overview of technology, research, feasibility, regulations and safety of battery systems in maritime applications [20]. Figure 1.8 shows the different uses of batteries recognized by EMSA:

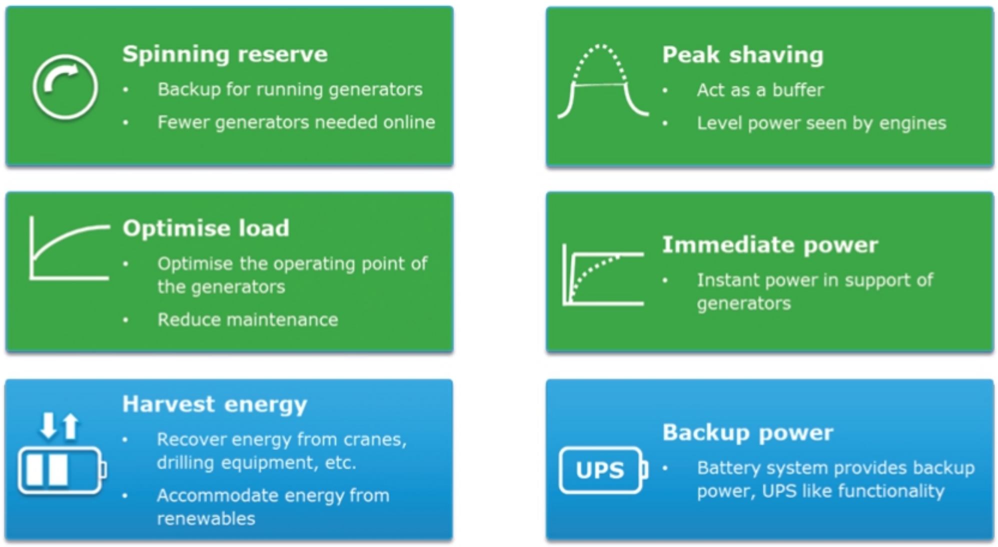


Figure 1.8: Funcitonal roles of battery systems onboard ships [20].

- **Spinning reserve:** This mainly applies to diesel electric dynamic positioning vessels. Using a battery-provided spinning reserve allows the vessel to reduce the number of generators running required to maintain the spinning reserve. It allows for lower fuel consumption, lower emissions, higher efficiency of running engines and lower maintenance costs due to reduced running hours [24].
- **Peak shaving:** Battery energy storage systems also provide additional electric power required for short-term additional power demands. In conventional propulsion systems, it is necessary to run additional diesel generator to meet those demands. Using a battery instead, the additional generator can be turned off, therefore saving on fuel and wear and tear of the engine. Vessels equipped with heavy electricity consumers such as cranes or electric-driven thrusters will benefit from this arrangement [24].
- **Optimise load:** This concept is recognized also like *Load Leveling*. BESSs are used to keep the load of the diesel engine/generator constant, allowing them to operate at maximum efficiency by charging when the load demand is low, then contributing to the fulfilment of the load request [26].
- **Immediate power:** This concept is recognized also like *Ramp Levelling* and is used to enhance dynamic engine performance [26]. Gas engines and other energy sources are not capable of handling fast load variations. In such cases, a BESS can be utilized to take care of the variations, while the main energy source produces slowly varying power [24].

- **Harvest energy:** This service consists of a time shift of energy generation by storing energy from photovoltaic panels, and/or other renewable sources such as wind, and then releasing it when they are absent. In addition, regenerative power improves ship energy management, using BESS to store energy obtained from the regenerative process of double speed engine braking (e.g., ships with cranes or large winches) [26].

- **Backup power:** BESS allows the vessel to stop diesel engines when onboard electricity consumption is low (e.g., in port). The generators are started only to charge the batteries. It is possible to reduce emissions because the diesel engines operate at optimal load (with maximum efficiency and minimum specific fuel consumption) during charging [24].

Additionally, other authors acknowledge the service of the **zero-emission mode** [26], that utilizes full electric propulsion for a limited time, reducing ship noise and vibration. Here, a BESS allows for switching-off all the onboard internal combustion engines. However, when fully powered by its own BESS, these ships are subject to specific technical limitations, such as max speed and maximum operation time, depending on battery power and energy content.

Among the commercially available battery technologies that can be used to provide services to boats, considering that is not feasible or possible to cover the entire spectrum of all technologies, because for any given chemistry there is most often a wide range of products representing different levels of quality and performance, information on some of them is provided [20]:

- **Lead-acid:** Lead-acid batteries are supplied worldwide by a large supplier base at a very low cost. Automotive batteries and industries where standby electrical power is critical are the biggest markets. It is considered to be very safe, since the electrolyte and active materials are not flammable; although the batteries are known to produce hydrogen under charging.The main drawbacks for such batteries are the specific energy and energy density. The power density is considered high for lead acid battery but is also here outperformed by Li-ion.

- **Nickel Cadmium (Ni-Cd):** Nickel cadmium batteries are mature technologies, that are widely used in UPS applications, for example. This technology is economically priced and presents the lowest per cycle cost. Good ventilation is important in the room to avoid explosive concentration of hydrogen. Ni-Cd have memory effect that causes a loss of capacity if not given a periodic full discharge cycle.

- **Nickel Metal Hybride (Ni-MH):** Nickel metal hybride has become one of the most readily available rechargeable batteries for consumer use and is used for the most at the same applications as Ni-Cd. It provides 40 percent higher specific energy than the standard Ni-Cd. Good ventilation is important in the room to avoid explosive concentration of hydrogen. The battery is more delicate and trickier to charge than Ni-Cd. High self-discharge is of ongoing concern, and devices with a Ni-MH battery gets “flat” when put away for only a few weeks.

- **Sodium Sulphur (Na-S):** The sodium sulphur batteries, are manufactured from cheap and plentiful raw materials and has higher specific energy compared to Li-ion batteries. However, the manufacturing processes and the need for insulation, heating and thermal management make these batteries quite expensive and counteracts the benefits, because these type of batteries need molten sodium-metal as a anode, making it necessary to operate at 300°C.

- **Lithium-ion (Li-ion):** Lithium-ion batteries can consist of different material and chemistries in the electrodes and the electrolyte, as well as manufacturing processes and related materials.

Has the highest specific energy of commercially available batteries and highest energy density of commercially available batteries. Some disadvantages are flammable electrolyte and potentially limited availability of materials. Among the different **cathodes** and **anodes** of lithium-ion batteries are shown in the table 1.2:

Table 1.2: Lithuim-ion batteries types [20].

Lithium-ion Battery	
Cathode Chemistries	**Anode Chemistries**
Nickel Manganese Cobalt Oxide (NMC)	Graphene
Nickel Cobalt Aluminium (NCA)	Titanate
Lithium Cobalt Oxide (LCO)	Silicon
Lithium Manganese Oxide Spinel (LMO)	
Lithium Iron Phosphate (LFP)	

- **Li-polymer:** Also known like *solid-state battery*, this type of technology gives freedom in design of the battery geometry and improvement of the packing efficiency of the cells. It facilitates a long cycle life and offers the possibility of employing high-voltage cathodes. All these effects increase the practical battery energy density. The biggest challenge however is regarded as the volume changes of the electrodes during charging and discharging, that causes loss of contact between the electrode and the electrolyte. These effects all make ion conductivity low. An polymer interlayer between the electrodes and the solid electrolyte has been proposed to overcome these challenges. Some disadventagse of this technology are its high production cost and low lifetime.

- **Lithium-Sulphur (Li-S):** Like conventional lithium-ion batteries, these batteries use Li+ ions as energy carriers. They react with sulfur at the cathode giving some high theoretical values for specific energy and energy density. Hence these batteries will be very attractive for marine applications. Safety concerns related to these batteries, like dendrite formation, have been improved. However, this compromises the specific energy and energy density and increases the cost dramatically. Other obstacles to overcome are high electrical resistance, capacity fading, self-discharge, mainly due to the so-called shuttle effect.

About the different types of batteries described above, there is a technical comparation in the table 1.3.

Table 1.3: Energy Storage Systems Comparison [22]

Battery	Energy Density (Wh/kg)	Volumetric Energy Density (Wh/L)	Cost (USD/kWh)	Lifecycle	Efficiency (%)
Lead-Acid	20-40	60-110	165	500-1000	50-95
Ni-Cd	20-60	60-150	-	2000-2500	70-90
Ni-MH	50-70	140-300	146	500-2000	89-92
Na-S	~120	150-300	400	>1500	86-92
Li-ion	50-250	250-675	200-600	400-9000	85-99
Li-polymer	~150	250-730	-	-	-
Li-S	400	350	400-500	1500	-

Different ships have different operating profiles and batteries must respond to specific energy and power demand while also having in consideration the desired/expected life-cycle for the battery [20]. In this scenario, lithium batteries are superior to the other mentioned batteries in terms of energy density, lifecycle, and efficiency [22]. Lithium batteries have been gaining traction throughout the years, with many applications including portable electronics such as mobile phones, and automobile applications such as electric vehicles. Also are slowly being adopted for aerospace, military and marine applications. Lithium batteries are highly versatile energy storage devices that can be altered to meet specific demands, making them one of the best choices for marine application.

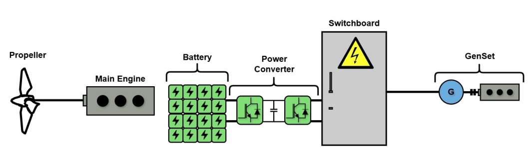
Considering the incorporation of batteries into the propulsion system of vessels, different propulsion architectures are recognized that provide a different use for batteries, even without considering them, as is the case with conventional propulsion architecture [20] [21]. This can be seen in the figures 1.9, 1.10 and 1.11.

a) Diesel-mechanic engine conventional system

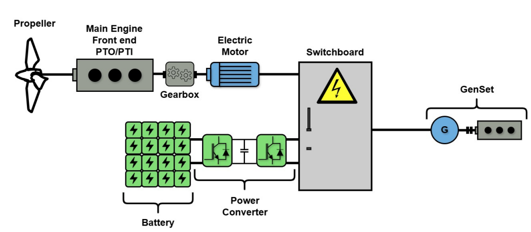
Figure 1.9: Traditional diesel-mechanic propulsion of a large merchant vessel.

In general, the configuration shown in Figure 1.9 is the most common installation on large ocean-going vessels [21]. It consists of a two-stroke main engine responsible for propulsion, while the ship's load is covered by auxiliary engines/sensets installed on board the vessel.

b) Diesel-mechanic engine hybrid system



(a) Semi-hybrid diesel-mechanic propulsion (large merchant vessel) with hybrid electric grid.



(b) Full-hybrid diesel-mechanic propulsion of a large merchant vessel with front end PTO/PTI from RENK.

Figure 1.10: Semi-hybridization.

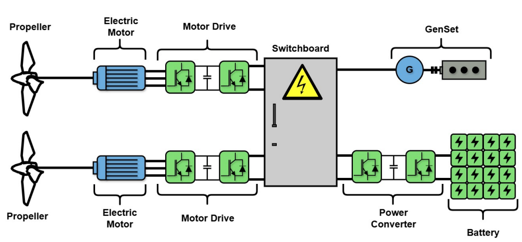
In Fig. 1.10a, a battery is included in a traditional system where the electric grid is separated from the propulsion of the vessel [21], so Iin this case, hybridization exists but is related to the vessel's electrical network, not to the propulsion system. The battery will in such a solution be effective for smoothing the connected hotel electrical loads and contribute to handle large load steps or peaks [20]. When large load steps are increased, the number of auxiliary motors also increases. In some cases, the load can regenerate energy, for example, in crane operations. In such cases, the battery can be used to capture the energy.

In 1.10b, a battery is included in combination with a shaft generator power take-out (PTO) and power take-in (PTI) [21]:

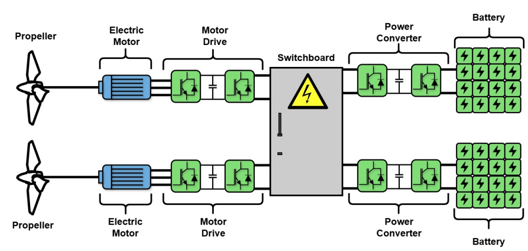
- **Power Take-Out (PTO):** The electrical/mechanical hybrid solution allows electricity to be generated by the main engine.
- **Power Take-In (PTI):** The electrical/mechanical hybrid solution allows propulsion power to be produced by generator sets and batteries. Additional power is possible (boost mode) when the main engine and PTI motor are running in parallel.

In addition, a ship with a hybrid electric/mechanical propulsion system may have a direct current distribution system, which can adjust the speed of the primary generator engines to the optimum fuel level depending on the load. This will reduce fuel consumption and minimize environmental impact [20].

c) Hybrid/electric propulsion system with electric motors



(a) Full-hybrid diesel-electric propulsion with batteries in a hybrid system.



(b) Pure electric propulsion with batteries as the primary energy source.

Figure 1.11: Propulsion system based on electric motors, with different power sources.

A full-hybrid diesel-electric solution, as is shown in fig. 1.11a, reduces the noise and vibration level on the ship [20]. Supplementary to be a source of energy for propulsion, the batteries will also smooth the load variations on the generator sets. This type of solution can for instance facilitate the use of zero emission operation when entering a harbour. Another version of this solution is with the batteries distributed into the drives directly [20]. In this way, each propulsion unit is independent of a common power source, which could be valuable for vessels that require highly reliable propulsion thrust.

In the fig. 1.11b, the propellers are connected to electric motors, which are driven by the energy stored in a battery system that is typically charged from shore [21]. In some battery-electric systems, a smaller diesel generator is sometimes included to ensure operation if the batteries fail to charge or to enable longer voyages, i.e. when the vessel has to dock [21]. For such a solution the batteries will be charged through an AC/DC converter. The converter can either be located on board the vessel or on shore. The fig. 1.11b shows two independent battery systems that deliver power to the thruster. This is according to class rules, which require that two independent battery systems are installed to provide propulsion power in case one of the systems fail [20].

Electrification of ship's propulsion is increasingly recognised as a core part of the maritime industry's future [24]. According to [27], the adoption of alternative fuels in the global fleet by number of ships is dominated by Liquefied Natural Gas (LNG) together with hybrid/battery ships, as is shown in Table 1.4.

Table 1.4: Ships 2022.

Alternative Fuel Ships		
	In operation during 2022	**Ordered**
LNG	926	534
Hybrid/battery powered ships	396	417
Total	1349	1046

In Table 1.4, can be seen that the proportion of hybrid/battery propulsion systems has been increasing since 2022. However, it should be noted that the use of batteries as a source of propulsion for ships is limited to certain types of ships, the main ones being ferries, dynamic positioning ships and platforms, dredging, short-range ships, wind farm support vessels, and tugs. For ships that travel most of the time at constant speed, such as container ships, it make no sense to replace a diesel mechanical drive by hybrid concepts, because the diesel-mechanical drives are optimized for this operation and then getting a lower consumption than a diesel electric or hybrid system [5].

Harbour tugs are a sector of the maritime industry with great potential for the implementation of hybrid propulsion, as they experience significant variations in power demand during operations [6].

Ports are imposing more and more stringent regulations on exhaust and noise emissions. Those demands for cleaner operation can be easily met by the implementation of batery energy storage systems (BESS) (with hybrid-electric propulsion). Using a combination of electric power, BESS and combustion engines, a hybrid tug optimizes engine loading, resulting in lower specific fuel consumption, higher efficiency, lower emissions and lower fuel consumption.

1.1.5 Tugboats

Figure 1.12: Tugboat

Tugboats, as can be seen in fig. 1.12, are ideal candidates for hybridization due to their operational profiles, which involve prolonged idling, low-speed maneuvering, and frequent speed changes that lead to inefficient fuel consumption [22]. Among vessel types, tugboats—used for towing and maneuvering large ships—are among the highest emitters per unit of energy due to their highly variable load profiles [5], since they spend a lot of time at 10-20% load (inefficient for diesel engines) and very short peaks at 100%. This technically justifies the use of batteries.

1.2 Challenges and Research Opportunities

Several studies have examined hybrid propulsion systems [9], identifying the main topics of study related to hybridization, which are listed below [24]:

- Optimal sizing of the batteries.
- Optimal control strategy for the parallel operation of a battery and diesel generators.
- Control and safe operation of lithium battery packs.
- BESS control system architecture.
- Design of a battery management system (BMS).
- Safety assessment of the BESS, its elements and safeguarding measures required by the classification societies.
- Safety assessment of the BESS, its elements and safeguarding measures required by the classification societies.

- Degradation of Li-ion batteries.
- Battery application for large vessels.
- Reduction of fuel consumption and emissions.

Some studies related to hybridization are listed below, with emphasis on those related to tugboat hybridization:

- The feasibility of hybrid propulsion concepts is examined for different types of vessels, including tugboats [5], showing that there are definitely opportunities to save energy by using hybrid drive systems on ships, even return the much higher investment by savings in fuel.
- A genetic algorithm to obtain the optimal operating points regard with series hybrid configurations in power generation, especially for the case of heavy duty hybrid drive applications [19].
- A marine hybrid propulsion system, focusing on vector control of the electric motor during different modes and verifying the control feasibility [23].
- An analytical design methodology for BESS in hybrid marine vessels [25].
- The optimization of hybrid propulsion system design for a tugboat has been explored, presenting a methodology that streamlines powertrain component sizing and control, minimizing costs for a specific operating profile [28].
- A coordinated control strategy for a variable-speed hybrid tugboat have been presented to improve fuel economy. The proposed strategy, validated through simulations and a smallscale experimental testbed, showed reduced costs and lower CO2 emissions [29].

However, the challenge lies in finding a clear and detailed methodology for the hybridization of vessels such as tugboats that provides a path from the conceptual phase to the technical details and subsequent implementation, taking into account technical and economic aspects.

1.3 Thesis Objectives and Outline

The overall objective of this work is to develop a mathematical model of the battery bank for a hybrid tugboat, involving a methodology for the design and sizing of the battery bank, analyzing the technical and economic feasibility of its implementation. This mathematical model can be tested with DC/DC converter models and other elements within the hybrid powertrain.

Specific objectives include:

- Model the operation of a conventional tugboat
- Define hybrid architecture.
- Propose the service power of the electrical system.
- Preliminary design of the battery bank based on technical and economic criteria.
- Simulation of the proposed cell and battery bank.

1.4 Methodology

The methodology to follow for the development of this work is:

- **State of Art:** Review of existing hybrid propulsion system types and battery technologies.
- **Mathematical Analysis and Modeling:** Theoretical study of the tugboat's load profile and the implementation of batteries to support the vessel's operation.
- **Results:** Simulation of the models obtained through Plecs or MatLab to complete the conceptual analysis and analyze the results related to the battery bank.

1.5 Scope and Limitations

The limitations of the work are:

- **Architecture Layout**: As discussed in section , there are different ways to make batteries supply power to the propulsion system and internal loads of the ship. Based on this, a semi-hybrid diesel-mechanical propulsion system will be established for the tugboat and compared with the tugboat's conventional propulsion system.
- **Bill of Components (BOM)**: Within the list of components of the hybrid propulsion system, the associated cost of each will be determined, estimating the costs of those components for which no commercial information is available.
- **Economic analysis**: Although information on costs related to the hybrid propulsion system will be added, it should be noted that inflation or deflation will not be taken into account in this exercise. Furthermore, this economic assessment does not consider the cost of the port charger.
- **Simulation considerations**: Based on the data provided on the tugboat's operational load profile, simulations will be performed on the sized battery bank. For power converter simulations, simulations will be performed with a shorter duration, given the amount of data required to analyze the battery bank's discharge profile. The thermal effects of the semiconductors used shall be disregarded.

In conclusion, this work will contribute to the analysis of the different initial stages of hybridizing a tugboat, considering the essential technical aspects that allow us to understand the benefits of this conversion, with an emphasis on the electrical stage, specifically the batteries.

1.6 Chapter Summary

Chapter 2

Identification of essential technical and operational characteristics of a tugboat and analysis of the associated operational load profile.

Chapter 3

Hybridization analysis and new operational services that a hybrid tugboat can offer based on its cargo operational profile.

Chapter 4

Design and sizing of battery banks, considering different technologies, technical variables, and economic analysis.

Chapter 5

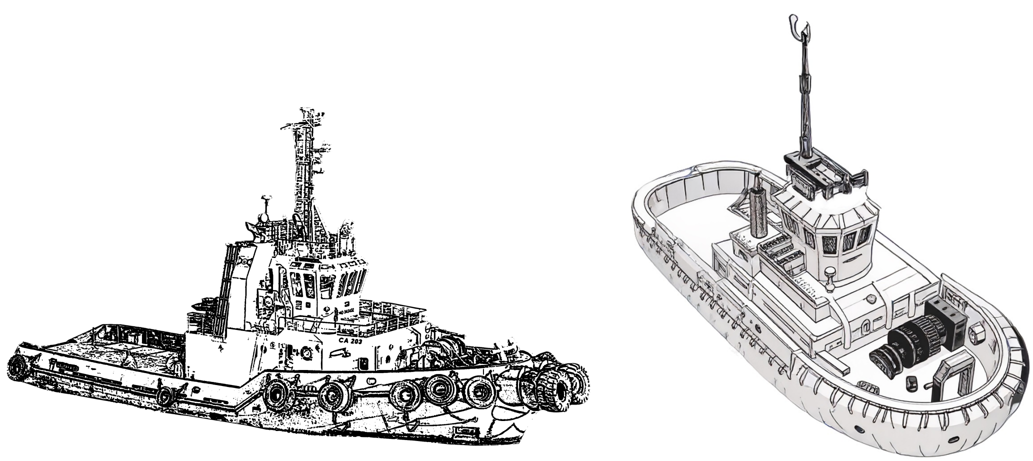
Simulation of the battery bank selected for the vessel, considering the mathematical model of a cell.

Chapter 2

Tugboats: Description and Operation

Tugboats are small but powerful vessels designed to assist in docking ships or moving structures without their own propulsion. Due to their robust engineering and high power, this chapter analyzes their technical and operational characteristics based on a real service profile in Chile, in order to identify the key aspects necessary for their hybridization.

2.1 Overview



(a) Harbour Tug.

(b) Escort Tug.

Figure 2.1: Different types of tugboats.

A tugboat is technically defined as a type of vessel specifically designed to assist other ships in moving by providing a controlled amount of force. They are essential vessels in ports, as is shown in fig. 2.1, where they assist in the movement of floating objects and in emergency tasks [31].

Maritime towing is legally defined as the action of pulling a vessel or floating device that lacks its own propulsion or navigational autonomy. This activity has a dual function: on the one hand, precision assistance in port maneuvers and, on the other, deep-sea towing or salvage. Both functions require specific design engineering and system redundancy to manage the different operational risks of each environment [33].

The importance of tugboats has grown exponentially with the phenomenon of naval gigantism [34]. Large modern ships, such as cruise ships and oil tankers, have tremendous momentum (inertia) that prevents them from stopping quickly or maneuvering effectively on their own in narrow channels or docks [32].

The tugboat is essential for managing the ship's inertia and ensuring operational safety during maneuvers and contingencies [33]. Operational efficiency depends on fuel consumption relative to load, factors that are conditioned by the configuration of the propulsion and energy system. The efficiency of conventional propulsion systems varies between 38% and 41% for a power range between 100 kW and 2,500 kW. For vessels with higher power ratings, efficiency increases to 48% [17] [18].

2.1.1 Types of tugboats

The classification of tugboats is primarily based on their operational domain and assigned primary mission. The usage and functions of tugs vary from port to port, as different ports have different requirements and intakes.

a) Harbour Tugs

Figure 2.2: A harbour tug in the port of Algeciras [36].

These vessels are specialized in assisting large ships entering and leaving port limits, as can be seen in fig. 2.2. Their design prioritizes agility and the ability to apply force in any direction with maximum efficiency [31].

- **Design:** Typical dimensions vary in length between 20 and 30 meters, with drafts of 3.0 to 4.5 meters. Their empty displacement speed ranges from 5 to 13 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 400 - 3,000 hp (300 and 2200 kW) [35].

b) Offshore/SAR Tugs

Figure 2.3: SAR BALDER, an offshore Tug built in 2006 [38].

An offshore or *search and rescue* (SAR) Tug, is designed to operate in the open sea, these tugboats are characterized by their robustness, greater autonomy, and ability to withstand severe ocean conditions. An example of this vessel is shown in fig. 2.3.

- **Design:** Typical dimensions vary in length between 25 and 70 meters, with drafts of 4.5 to 6 meters. Their empty displacement speed ranges from 5 to 13 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 1500 - 6,000 hp (1120 - 4470 kW) [35].

c) Escort Tugs

Figure 2.4: Robert Allan Rastar 3200 Escort Tug [37].

These tugboats perform a critical safety mission. Their main objective is to provide preventive escort to ships carrying hazardous materials (such as LNG or crude oil) in high-risk areas, standing ready to counteract any possible loss of control of the assisted vessel.

- **Design:** Typical dimensions vary in length between 40 and 80 meters, with drafts of 5 to 6 meters. Their empty displacement speed ranges from 15 to 16 knots (9.3 - 24 km/h). The power of these tugboats can vary considerably, generally between 6,000 -20,000 hp (4470 - 14,900 kW) [35].

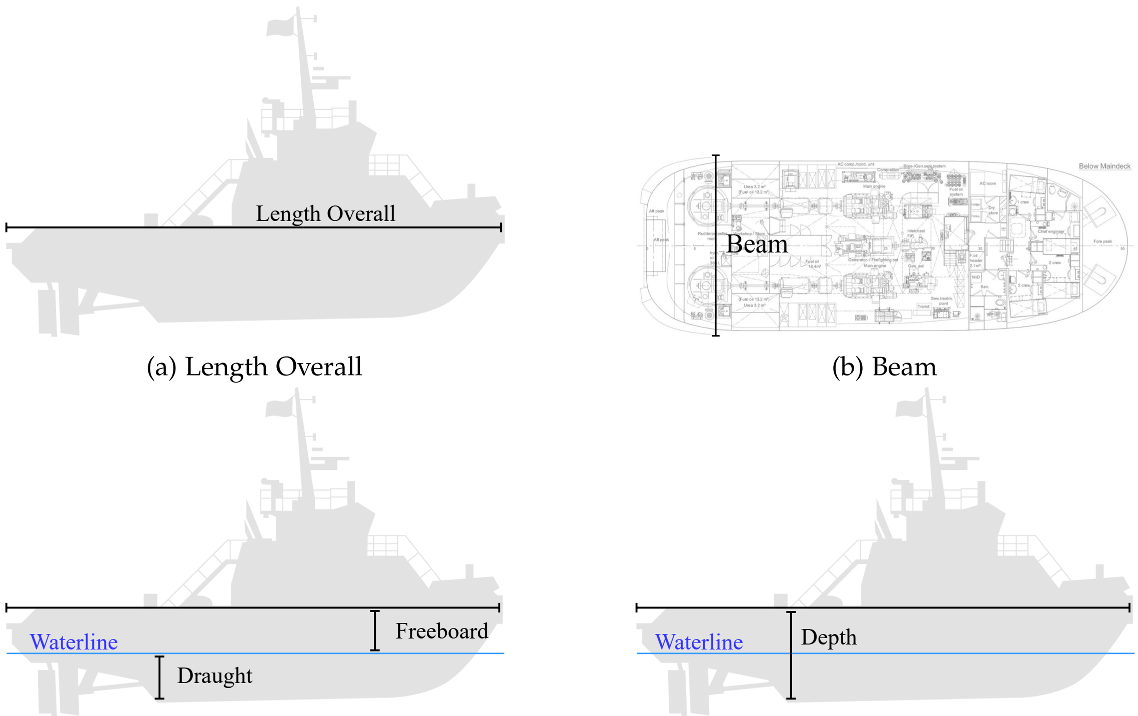
d) Tugs Comparation

Below, in the table 2.1, there is a summary of key operational and dimensional characteristics:

Table 2.1: Operational Classification and Dimensional Parameters of the Tugboat

Type of Tugboat	Operation	Design objective	Typical BP
Harbour Tug	Protected maritime and port areas such as bays or estuaries.	Maneuverability, Reduced Draft	40-80
Offshore/SAR Tugs	Open Ocean	Autonomy, Stability, SAR Power	60-200+
Escort Tugs	High-Risk Areas (LNG, Crude Oil)	Rollover Control, Side Pull Capacity	60-120

2.1.2 Dimensions



(a) Length Overall

(b) Beam

(c) Draught and freeboard

(d) Depth

Figure 2.5: Physical dimensions of the tugboat.

a) Length Overall

The overall length is measured from the bow to the stern of the vessel in a straight line, excluding bowsprits, rudders, outboard motors and their supports, cranks and other accessories, fixed elements, and extensions [45], as is shown in figure 2.5a.

b) Beam

A tugboat's beam is the distance between the two widest points of the hull [46], as is shown in the figure 2.5b. It is usually measured at the waterline, although other specifications such as maximum beam or deck beam may be used. Beam is a critical dimension in naval architecture, affecting the stability, maneuverability, and overall performance of the vessel [47].

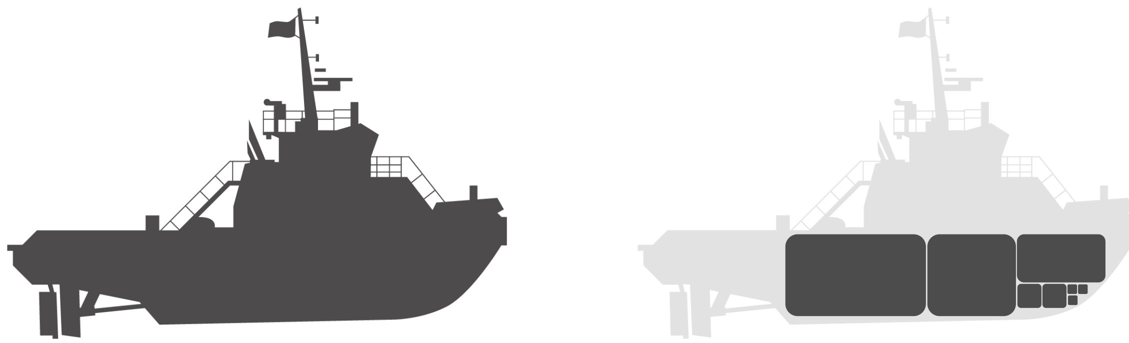
c) Draught

Draught or Draft is the vertical distance from the moulded base line amidships to the actual waterline [12], as is shown in the figure 2.5c.

d) Depth

Depth or molded depth of a vessel is measured at the center of the length, from the top of the keel to the highest continuous deck on the side [12], as is shown in the figure 2.5d.

2.1.3 Volumen and Weight



(a) Gross Tonnage Tug (b) Net Tonnage Tug

Figure 2.6: Total volume and available volume inside the tugboat.

a) Gross Tonnage (GT)

Gross Tonnage (GT) measures the "total inside space" of the ship, including all areas like cargo holds, engine rooms, and crew spaces. It's about "volume", not weight [12] [48], as is shown in figure 2.6a.

- Under vessel measurement rules of various nations, the Panama Canal and the Suez Canal gross tonnage is a measure of the internal volume of space within a vessel in which 1 ton is equivalent to 2.83[m^3] or 100[ft^3].
- **ICTM 1969 Agreement**: Introduced a standardized numerical value based on a logarithmic function of the total molded volume (V). The official formula is:

$$GT = K_1 \cdot V$$

Where V is the volume in [m^3] and the coefficient K_1 is calculated as $0.2 + 0.02 \cdot \log_{10}(V)$. Under the ICTM, a unit of volume is worth more on a large ship than on a small one, so there is no fixed definition of a "ton".

b) Net Tonnage (NT)

Net Tonnage (NT) is focuses on the space available for cargo [12] [48], as is shown in figure 2.6b.

- Net tonnage according to the national and canal rules is derived from gross tonnage by deducting an allowance for the propelling machinery space and certain other spaces.
- Net tonnage according to ICTM is a logarithmic function of the volume of cargo space, the draft-to-depth ratio, the number of passengers to be accommodated, and the gross tonnage.

c) Deadweight (DWT)

DWT refers to weight and indicates how much the ship can safely carry. It is expressed in long tons of 2,240 pounds (1,016.0469088 kilograms) [49].

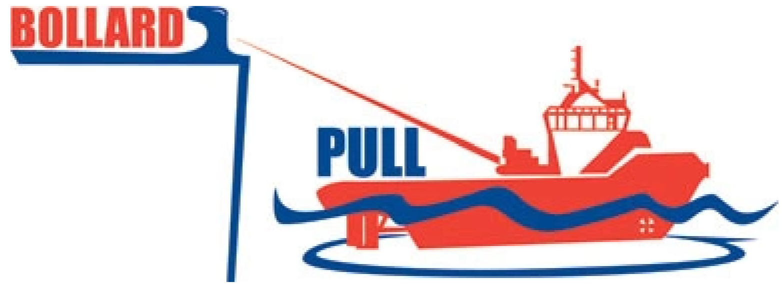


Figure 2.7: The bollard pull test to measure a vessel's maximum towing capacity [32].

d) Displacement Tonnage

Displacement tonnage is used to define the size of naval ships. It refers to the weight of the volume of water displaced by a vessel in normal seagoing condition [49].

2.2 Technical Characteristics of the Propulsion System

To understand the work of a tugboat as a maritime unit, it is essential to analyze both its essential operational characteristics and the elements of its propulsion system that enable it to function.

2.2.1 Bollard Pull - BP

Bollard pull (BP) is the fundamental metric for quantifying a tugboat's pulling power.

Equivalent to automotive torque, BP measures the static traction of the vessel. The process involves securing the vessel to a fixed point and recording the tension of the line using load cells at full thrust, as is shown in the figure 2.7.

The BP measures load capacity in units such as kN or tf. All tugboats require a fixed point certification issued by classification societies (such as ABS or BV) to validate their operability [34].

The tugboat's design seeks to maximize its pulling power in relation to its dimensions. A common method of achieving this is to install a nozzle around the propeller, thereby optimizing propulsion efficiency. [34].

2.2.2 Propulsion Engineering for Maneuvering

The maneuverability of a tugboat, which directly influences its operational efficiency and safety, is intrinsically linked to the design of its propulsion system.

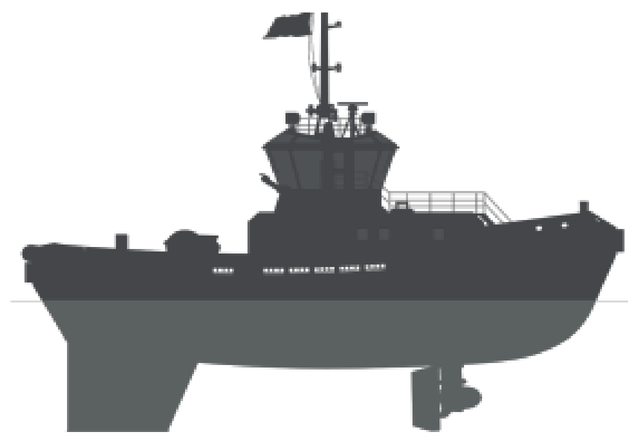
a) Conventional Propulsion

Figure 2.8: Conventional propulsion: A tug fitted with fixed propellers, single or twin screw (left or right handed) and single rudders with fixed nozzles [50].

It uses a standard diesel engine connected to a propeller, as is shown in the figure 2.8. It is a proven and simple system, with good efficiency in linear towing tasks within ports and coastal areas. However, it offers limited lateral maneuverability compared to more advanced systems [43].

b) Azimuthal Stern Drive (ASD)

(a) Azimuth Stern Drive (ASD): A tug fitted with two azimuth thrusters in nozzles at the stern, capable to direct all its thrust in any direction [50].



(b) ASD Tractor: A tug with azimuth thrusters located generally forward off the midship and a rudder-shaped fin at aft side. [50].

Figure 2.9: ASD Tugs representations.

These are rotary thrusters, where a motor is capable of directing all its thrust in any direction [43]. Its two independent propellers maximize maneuverability, allowing the ship to turn 360° on its own length [44], as is shown in the figure 2.9a. The only difference with conventional tugboats are the modifications to the stern and structure to house the propellers and withstand the forces exerted by the nozzle rudder, which are different from those exerted by a conventional rudder and propeller [44].

Z-Drive propulsion systems optimize power transmission and provide superior maneuverability when towing thanks to their efficient mechanical configuration [43].

An ASD Tractor Tug generally has two fixed pitch azimuth propellers in the center of the boat or slightly forward of the center of the boat, as is shown in the figure 2.9b.

To compensate for the loss of directional control caused by very forward-mounted thrusters, a stern fin is installed to stabilize the trajectory [51].

c) Voith-Schneider Propulsion (VSP)

Figure 2.10: Voith-Schneider Propeller (VSP) propulsion: A tug with VSP generally located forward off the midship and a rudder-shaped fin at aft side [50].

This system, as is shown in the figure 2.10, offers omnidirectional thrust and total steering control, resulting in formidable positioning accuracy. These harbor tugs have been equipped with *VoithSchneider* propellers, a unique type of propulsion system consisting of oscillating blades that allow them to have total, targeted control over the vessel without having to adjust the orientation of the main engine or propellers [43].

This propulsion system allows for highly precise operation in any direction due to its agility, making it ideal for rescue operations as well as in ports crowded with high-density maritime traffic [43].

d) Propulsion Comparation

The transition from a conventional system to an azimuth system, and even more so to a Voith-Schneider system, involves an increase in engineering complexity and acquisition cost, but this is fully justified by the need to operate with high precision and safety in confined spaces or under adverse weather conditions. The design of these advanced systems is crucial for safety, as they allow the tugboat to apply the required force (BP) without compromising lateral stability and minimizing the risk of capsizing.

Table 2.2 compares the different types of propulsion discussed in this document.

Table 2.2: Comparison of Propulsion Systems in Tugboats and Performance.

Propulsion System	Advantages	Maneuverability	Typical Application
Conventional	Low Cost, Proven Design	Limited (Good in a straight line)	Single Coastal/Offshore Trailer
ASD	High BP, Axis Rotation, Port Agility	Good	Port Towing and Escort
VSP	Omnidirectional Thrust, Extreme Stability	Very Good	Congested Ports

2.2.3 Main Diesel Engine

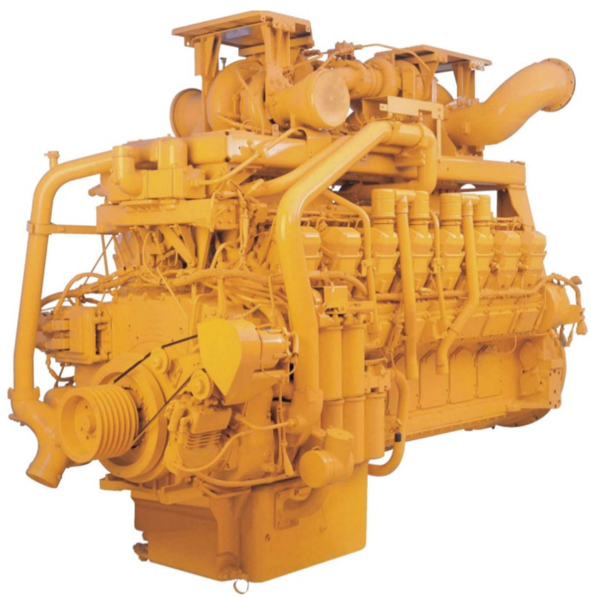


Figure 2.11: Main diesel engine for a tugboat propulsion system.

Diesel engines for propulsion, as is shown in figure 2.11, are designed to increase efficiency and keep costs to a minimum, whether in port or at a terminal. They are characterized by high power density, acceleration, and static traction.

The system supports continuous variable load and an emergency overload of 10% (maximum 1h/12h), without exceeding a total limit of hours per year.

BHP represents the gross power of the propulsion engine prior to mechanical losses in the transmission system and axle [52]. It influences maneuverability when docking and handling in adverse conditions.

BHP measures the gross power of the engine; however, the actual useful power is reduced by transmission losses and hydrodynamic resistance.

2.2.4 Conventional Propellers

a) Fixed Pitch Propellers

(a) Fixed pitch propeller.

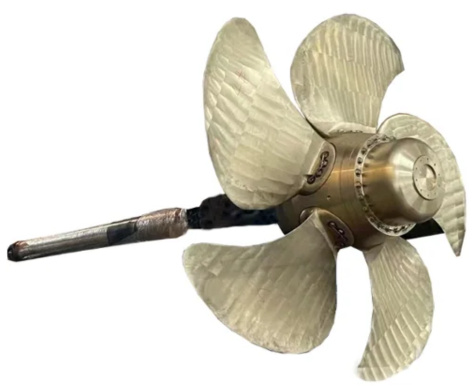
(b) Fixed pitch propeller with nozzle.

Figure 2.12: Fixed pitch propellers configuration.

These types of propellers, as is shown in the figure 2.12a, maintain their configuration unchanged, meaning that their blades do not move on their axis [35].

Also, there are fixed pitch propellers with improved propulsion efficiency thanks to the nozzle that aids vector thrust [35], as is shown in the figure 2.12b.

b) Variable Pitch Propellers



(a) Variable pitch propeller.

(b) Variable pitch propeller with nozzle.

Figure 2.13: Variable (controllable) pitch propellers configuration.

Unlike fixed-pitch propellers, in a variable pitch propeller (also known like controllable pitch proppeller) each of the blades can rotate on its own axis [35]. An example can be seen in the figure

2.13a, where each blade is attached to the central shaft by means of an anchor that allows it to rotate.

In some cases, the propulsion efficiency can be improved thanks to the nozzle that aids vector thrust [35], as is shown in the figure 2.13b.

e) Propellers Comparation

Variable pitch propellers are more efficient than fixed pitch propellers because the adjustment of the blades allows for maximum power or variable speed, which is not the case with fixed pitch propellers that are designed for specific ordinary operating conditions.

It should be noted that variable pitch propellers do not provide as much thrust for reverse navigation as fixed pitch propellers do, which limits the tasks that tugboats can perform.

2.2.5 Special Propellers

(a) Schottel system. (b) VSP propeller.

Figure 2.14: Special Propellers.

a) Schottel system

In this system, the propeller positioned at a 90º angle to the vertical, fixed to the shaft is a nozzle inside which the propeller rotates, as is shown in the figure 2.14a. The entire assembly can rotate 360º on the vertical axis, providing excellent maneuverability and allowing movement in any direction.

b) Voith-Schneider System (Cycloidal Propeller)

This propeller system consists of a rotor that rotates on a vertical shaft fixed to the hull approximately at its pivot point, as is shown in the figure 2.14b. It is equipped with several blades that pivot on vertical axes controlled from the bridge by a device called a steering control. This device is responsible for directing the thrust in the opposite direction to the desired thrust. The mechanism is synchronized to center the thrust at a single point.

2.3 Naval Maneuvers and Assistance Techniques

Towing operations are governed by strict procedures that seek to maximize safety in dynamic and often restricted environments.

2.3.1 Docking and Undocking Procedures

The assistance of tugboats is crucial during docking and undocking maneuvers. The undocking maneuver can be as laborious as the docking maneuver and must be planned in advance. The objective is to leave the assisted vessel clear of the dock, in a safe position that allows the rudder and propeller to be effective so that the vessel can gain control and momentum.

a) Operational Coordination

The maneuver is initiated by request via VHF radio. The bow and stern operating personnel receive instructions to unmoor the lines.

b) External Agent Management

It is essential to consider the effects of winds and currents. Strong winds from outside can add to the effects of the engine in reverse during undocking, creating a dangerous situation that can violently push the stern or bow toward the dock. The ability of the highly maneuverable tugboat to apply instantaneous lateral force reduces the time the assisted vessel is exposed to the combined and unfavorable action of wind and propulsion, mitigating the risk of collision. If the wind action is severe, tugboat assistance is mandatory, especially if the actions of the assisted vessel's capstan are insufficient.

2.3.2 Tugboat Performance

The assistance of tugboats is crucial during docking and undocking maneuvers. The undocking maneuver can be as laborious as the docking maneuver and must be planned in advance. The objective is to leave the assisted vessel clear of the dock, in a safe position that allows the rudder and propeller to be effective so that the vessel can gain control and momentum.

Tugboats use various configurations to apply controlled force:

a) Tugboat Working on a Beam or Over a Cable

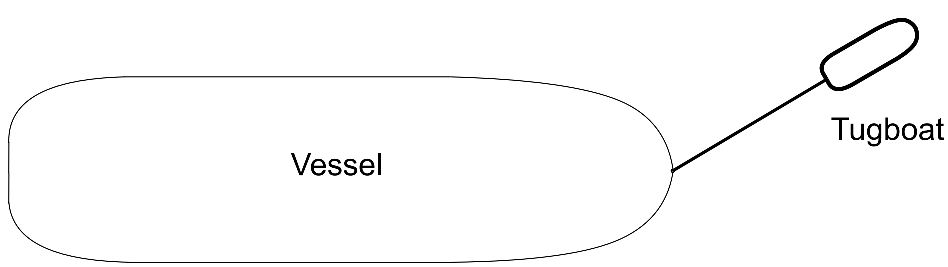


Figure 2.15: Arrow maneuver or rope maneuver [35].

Line towing allows the tugboat to assist the ship from a distance, adjusting its function (pull or hold) according to the connection point of the rope, as is shwon in the figure 2.15. Trailing-line towing optimizes power transfer and prevents collisions, but its large space requirement prevents use in restricted areas [35].

b) Bow-Supported Tugboat (Ram Locking)

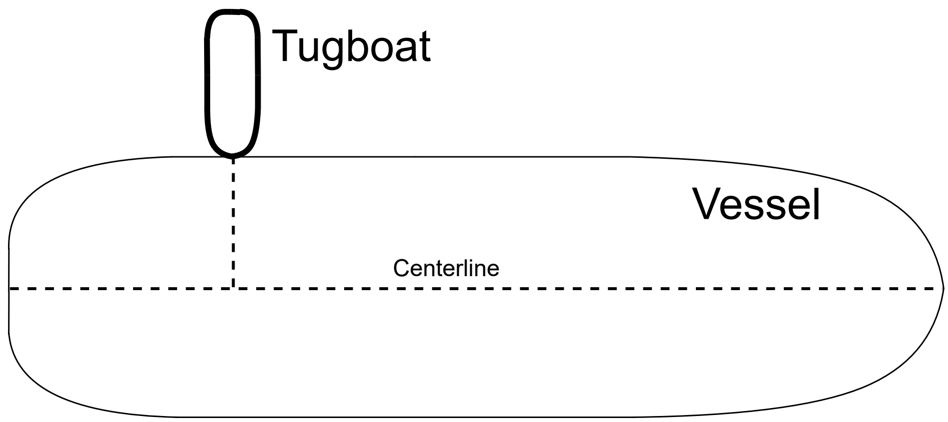


Figure 2.16: Maneuver supported by the bow [35].

In this procedure, the tugboat rests its bow on the side of the vessel it is assisting and pushes it in a direction that is essentially perpendicular to the centerline (the straight line running from stern to bow), as is shwon in the figure 2.16. It is common in this procedure for the tugboat to be secured to the ship with 1, 2, and 3 mooring lines, which prevents relative slippage between the two vessels during the maneuver [35].

c) Tugboat alongside

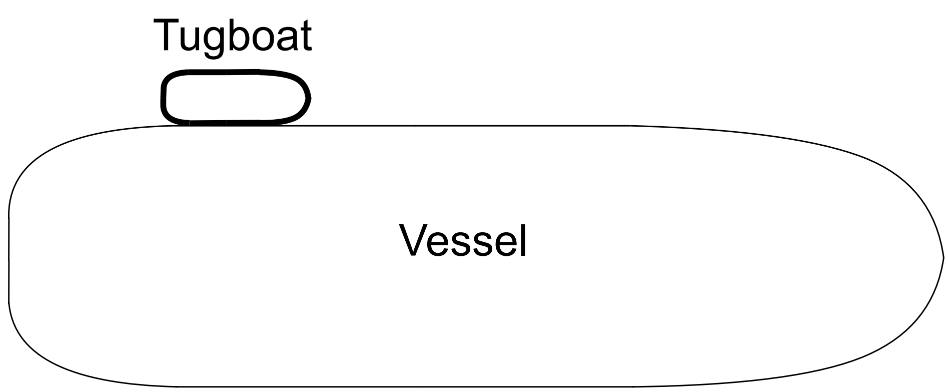


Figure 2.17: Alongside Maneuver [35].

In this procedure, the tugboat is positioned alongside the ship and roughly parallel to it, moored to the ship by several ropes that ensure the transmission of force, as is shown in the figura 2.17. This procedure is generally used to maneuver ships that do not have sufficient propulsion, in confined spaces and in very calm waters [35].

2.4 Service Maneuvers of a Tugboat

The following is a description of the actual work performed by a tugboat in Chilean ports, including its history, technical characteristics, and finally, its performance during service.

2.4.1 Operational Profile

Figure 2.18: TUG route to the Puerto Montt's pier.

Next, an analysis of a tugboat's operational load profile of a tugboat in Chile is performed, considering that it has a 10-hour work schedule, performing this task in Puerto Montt. Figure 2.18 shows the route taken by a tugboat from the shipyard to the dock in Puerto Montt. The route taken by the tugboat is divided into three different sections.

- Blue: Low-speed route (below 6 knots), tugboat departure from shipyard.
- Yellow: Nominal propulsion speed, tugboat moves toward the dock to perform operational maneuvers.
- Red: High-power section, operational maneuver performed by a tugboat assisting a ship.

Below are some of the features associated with this tugboat:

- **Type of Tugboat**: This tugboat is considered a **harbor tugboat**, as it is focused on towing and pushing large ships that want to enter or leave Puerto Montt.
- **Dimensions**: Table 2.3 associated with the dimensions of the tugboat to be analyzed is added.

Table 2.3: Dimensions

Dimensions Tugboat	
Length overall	25 m
Beam	10.5 m
Draught	3.9 m
Depth	4.6 m

- **Volumen and Weight**: Table 2.4 contains information related to the weight and volume of the tugboat.

Table 2.4: Volume and Weight

Dimensions Tugboat	
Gross Tonnage	269 tons
Deadweight	220 tons

- **Technical Characteristics of the Propulsion System**: The tugboat has an **ASD propulsion** system, which incorporates **two Schottel propellers**. Information is shown in Table 2.5 below.

Table 2.5: Machinery

Machinery Tugboat	
Bollard Pull	65.4 tons
Main Engines	2 x Cat 3516 B
Total BHP	4894 HP (3650 kW)
Propulsion	2 x Schottel SRP 1215

- **Naval Maneuvers and Assistance Techniques**: According to the service provided by the tugboat, it is established that in the red section of Figure 2.18, the tugboat can perform operational maneuvers pulling a vessel from the end of a rope in arrow mode or **over a cable** and also perform a **alongside** maneuver, assisting ships that need to enter the port or leave the dock to set sail.

Based on the route to be analyzed and the maneuvers that the tugboat must perform at the dock, there are two operating profiles for each maneuver, as it can see in the figure 2.19 and figure 2.20.

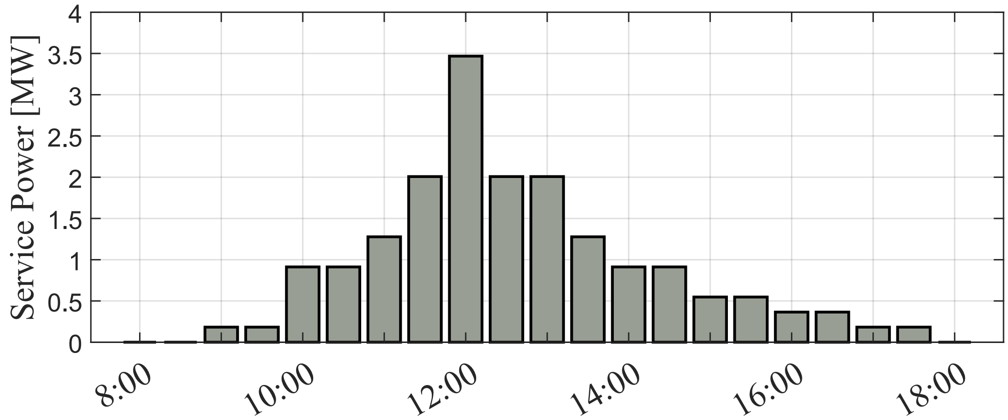


Figure 2.19: Operational profile of a conventional tugboat: one maneuver.

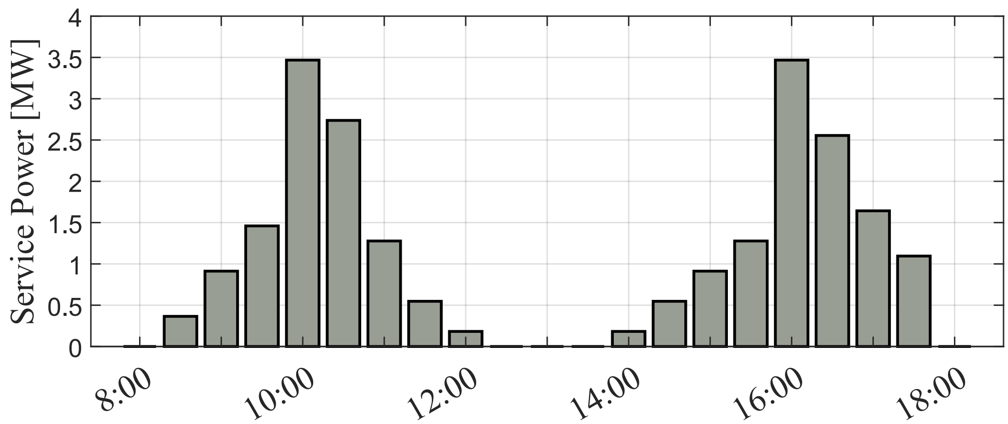


Figure 2.20: Operational profile of a conventional tugboat: two maneuvers.

Figures 2.19 and 2.20 represent the service maneuvers performed along the entire route, shown in Figure 2.18. These graphs demonstrate the power delivered by the tugboat's propulsion system.

2.4.2 Brake-Specific Fuel Consumption (BSFC)

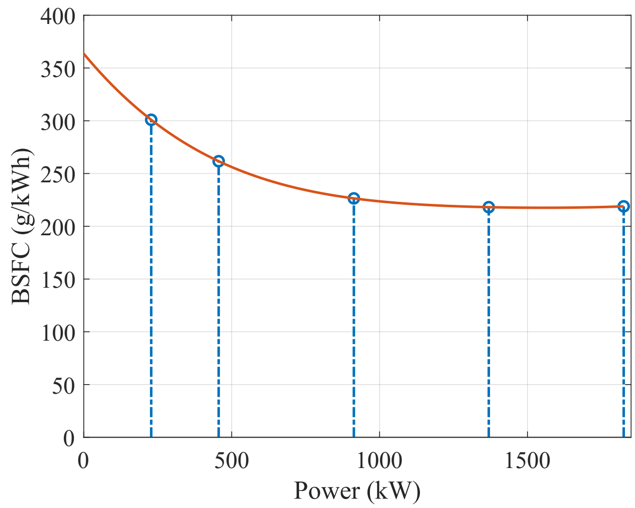


Figure 2.21: Brake-specific fuel consumption of a diesel engine model Cat 3516 B.

To analyze the operating profile, it is necessary to introduce the concept of **brake-specific fuel consumption (BSFC)**. BSFC is a measure of an engine's fuel efficiency, calculated as the amount of fuel consumed per unit of power produced over a specific time [9].

The BSFC as a function of output power is illustrated in fig. 2.21 for a typical engine used in tugboats. The engine achieves maximum efficiency at rated power, where BSFC is minimal. In contrast, operating at low power increases consumption per unit of energy, decreasing efficiency.

The relative efficiency evaluates how the conventional engine is working taking as a reference the optimal efficiency at the nominal operating point as shown in eq. (2.1) [9]. At this operating point the relative efficiency is 100% reducing its value if the engine works at lower levels of power.

$$\eta_{\text{rel}} = \frac{\text{BSFC}_{\text{nom}}}{\text{BSFC}} \tag{2.1}$$

With the information provided by the engine in Figure 2.21, it is then possible to analyze how relative efficiency varies based on the load profile. This is shown in Figures 2.22 and 2.23.

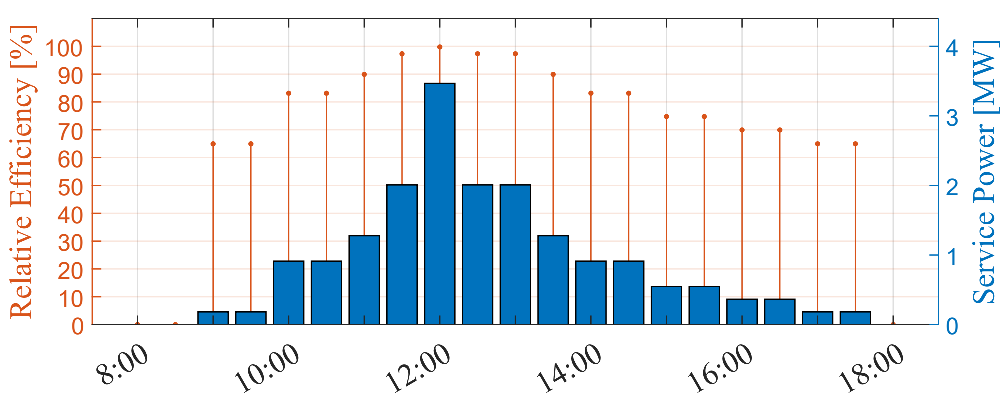


Figure 2.22: Relative efficiency of a convencitonal tugboat: one maneuver.

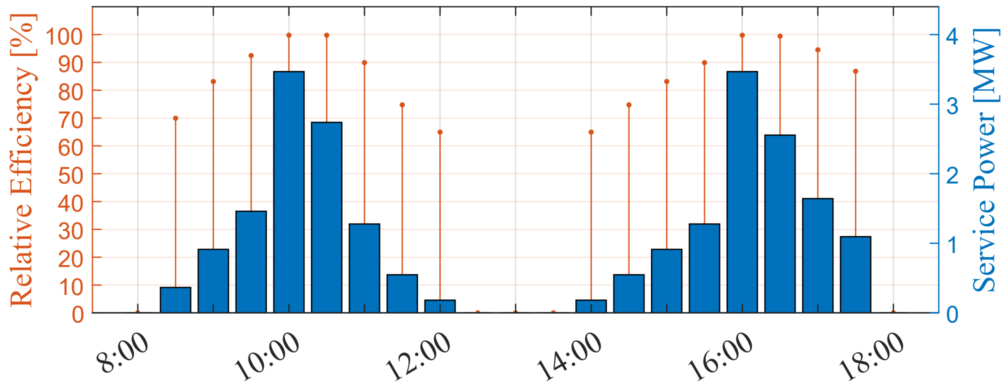


Figure 2.23: Relative efficiency of a convencitonal tugboat: two maneuvers.

It can be observed in figures 2.22 y 2.23 that as the tugboat operates close to the nominal power point of its diesel engines, its relative efficiency improves. This indicates optimal fuel consumption during those periods. Conversely, at lower operating points, the engine consumes more fuel relative to the energy produced.

2.4.3 Generated Emissions and Fuel Consumption

To calculate emissions, the following formula relates brake power (P_b), BSFC, usage time (t_u), and a carbon factor (CF):

$$\text{Emissions} = \frac{\text{BSFC} \cdot P_b \cdot t_u \cdot \text{CF}}{10^{-6}} \quad [\text{t CO}_2] \tag{2.2}$$

With equation (2.2), you can find out the amount of emissions produced by the tugboat for each load profile. This can be seen in Figures 2.24 and 2.25, where fuel consumption per hour of tugboat service can also be represented, considering the density d of diesel, which in this case corresponds to 832.5 [g/L] and fuel consumption is related to emissions as follows:

$$\text{BSFC} = \frac{d \cdot \text{Fuel}}{P_b} \longrightarrow \text{Emissions} = \frac{\text{Fuel} \cdot d \cdot t_u \cdot \text{CF}}{10^{-6}} \quad [\text{t CO}_2] \tag{2.3}$$

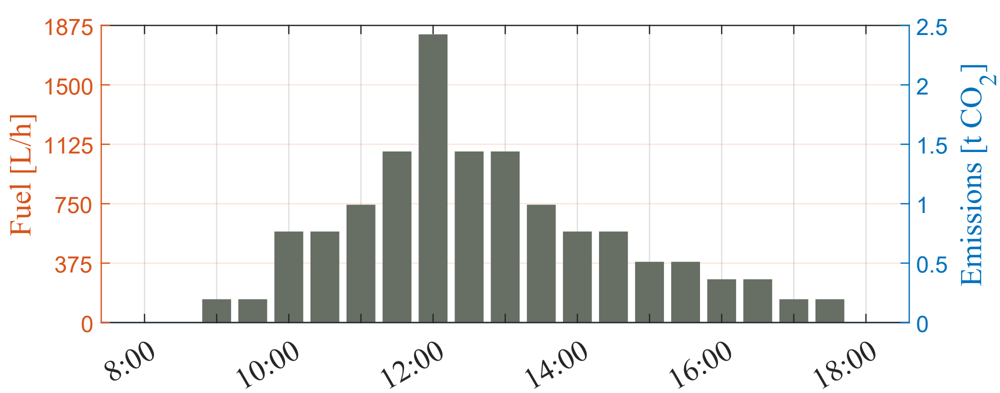


Figure 2.24: Fuel consumption and emissions with respect to the tugboat's operating profile during a maneuver.

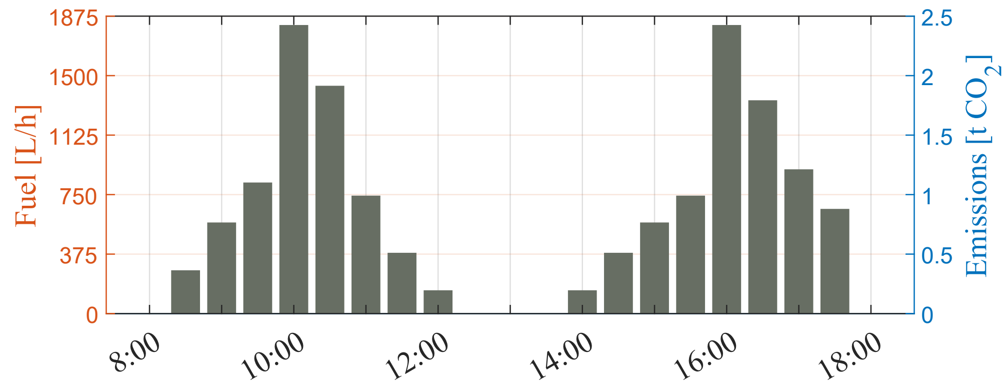


Figure 2.25: Fuel consumption and emissions with respect to the tugboat's operating profile in two maneuvers.

Based on the information obtained in Figures 2.24 and 2.25, it is possible to determine the fuel consumption of a conventional tugboat during a day of service for one and two maneuvers, identifying the daily consumption levels at low, medium, and high power.

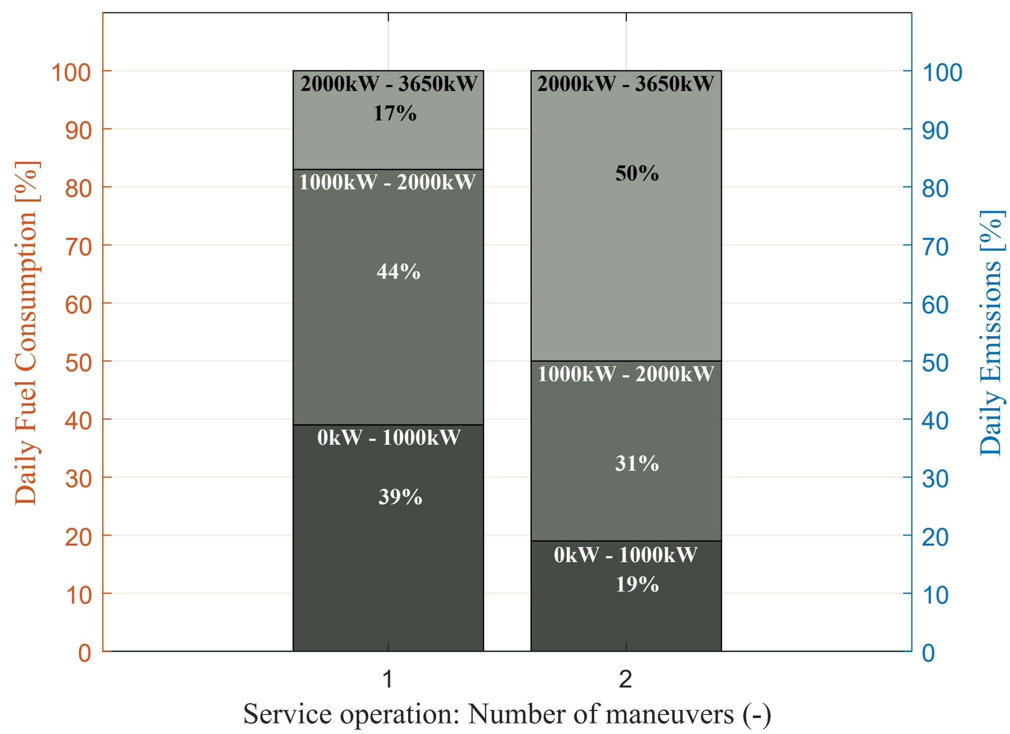


Figure 2.26: Percentage of emissions per day based on power ranges.

Fig. 2.26 shows the accumulated fuel consumption and CO_2 emissions generated for one and two maneuvers. Each service operation is divided into three segments:

- The lower segments correspond to propulsion system power levels with a relative efficiency below 90% (0kW - 1000kW).
- The middle segments correspond to relative efficiencies ranging from 90% to 98% (1000kW - 2000kW).

- The upper segments represent power levels with the highest relative efficiency, ranging from 98% to 99.8% (2000kW - 3650kW).

At low power levels, fuel consumption can account for up to 38% and 11% of the total daily fuel usage, for one and two maneuvers, respectively. Resulting in high emissions due to the diesel engine operating far from its optimal efficiency point, emphasizing the variability of the load profile of this type of vessel and its impact on fuel consumption and emissions.

2.5 Chapter Summary

This chapter established the fundamental technical and operational framework required to understand the role of tugboats in the maritime industry, serving as the baseline for the proposed hybrid propulsion design.

The tugboat is defined as an essential assistance vessel designed to aid the mooring, berthing, and maneuvering of larger ships, particularly in the context of naval gigantism. The chapter classified these vessels into three primary categories based on their operational domain:

- **Harbour Tugs**: Designed for agility and restricted drafts within ports.
- **Offshore/SAR Tugs**: Built for open ocean robustness and long-range autonomy.
- **Escort Tugs**: High-power vessels specialized in safety for hazardous cargo in high-risk zones.

A critical analysis of propulsion engineering was presented, highlighting Bollard Pull (BP) as the primary metric for quantifying a tug's static pulling force. The comparison of propulsion systems revealed distinct trade-offs:

- **Conventional Propulsion**: Offers linear efficiency but limited lateral maneuverability.
- **Azimuth Stern Drive (ASD)**: Utilizes rotatable thrusters (Z-Drive) for 360-degree maneuverability, a standard for modern port operations.
- **Voith-Schneider (VSP)**: Provides omnidirectional thrust and extreme precision for congested environments.

The chapter concluded with a case study of a tugboat operating in Puerto Montt, Chile. By analyzing the Brake-Specific Fuel Consumption (BSFC), the study identified a significant inefficiency in traditional diesel configurations:

- Diesel engines achieve optimal efficiency (low BSFC) only at high power ratings.
- The operational profile of a tugboat is highly variable; during low-load periods (e.g., transit or standby), the engines operate at low relative efficiency, leading to higher fuel consumption per kWh produced.
- Data indicated that in low power ranges (180kW – 1270kW), the relative efficiency drops significantly, yet this range can account for up to 38% of daily fuel consumption in single-maneuver scenarios

This analysis demonstrates that the intermittency of the load profile and the reliance on oversized diesel engines for low-load tasks result in excessive greenhouse gas emissions, justifying the need for the hybrid energy storage system proposed in this thesis.

Chapter 3

Hybrid Tugboats: Requirements and Performance

This chapter presents the fundamental concepts associated with tugboat hybridization, its most common technological configurations, and the operational benefits associated with the incorporation of battery banks. The objective is to establish the theoretical and technical basis for the design process carried out in subsequent chapters.

Given the intermittent nature of the load profile and the dependence on diesel engines, which are oversized for low-load tasks, the implementation of the hybrid energy storage system proposed in this thesis is analyzed.

3.1 General Power Sources and Energy Storage System

In general, the hybrid tugboat relies on a triad of power sources:

- Main Propulsion Engines,
- Auxiliary Generators,
- Battery Energy Storage System (BESS).

The functions of these components have changed significantly compared to conventional designs, and the aim is to contribute to both propulsion and hotel loads.

3.1.1 Main Propulsion Diesel Engines

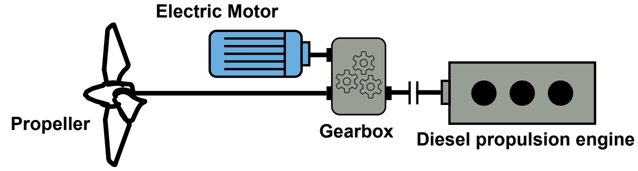


Figure 3.1: Main diesel engine used in hybrid vessels.

In hybrid vessels, the main engines are only started when high power is actually needed, as the electric motor handles low-speed maneuvers and idling. This reduces their operating hours, which increases their operational efficiency and reduces their emissions [21] [64].

As shown in Figure 3.1, The mechanical connection via a reduction gearbox allows an electric motor to be coupled [8] [21], enabling the main diesel engine to transmit its full rated power to the shaft, complementing the work of the electric motor. [64].

3.1.2 Auxiliary Diesel Generators

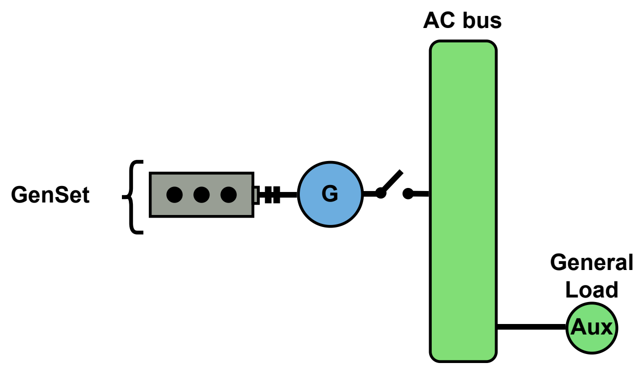


Figure 3.2: Main generator inside a hybrid vessel.

In a conventional tugboat, auxiliary generators (generator sets) power the ship's service loads [53]. In a hybrid tugboat, they can play a role in both powering the hotel loads and taking over propulsion.

They are usually arranged in banks to provide different service powers (for example, four small generators instead of two large main generators) [94] [63], which makes it possible to use only one generator for low energy demand and the rest for higher demands.

As shown in Figure 3.2, auxiliary diesel generators can be used to [28]:

- Deliver power to the propellers via a series connection when less power than the rated power of the main diesel engines is required.
- Without connecting to the propellers, it allows the batteries to be charged when there is no possibility of charging them in port.
- Power supply to the tugboat hotel loads.

3.1.3 Battery Bank

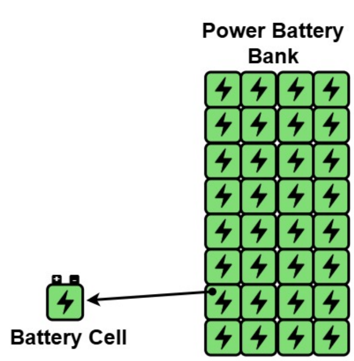


Figure 3.3: Battery bank made with multiple cells.

The function of the BESS shown in the Figura 3.3 is not only to store energy; it also complements the operation of the main diesel engines. As mentioned in section 1.5, these are some of the complementary functions that the BESS can provide in a hybrid tugboat.

- **Peak Shaving:** The use of a BESS solves the inefficiencies of diesel engines when faced with sudden changes in power. Instead of forcing the engine to accelerate rapidly—which causes thermal stress and polluting emissions (black smoke)—the battery supplies energy instantly.
- **Spinning Reverse:** If the active diesel generator fails, the BESS instantly takes over the load, preventing a blackout and saving fuel and operating hours for the main diesel engine.
- **Optimise Load:** The integration of a battery energy storage system (BESS) allows diesel engines to always operate at their maximum efficiency point, regardless of speed demand:
  - Low propulsion demand: Excess energy generated by the engine is used to charge the batteries.
  - High propulsion demand: Batteries provide additional power (hybrid propulsion), preventing the engine from being overloaded or taken out of its optimal range.

3.2 Classification of Hybrid Propulsion Systems

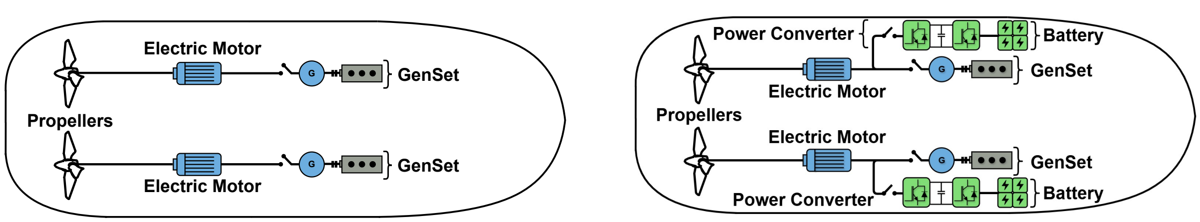
The maritime industry has evolved distinct topological approaches to hybridization, each offering specific trade-offs between complexity, cost, redundancy, and efficiency.

3.2.1 Series Hybrid Architecture

In a hybrid series configuration, the mechanical connection between the internal combustion engine and the propeller shaft is completely decoupled, as is shown in 1.4a. The vessel is propelled exclusively by electric motors [56]. The main diesel engines function strictly as generator sets (gensets). They produce electrical energy and fed into a common distribution bus. This bus supplies power to variable frequency drives (VFDs) that control the electric propulsion motors.

This system eliminates the inefficiency of diesel engines at low revs, allowing them to always operate at peak performance or delegate the task entirely to the batteries when the load is low [55].

a) Electric shaft



(a) Electric shaft propulsion system. (b) Electric shaft with electric energy storage system.

Figure 3.4: Electric Shaft configuration system.

The generator transforms the mechanical power of the combustion engine to electrical power. The electric motor on the right transforms the electric power back to mechanical power which is then supplied to the propulsor, as is shown in figure 3.4a. This system is called "electric shaft", as there is no control input to manipulate the power transmission [55].

The electric propulsion system uses the fixed properties of the generator and motor to establish constant power transmission between the combustion engine and the propeller, based on defined voltage, current, and frequency parameters [55].

Wired connections make it easy to integrate multiple motors and generators, eliminating the rigidity of mechanical shafts; however, the system is vulnerable to overload: if the holding torque of the synchronous machines is exceeded, the system becomes desynchronized, losing power and risking mechanical integrity (cogging torque) [55].

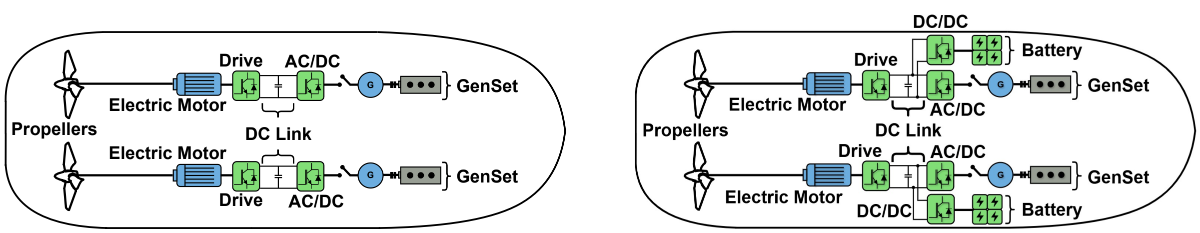
The electric shaft system can also be extended with an electric energy storage, to provide additional power during short mission segments, as is shown in figure 3.4b.

The integration of batteries not only improves consumption, but also allows for a physical redesign of the system:

- **Downsizing:** The combustion engine, electric motors, and wiring can be smaller and lighter, as the storage system compensates for any shortcomings.
- **Electrical Management:** The storage system compensates for reactive power and, through power electronics, eliminates the risk of desynchronization (cogging torque) mentioned above.

However, as a vulnerability, the system relies entirely on this compensation to reach its rated power. Therefore, any electrical failure would immediately reduce propulsion capacity [55].

b) DC-Link Topology



(a) Turboelectric propulsion system.

(b) Turboelectric with energy storage system.

Figure 3.5: Serial connection with built-in DC link.

Mechanical power is converted to electricity, fed into a DC bus (often 1000V DC), and then inverted back to AC for the propulsion motors, as is shown in figure 3.5a. Power converters enable independent operation between the motor and the propeller, allowing various fixed-pitch rotors to be managed autonomously [55]. Additionally, the generator set can be resized by adding a BESS connected to the DC link, as shown in Figure 3.5b.

The series hybrid system is ideal for operations involving frequent stops and low speeds (such as tugboats or ferries), but it loses competitiveness on routes traveled at constant speeds, as the system is between 10% and 15% less efficient than a direct mechanical drive due to multiple energy conversions (from mechanical to electrical and vice versa) [55].

3.2.2 Parallel Hybrid Architecture

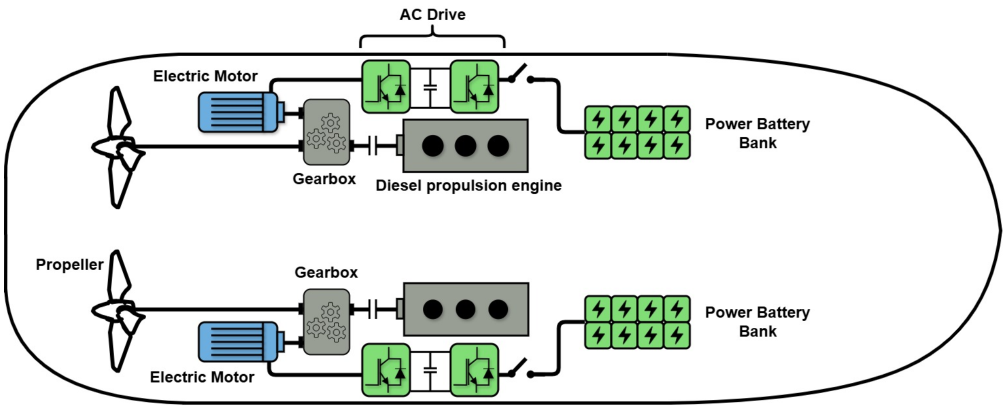


Figure 3.6: Parallel hybrid propulsion system.

In a parallel hybrid layout, both the electric motor and the combustion engine are mechanically connected to the propeller shaft with a gearbox [54].

In this setup, both the combustion engine and an electric machine (motor/generator) are mechanically connected to the propeller shaft, typically via a reduction gearbox with complex clutching arrangements, as is shown in 1.4b.

In this case, as shown in Figure 3.6, the main function of electrical storage is to supplement the power of the main diesel engine, allowing the latter to be sized only for average loads, not for peaks.

The following are the operating modes that can be found in the parallel hybrid configuration:

- **Mechanical Mode (PTO)**: The main diesel engine drives the propeller directly. This is used for heavy towing and high-speed transit. The electric machine may freewheel or act as a generator to charge batteries, mode called Power Take Off (PTO) [8]. This is shown in figure 3.7.

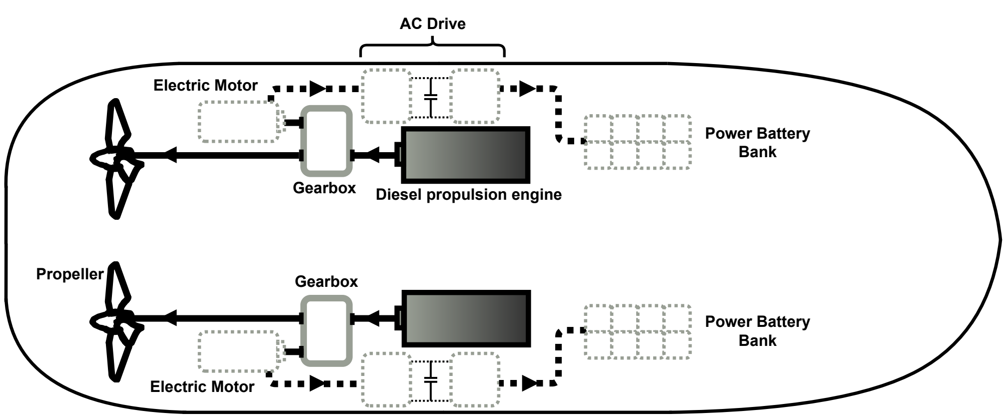


Figure 3.7: Energy flow in PTO mode.

- **Boost Mode (PTI)**: Both the diesel engine and the electric motor provide torque to the shaft simultaneously through the gearbox to the propeller [8]. This allows the vessel to achieve a total power output greater than the rating of the main engine alone, providing the ”peak” power needed for emergency ship assist maneuvers without requiring a larger main engine. This mode is also called Power Take In (PTI) and is shown in figure 3.8.

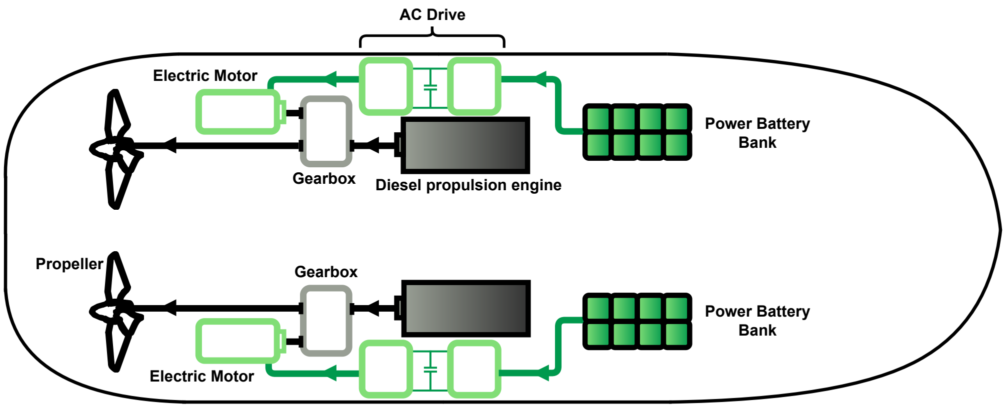


Figure 3.8: Energy flow in PTI mode.

- **Electric Mode (PTH)**: The diesel engine shuts down and the shaft is powered exclusively by the electric motor. This configuration allows for silent, emission-free navigation in ports (0-6 knots), as well as functioning as an emergency system that guarantees the boat's mobility at reduced speed in the event of main engine failure. This mode is also called Power Take Home (PTH) and is shown in figure 3.9.

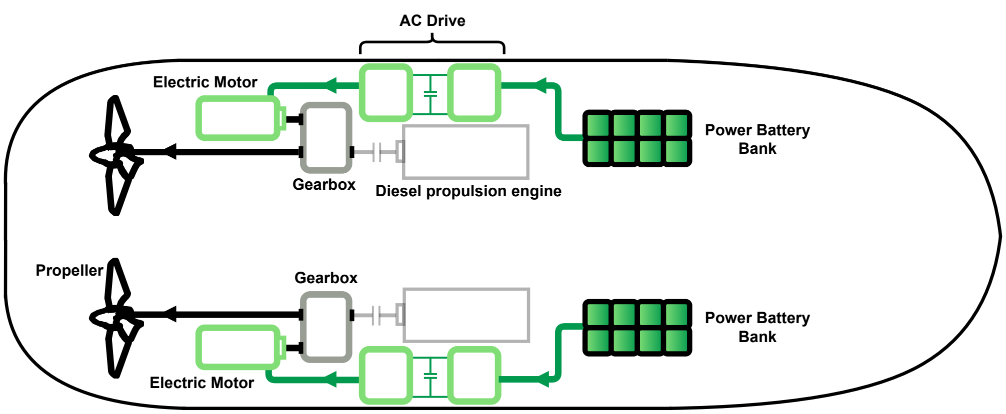


Figure 3.9: Energy flow in PTH mode.

3.2.3 Comparation

Table 3.1 below compares some characteristics between the aforementioned propulsion systems :

Table 3.1: Performances of hybrid propulsion systems [6]

	Series	**Parallel**
Sizing of elements/costs	To be fully efficient, the system must combine several power sources with rated power equal to a fraction of total power. It leads to supplementary costs volumes and weight.	The system allows optimizing the sizing of each of the propulsion systems. Supplementary costs and volume related to complex mechanical system (clutches, gearboxes).
Power and Efficiency	Efficiency limited by the multiplication of series systems (motor, generator, transformer, converters, motors, etc.). If multiple sources are used, combustion engines can operate at maximum efficiency.	Good efficiency related to the reduction in the number of components in the series (especially in combustion engine mode). The combustion engine operates at maximum efficiency.
Redundancy/reliability	The system is modular, and offers power sources redundancy.	Very high level of redundancy since the propulsion power can be provided independently by two independent systems.

The parallel hybrid architecture is currently the dominant configuration for high-bollard-pull tugs because it preserves the mechanical efficiency of the direct drive for peak loads.

3.3 Grid Distribution

Typically, electricity has been distributed to other consumers in AC and converted to direct current (DC) for consumers that need it – usually smaller onboard electrical systems, as opposed to the bigger systems such as electric propulsion that require AC [85]. Traditional ships use 440V or 690V AC grids.

But as fully electric and hybrid vessels have become increasingly popular, a new electrical integration concept that uses DC for electricity distributio has become a hot topic. With the integration of batteries (DC sources) and variable speed drives (DC internal), makes AC inefficient.

The architectural shift from AC to DC distribution is the technological enabler of modern hybrid vessels.

3.3.1 AC Link

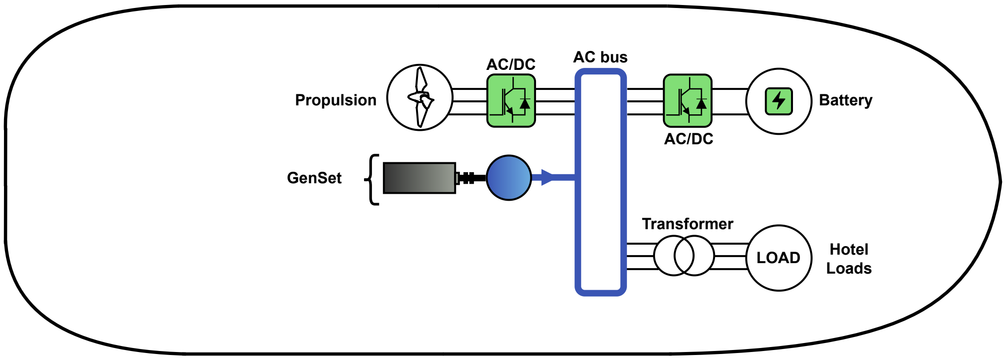


Figure 3.10: AC Link.

Historically, AC has been the standard on ships [58], but it imposes rigid restrictions on the diesel engine:

- **Mandatory synchronization:** Motors must rotate at a constant speed to maintain stable voltage and frequency in the electrical system.
- **Narrow efficiency range:** Due to this fixed speed, the engine only achieves its optimal fuel consumption (SFOC) in a very limited range, generally around 85% of its rated load.

The control strategy for electrical architectures consists of two parts [59]:

- **The control of the electrical fixed frequency** network aiming to provide robust power supply to all electrical users.
- **The control of the propulsion** aiming to drive the ship in a certain speed and direction.

For onboard power systems, voltages can range from 400 V in small ships to 11 kV in large vessels with total onboard power up to 100 MW. Even when docked in a port, a large vessel may have 10 MW of power requirement [57].

3.3.2 DC Link

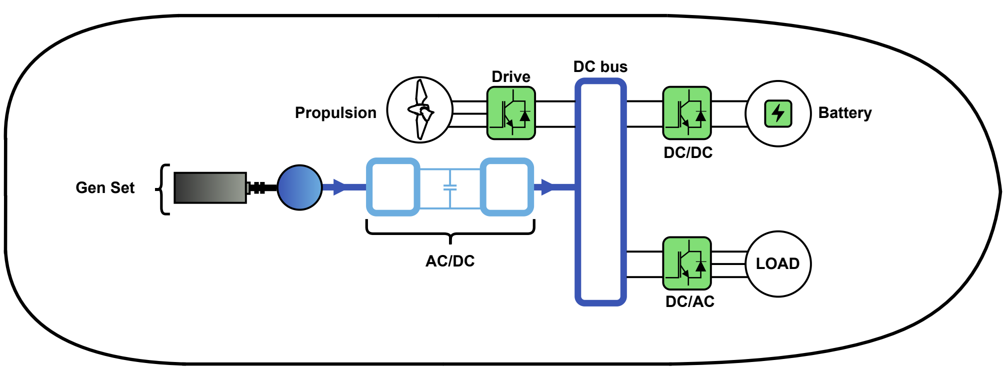


Figure 3.11: DC Link.

The Onboard DC Grid concept eliminates the need for bulky propulsion transformers and main AC switchboards. This can reduce the total space occupied by electrical equipment by between 30% and 75%, freeing up space in a tugboat's engine rooms [60].

DC networks allow the motor speed to vary according to the load, unlike the constant speed required in AC. This decoupling optimizes consumption at low demand, achieving fuel savings of up to 27% [60].

Hybrid tugboats typically use 1000 V DC networks. These are categorized as LVdc (up to 1000 V and 20 MW) or MVdc (above 1000 V and 20 MW) systems, using unipolar or bipolar bus architectures [30].

3.3.3 Comparation

In Table 3.2 there is a comparation between the two ways to electrify the shipboard power system.

Table 3.2: Comparation between AC and DC grids [30] [60]

	AC	**DC**
Shipboard Power System (SPS)	AC bus is a direct connection system where operational stability depends strictly on maintaining constant voltage and frequency from the generators.	The DC bus acts as a flexible interface that allows high-speed motors to be used and their speed to be varied without worrying about frequency synchronization issues.
Conversion Losses	In a traditional drive with an AC grid, power flows: Generator (AC) → Switchboard (AC) → Rectifier (DC) → Inverter (AC) → Motor.	With a DC grid, the "Rectifier" stage is centralized or integrated, and batteries connect directly to the DC bus via DC/DC converters. This eliminates multiple conversion steps.
Protective System	AC currents are by nature far simpler to break because of their natural zero crossing every half cycle.	DC circuit breakers do exists to some extent but are more complex, larger and more expensive than comparable AC circuit breakers.

3.4 Incorporation Grid Distribution

Electric propulsion and distribution systems can fulfill their role either jointly or separately [7] [53].

3.4.1 Segregated Power System

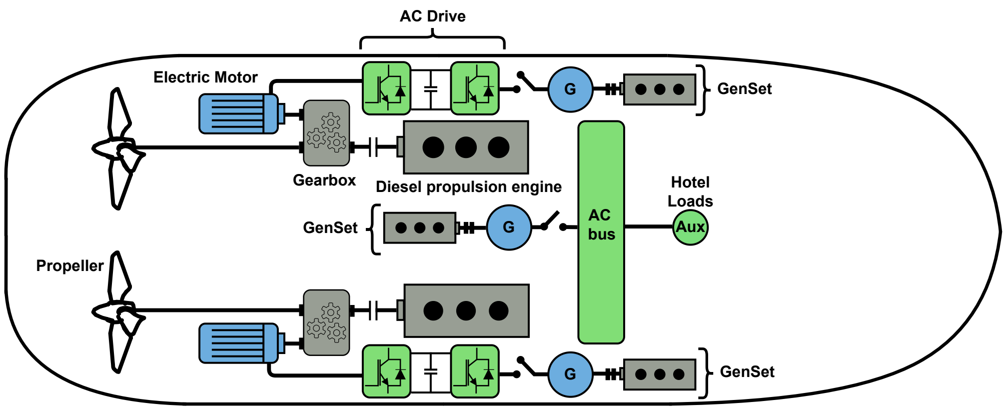


Figure 3.12: Segregated PowerSystem

The segregated power system (SPS) has two separate electrical systems, one dedicated to providing power for ship propulsion and the other reserved for feeding auxiliary loads [7].

In this configuration, energy flows linearly and mechanically:

- **Direct Drive:** The prime movers (diesel engines) drive the propellers through a physical connection of reduction gears and long shafts.
- **Function of the Gears:** Their role is to transform the high speed of the engine into a speed suitable for the propeller while transferring power, as well as to couple the electric motor with the diesel engine [53].

In this traditional configuration, propulsion generators and auxiliary generators are separated, which creates three main problems:

- **Incompatibility:** Operating at different voltages and frequencies, the systems cannot share power with each other.
- **Lack of Flexibility:** Excess power from the main engines cannot be used for ship loads, reducing operational efficiency.
- **Low Performance and Emissions:** Although 90% of installed power is for propulsion [7], auxiliary generators often operate far from their optimal load due to hotel loads, which unnecessarily increases consumption and pollution.

3.4.2 Integrated Power System

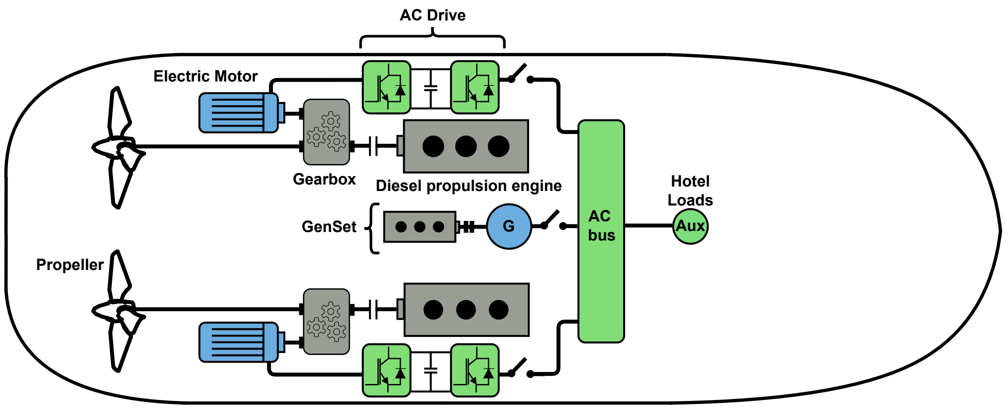


Figure 3.13: Integrated PowerSystem

In this architecture, the total power required by the propulsion system and service loads is supplied by one or more generators connected to the medium-voltage bus and then distributed to the low-voltage bus via transformers [7].

The propulsion motors are fed from power converters, which are connected directly to the main busbar, and the service loads are fed from a low-voltage busbar, using stepdown transformers.

The integrated power system (IPS) system integrates generation and distribution into a common electrical network. This energy exchange dramatically improves operational flexibility and system availability compared to traditional configurations [53]. By actively managing the power delivered, IPS minimizes the use of generators, optimizing consumption and environmental impact.

IPS eliminates the separation between propulsion power and service power, creating a single, shared network [53]:

- **Unified Network:** All generated energy is centralized, powering both the propulsion engines and the ship's loads.
- **Intelligent Management:** Allows you to optimize how many generators should be turned on based on actual demand, preventing them from operating inefficiently.

Thus, the direct benefits come from improved operational flexibility, increased energy availability, optimized fuel use, and reduced pollutant emissions.

3.5 Service Maneuvers of a Hybrid Tugboat

A hybridization of the tugboat propulsion system seen in Section 2.4 is proposed. The general layout of the hybrid propulsion system is presented in Figure 3.14.

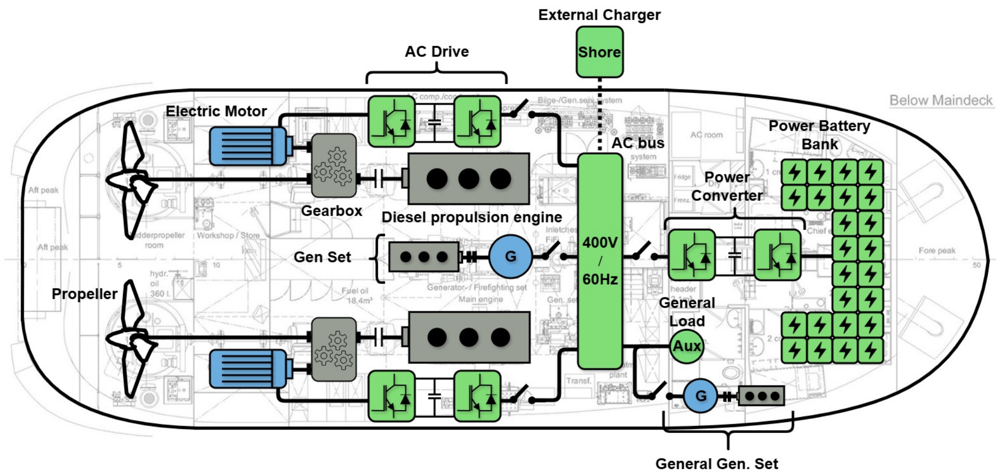


Figure 3.14: General Scheme

This hybrid propulsion system is designed to incorporate electrical and electromechanical elements without removing the original components inside the tugboat, specifically those related to the diesel engines of the conventional propulsion system.

Below are some of the features associated with this hybrid tugboat:

- **Classification of Hybrid Propulsion System:** The propulsion system is a parallel hybrid propulsion system, as it has a gearbox that allows the electric motor and the diesel propulsion engine to be coupled together in order to deliver power to the propeller either mechanically or electrically.
- **Grid Distribution:** The tugboat's internal electrical distribution network is AC, so as not to add an additional stage to the main diesel generator, thus avoiding complexity in control. In addition, the voltage level (400 V) is associated with the levels that can be found in port in order to charge the batteries, taking into account that such (slow charging) can be carried out during the remaining hours outside of service hours.
- **Incorporation Grid Distribution:** The electrical distribution network is considered integrated, given that there is no separation from the propulsion stage. The purpose of this is to enable the main generator to deliver power to the propellers if required.

3.5.1 Operating Modes

Based on the proposed design for the hybrid propulsion system shown in the previous section and its main characteristics, the following operating modes can be considered [65]:

a) Towing Mode

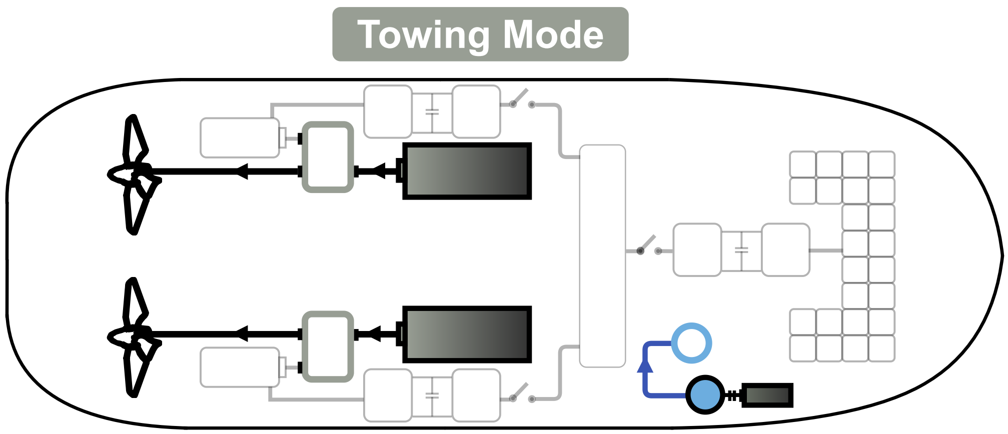


Figure 3.15: Towing Mode

In *Towing Mode*, the main engines start up and drive the rudder propellers, as is shown in Figure 3.15. The main generator is disconnected from the propulsion system and the auxiliary generator set powers the normal electrical system. This mode is intended for use during thrust/traction operations and free navigation at speeds above 8 knots. It should also be noted that PTO and PTI functionalities are not considered, given the modes of operation explained below.

b) Free Sailing Mode

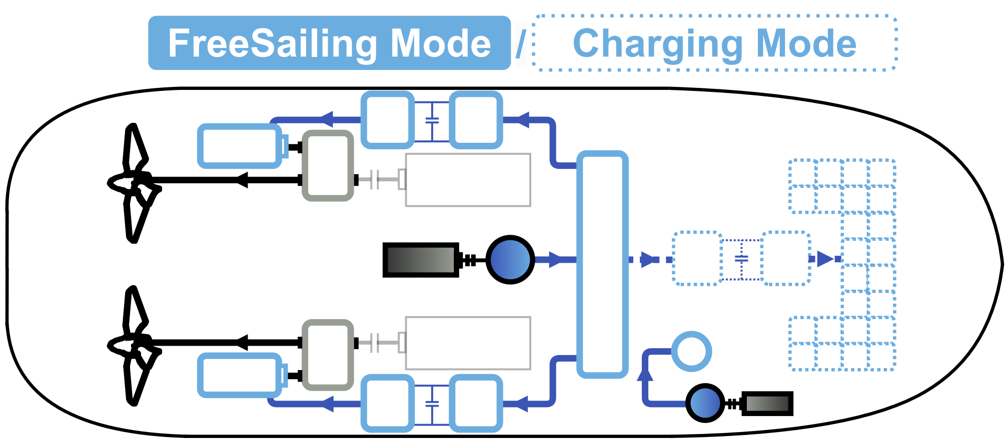
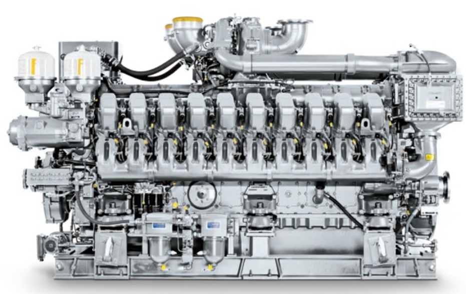


Figure 3.16: FreeSailingMode

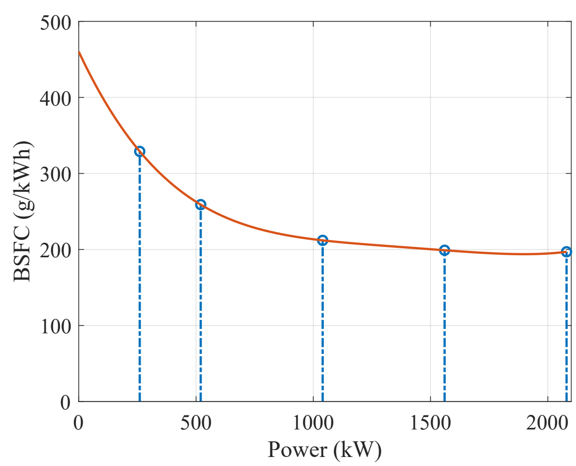
In *Free Sailing Mode*, the main generator set powers the electric propulsion motors that drive the rudder propellers, while the main engines are not running and the auxiliary generator set also powers the standard electrical system, as shown in Figure 3.16. Free sailing mode can be used to maintain position, maneuver, and sail freely at speeds of up to 8 knots.

This mode also allows the batteries to be charged when the tugboat is docked, for example (*Charging Mode*). If the state of charge (SOC) of the battery banks drops to 60%, the main generator switches on to charge the batteries. As soon as the batteries are charged to an SOC of 80%, the generator switches off and the batteries can power the vessel if necessary.

At medium operating power, the efficiency of the system is measured by comparing the consumption of the diesel engines of a conventional tugboat with the consumption of the main generator under the same loads.
For this purpose, we considered the use of an MTU 16V 4000 P63 engine, whose BSFC profile can be seen in Figure 3.17.



(a) Diesel engine used in the main generator.



(b) Brake-specific fuel consumption of a genset model MTU 16V 4000.

Figure 3.17: Proposed main generator for use in hybrid tugboat.

c) Standby Mode

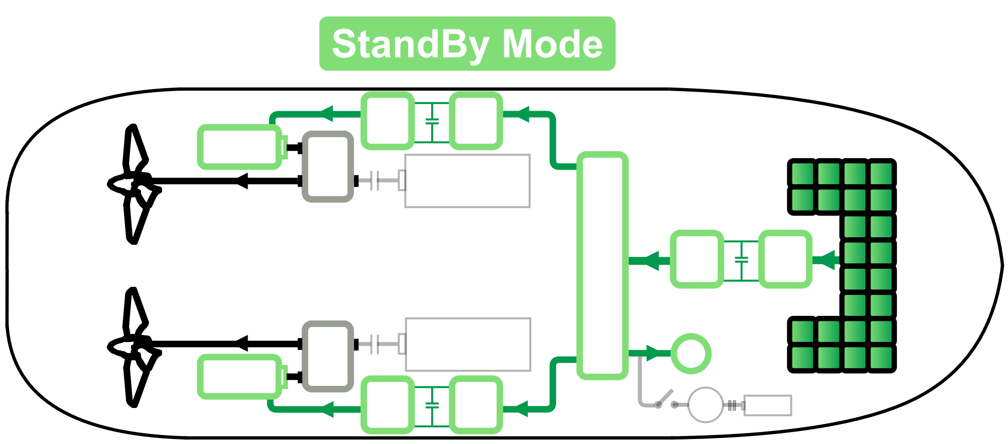


Figure 3.18: StandBy Mode

In *Standby Mode*, the diesel engines will shut down and the battery will power the normal electrical system and the electric propulsion motors that drive the rudder propellers, as is shown in Figure

3.18. *Standby Mode* can be used to maintain position, maneuver, and navigate freely at speeds of up to 5 knots, thus complying with the PTH functionality. In addition, the battery bank may provide **peak shaving** and **spinning reverse** functions.

3.6 Operational Profile

Based on the operating modes described above and the tugboat load profiles seen in section 2.4.1, operating profiles for the hybrid tugboat are proposed below, considering battery banks with different nominal powers.

3.6.1 One Maneuver

Based on the profile in Figure 2.22, two hybridization alternatives are proposed that integrate battery banks with capacities of 15% and 25% of the propulsion system's nominal power. This can be seen in Figures 3.19 and 3.20, respectively.

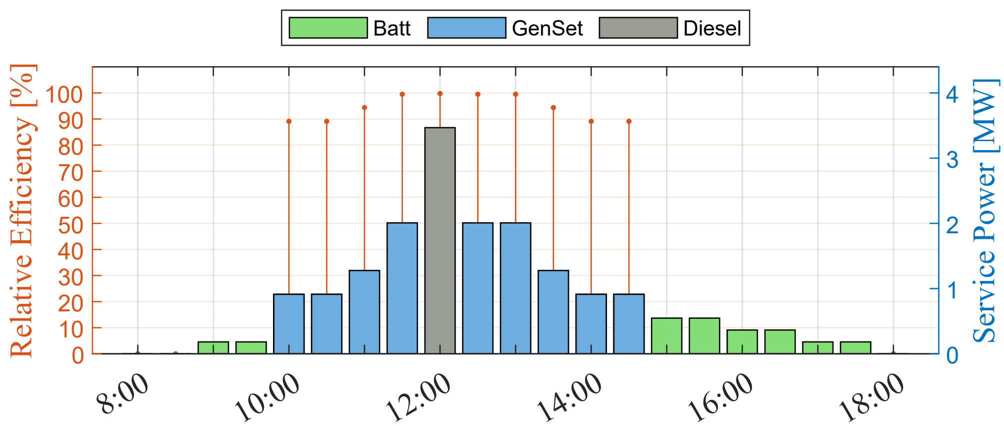


Figure 3.19: Operational profile and relative efficiency of a hybrid tugboat for one maneuver and a battery banks with a capacity of 15% of the propulsion system's rated power.

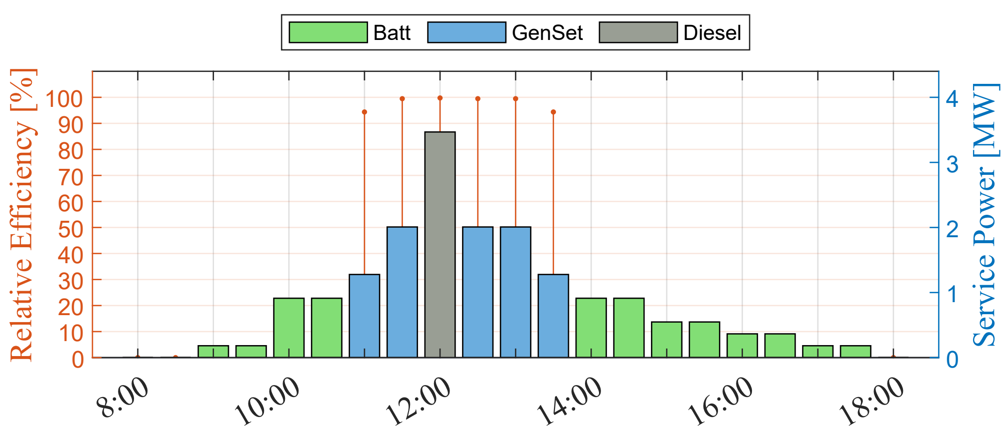


Figure 3.20: Operational profile and relative efficiency of a hybrid tugboat for one maneuver and a battery banks with a capacity of 25% of the propulsion system's rated power.

By visually comparing the graphically represented load profiles, it can be seen that increasing the nominal power of the bank reduces the use of *Free Sailing Mode*, without altering *Towing Mode*.

3.6.2 Two Maneuvers

Following the above procedure, the load profile of the service maneuver (Fig. 2.23) was hybridized with battery banks at 15% and 25% of the nominal power, the results of which are illustrated in Figures 3.21 and 3.22, respectively.

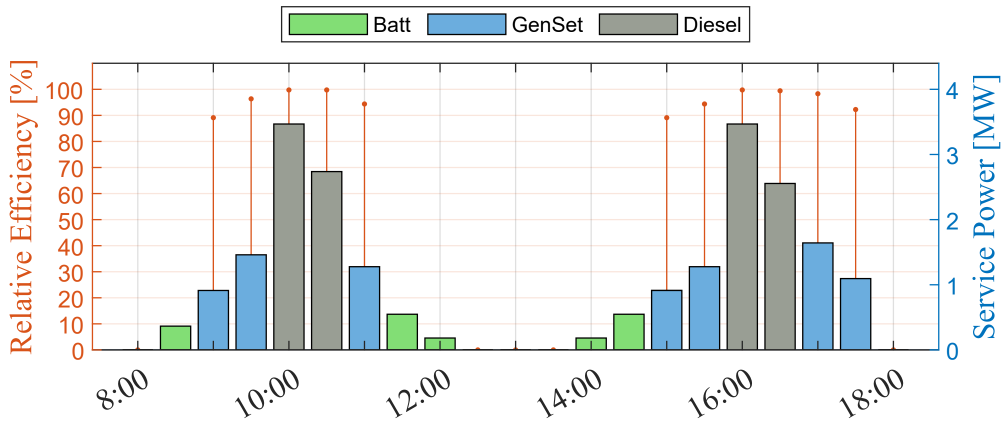


Figure 3.21: Operational profile and relative efficiency of a hybrid tugboat for two maneuvers and a battery banks with a capacity of 15% of the propulsion system's rated power.

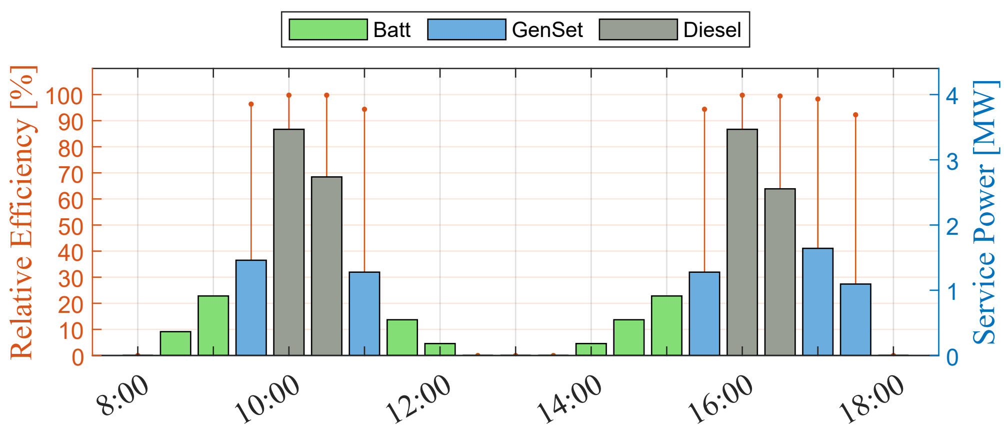


Figure 3.22: Operational profile and relative efficiency of a hybrid tugboat for two maneuvers and a battery banks with a capacity of 25% of the propulsion system's rated power.

The figures 3.19, 3.20, 3.21 and 3.22 show that each operating mode is executed in the operating profile depending on the power levels requested by the service. This leads to improvements in the relative efficiency of the engines, since the modes used by the engines were selected to achieve optimal BSFC for those power levels, which also improves fuel consumption and reduces CO_2 emissions.

3.7 Generated Emissions and Fuel Consumption

The hybrid tugboat's operating modes reduce fuel consumption and emissions, especially during periods of service when batteries are used. It can also be seen that *Free Sailing Mode* significantly reduces fuel consumption levels thanks to the use of the main generator.

3.7.1 One Maneuver

Based on the load profiles of a hybrid tug maneuver, the fuel consumption/emissions generated for that maneuver are shown in the graph below, as seen in Figures 3.23 and 3.24.

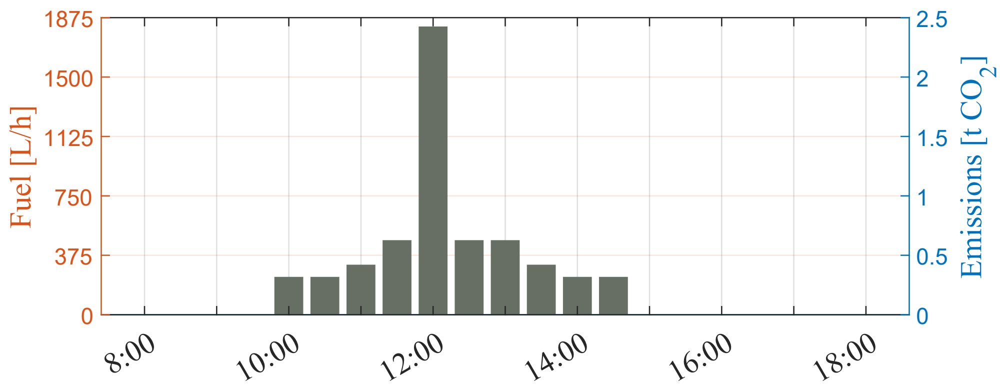


Figure 3.23: Fuel consumption and emissions with respect to the operating profile of the hybrid tugboat during a maneuver and with a battery bank at 15% of the nominal propulsion power.

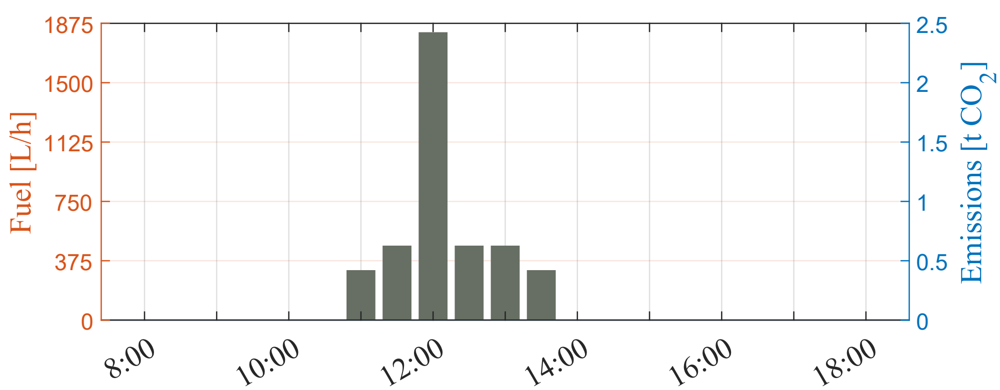


Figure 3.24: Fuel consumption and emissions with respect to the operating profile of the hybrid tugboat during a maneuver and with a battery bank at 25% of the nominal propulsion power.

Comparing figures 3.23 and 3.24 with figure 2.24, a considerable reduction in fuel consumption and emissions generated can be observed. This reduction increases as the battery bank operates at a higher nominal power (25%).

3.7.2 Two Maneuvers

As in the previous case, Figures 3.25 and 3.26 show the fuel consumption and emissions generated by two maneuvers of the hybrid tugboat.

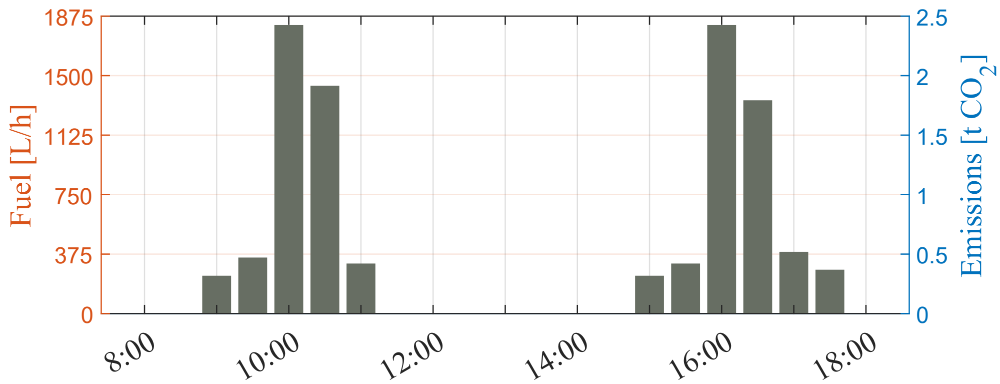


Figure 3.25: Fuel consumption and emissions with respect to the operating profile of the hybrid tugboat during two maneuvers, considering a battery bank of 15% of the nominal propulsion power.

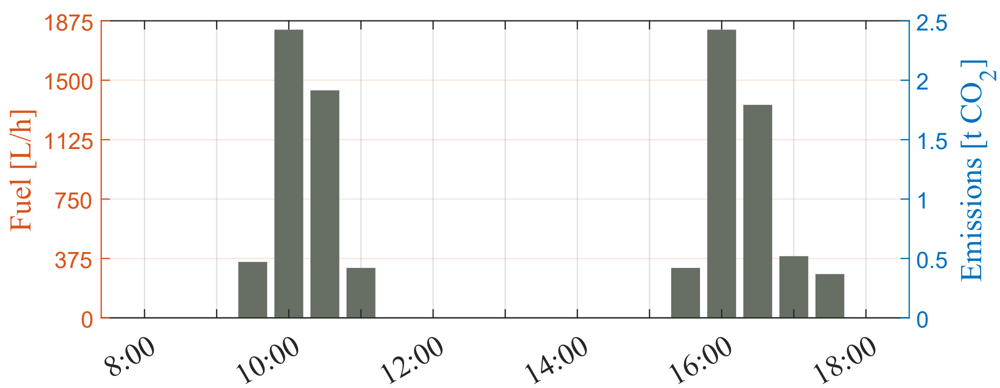
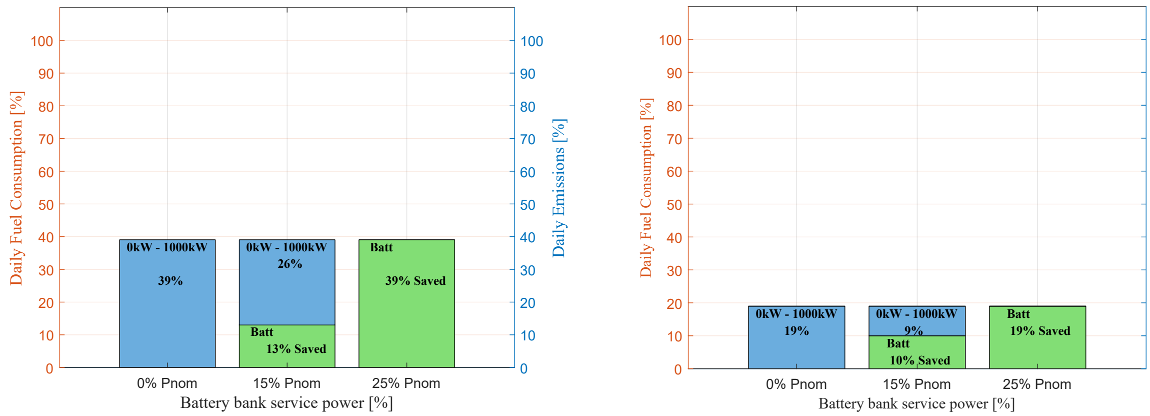


Figure 3.26: Fuel consumption and emissions with respect to the operating profile of the hybrid tugboat during two maneuvers, considering a battery bank of 25% of the nominal propulsion power.

With regard to battery bank capacity, a comparison is made of how increased capacity improves operation compared to conventional tugboats.

3.7.3 Comparation

Based on the results shown by the hybridization, a comparison is made of the contribution of each battery bank to the reduction of fuel consumption and CO_2 emissions. This can be seen in Figure 3.27.



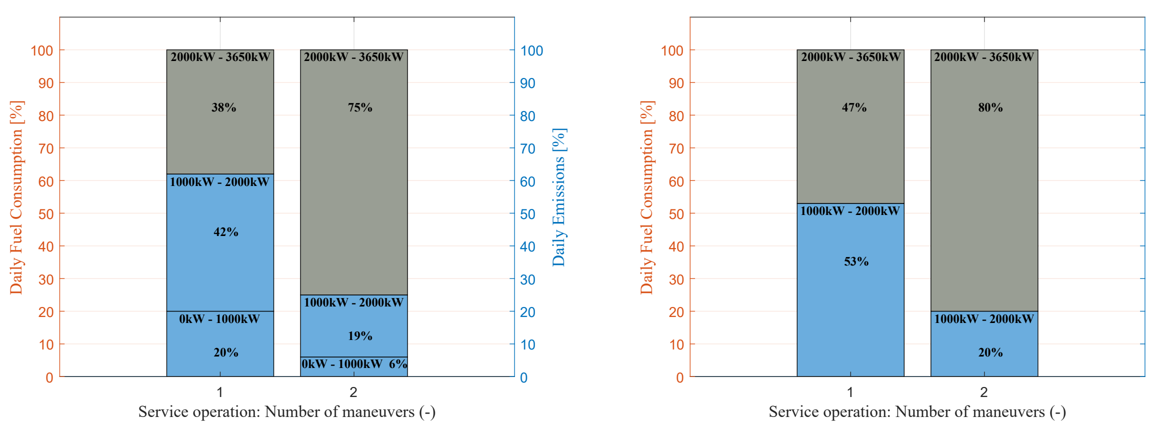
(a) Contribution of the battery bank (green) for different service powers in one maneuver.

(b) Contribution of the battery bank (green) for different service powers in two maneuvers.

Figure 3.27: Fuel savings and reduction in CO_2 emissions associated with increased service power from the battery bank.

Analysis of the data (Figures 3.27a and 3.27b) shows that the hybrid tugboat reduces fuel consumption compared to a conventional tugboat when power requirements are low, thanks to the use of the main generator (blue) and batteries (green). This demonstrates that, by having a battery bank with greater service power, the tugboat does not need to start up the main generator for minor tasks. As a result, there is lower consumption and a reduction in daily pollutant emissions.

Considering the possible use of the battery bank in the tugboat's low-power range, it is possible to observe how the percentages of daily fuel consumption and daily emissions generated are distributed for one and two maneuvers of the hybrid tugboat. This can be seen in Figure 3.28.



(a) Daily percentages based on power ranges, considering a battery bank with a 15% of the total propulsion power.

(b) Daily percentages based on power ranges, considering a battery bank with a nominal power equal to 25% of the total propulsion power.

Figure 3.28: Percentage of daily emissions and daily fuel consuption for differents battery sizes.

In this way, a comparison of the total daily fuel consumption (CO_2 emissions) between the conventional tugboat and the hybrid tugboat can be obtained for one and two maneuvers with the proposed architecture, considering a battery bank of 15% of the total propulsion power and another battery bank of 25% of the total propulsion power.

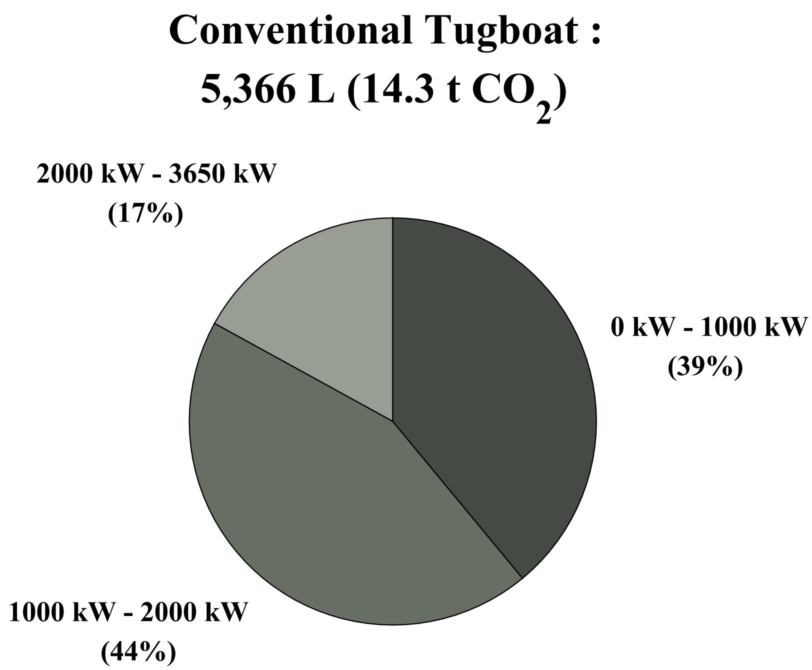
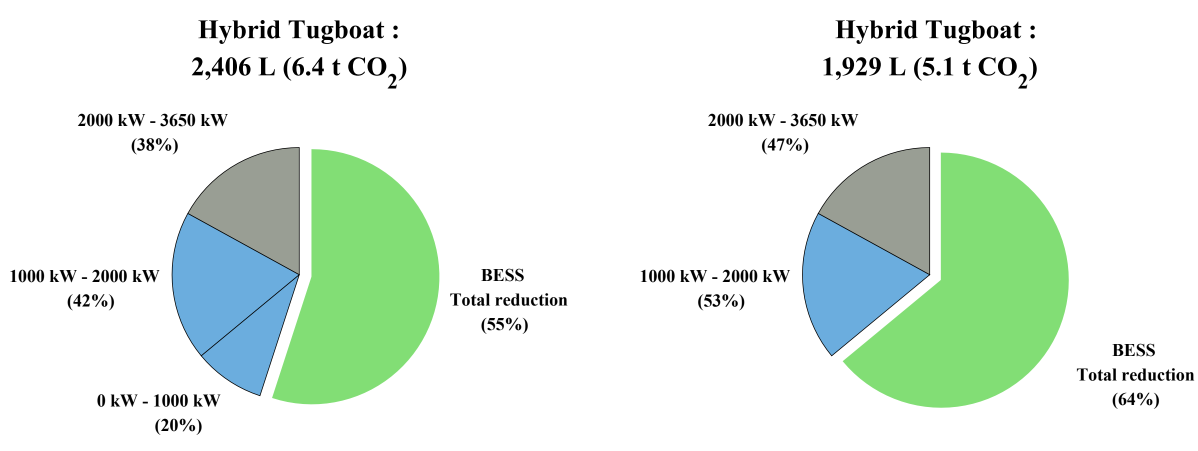


Figure 3.29: Total daily fuel consumption and daily emissions generated by a conventional tugboat during one maneuver.



(a) Daily fuel consumption and emissions of the hybrid tugboat (BESS at 15% of rated propulsion power).

(b) Daily fuel consumption and emissions of the hybrid tugboat (BESS at 25% of rated propulsion power).

Figure 3.30: Comparison between the total daily fuel consumption (total daily CO_2 emissions generated) of the hybrid tugboat for one maneuver.

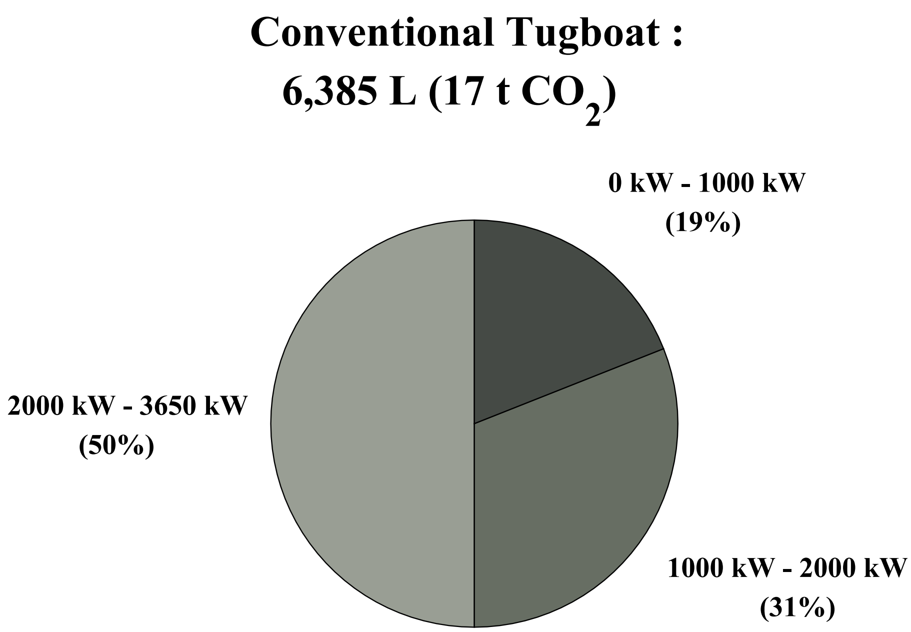
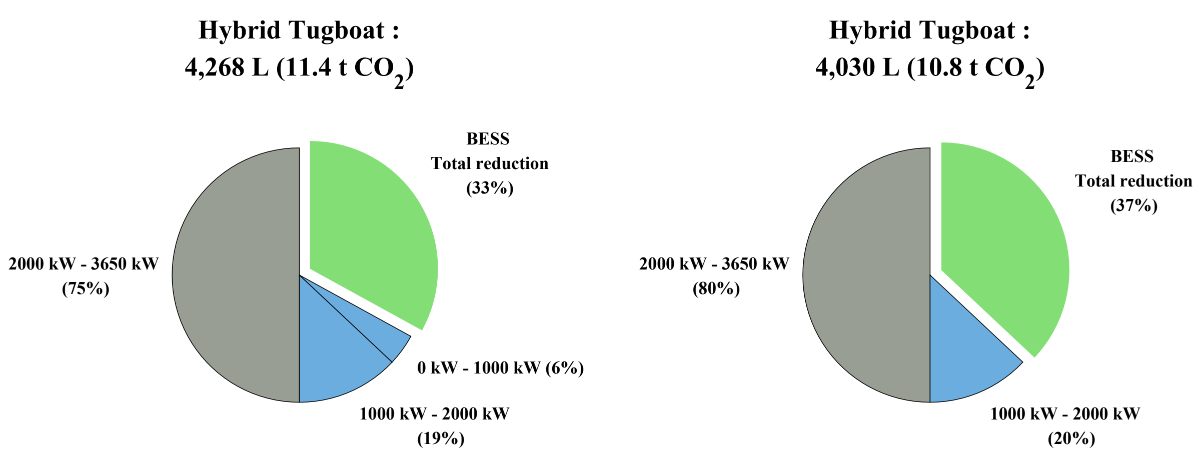


Figure 3.31: Total daily fuel consumption and daily emissions generated by a conventional tugboat during two maneuvers.



(a) Daily fuel consumption and emissions of the hybrid tugboat (BESS at 15% of rated propulsion power).

(b) Daily fuel consumption and emissions of the hybrid tugboat (BESS at 25% of rated propulsion power).

Figure 3.32: Comparison between the total daily fuel consumption (total daily CO_2 emissions generated) of the hybrid tugboat for two maneuvers.

The images above show how fuel consumption is reduced with the support of the battery bank and the use of the main generator at medium and low power levels (blue). Table 3.3 summarizes the improvements of hybridization with respect to the maneuvers of the conventional tugboat.

Table 3.3: Improvements in hybridization compared to a conventional tugboat.

Hybridization improvements		
	Batt Bank 15% P_{nom}	**Batt Bank 25% P_{nom}**
One Maneuver	55%	64%
Two Maneuvers	33%	37%

3.8 Chapter summary

This chapter establishes the theoretical and technical foundation for designing an layout for a hybrid propulsion system system. It primarily focuses on how different propulsion setups and electrical grids can improve efficiency and reduce the environmental impact of maritime operations.

- **Power Triad**: Hybrid tugboats mostly have three main energy sources: main propulsion engines, auxiliary generators, and the battery energy storage system (BESS).
  - **BESS Functions**: Beyond energy storage, the battery bank provides multiple functions such as peak shaving (management of high accelerations), spin reserve (prevention of blackouts), and load optimization (maintaining motors in their efficiency zones).
- **Propulsion Architectures**: Serial architectures (motor and propeller decoupled) and parallel architectures (motor and propeller connected to the shaft) are compared. The parallel configuration is currently the most common in hybrid tugboats, as it allows for variations in operating mode and does not rely on a generator to propel the propellers.
- **Grid Distribution**: A technological shift is taking place from traditional AC links to DC links. DC networks allow motors to operate at variable speeds. However, AC links have the advantage of not requiring a DC charger at the port, making it easier to charge hybrid tugboats.

Therefore, the operational advantages of the proposed hybrid system are the use of specific modes, such as towing, free sailing, and standby mode, to optimize efficiency. By using batteries for low-power tasks and the main generator for medium-power tasks, the tugboat can significantly reduce its daily fuel consumption and CO_2 emissions. Thus, the only decision left to make is whether to cover the low-power range fully or partially with the power delivered by the battery bank.

Chapter 4

Design of the battery bank for a hybrid propulsion system

This chapter presents the preliminary design process for the battery bank required for the tugboat's hybrid propulsion system. The energy capacity, power requirements, and operating constraints are determined, and then different storage technologies are evaluated from a technical and economic perspective.

4.1 Energy Capacity (kWh)

To calculate the net capacity required for the battery bank design, various factors that provide context must be considered [66] [67].

The fundamental energy requirement, denoted as E_{load} is calculated by integrating the power demand over the duration of the design mission.

$$E_{\text{load}} = \int_0^t (P_{\text{propulsion}}(t) + P_{\text{hotel}}(t))\, dt$$

For discrete phases, this summation becomes:

$$E_{\text{load}} = \sum_{i=1}^{n} P_i \times t_i$$

Where:

- P_i is the total electrical load at the DC bus (propulsion + hotel) in kilowatts (kW) for phase i.
- t_i is the duration of phase i in hours.

This value, E_{load}, represents the energy that must be delivered to the propellers and the hotel loads. However, calculating the installed battery capacity ($E_{\text{installed}}$) requires expanding this value to account for the physical and chemical realities of the energy storage system.

This transformation is achieved through the application of specific sizing factors.

a) System Efficiency Factor (η_{sys})

Energy stored in the electrochemical cells must traverse a electrical path to reach the propulsion motor. Each component in this chain extracts an efficiency penalty (heat loss).

1. **Coulombic/Voltaic Efficiency** (η_{bat}): Li-ion batteries are highly efficient, but internal resistance (R_{int}) causes I^2R losses during discharge. Typical round-trip efficiency is $92-96\%$; discharge efficiency is generally taken as 0.96 to 0.98 [70].

2. **DC/DC Converters** η_{conv}: Many hybrid architectures utilize DC/DC converters to decouple the battery voltage from the main DC bus voltage. These typically operate at 0.97 to 0.98 efficiency.

3. **Variable Frequency Drives (VFD)** η_{vfd}: This inverters for electric motors converting DC bus power to AC for the motors operate at 0.96 to 0.98 efficiency.

4. **Electric Motor** η_{motor}: Permanent Magnet (PM) machines are preferred for hybrid tugs due to their high efficiency at partial loads compared to induction motors. PM motor efficiency is typically 0.95 to 0.97.

To size the battery based on shaft power demand, the cumulative efficiency is:

$$\eta_{sys} = \eta_{bat} \times \eta_{conv} \times \eta_{vfd} \times \eta_{motor} \tag{4.1}$$

b) Maximum Depth of Discharge (DOD_{max}) and Reserve Margins

The Depth of Discharge (DOD) is the percentage of the battery's capacity that has been discharged relative to its total capacity. While Li-ion chemistries can be discharged to 100% (0% SOC), doing so drastically reduces cycle life and increases the risk of cell degradation.

- **Cycle Life Correlation**: The relationship between DOD and cycle life is non-linear. For example, an NMC cell might offer 2,000 cycles at 100% DOD but 5,000 cycles at 80% DOD.

- **Operational Reserve**: Marine safety standards often dictate a "reserve energy" capability. For a tugboat, this might mean retaining enough energy after a standard operation to perform emergency maneuvering or firefighting for a specified duration.

- **Voltage Sag**: At very low SOC (< 10%), the cell voltage drops rapidly. To maintain the required bus voltage and power delivery, the current must increase, leading to higher thermal stress.

Standard marine practice utilizes a Maximum DOD of 80%. This implies the battery operates between 10% and 90% SOC.

$$\text{Factor}_{\text{DOD}} = \frac{1}{DOD_{max}} \tag{4.2}$$

c) Battery Aging Factor (k_{aging})

Batteries degrade over time due to two primary mechanisms:

1. **Calendar Aging:** Loss of capacity due to chemical decomposition of the electrolyte and SEI (Solid Electrolyte Interphase) layer growth, occurring even when the battery is idle. This is temperature and SOC dependent.

2. **Cyclic Aging:** Mechanical stress on the electrode materials during ion insertion/extraction, leading to cracking and active material loss. This is driven by throughput (Ah) and C-rates.

End of Life (EOL) Definition: In the maritime industry, a battery system is considered to have reached its End of Life (EOL) when its capacity fades to 80% of its original nominal capacity (Beginning of Life, BOL). The vessel must be capable of performing its design mission on the last day of the battery's life (e.g., Year 10). Therefore, the battery must be oversized at installation so that the faded capacity still meets the requirement.

$$k_{aging} = \frac{1}{SOH_{EOL}} \tag{4.3}$$

d) Temperature Correction Factor (k_{temp})

Battery performance is inextricably linked to temperature. The standard rating is typically given at 25°C.

- **Low Temperature Effects**: At temperatures below 15°C, the viscosity of the electrolyte increases, and ion mobility decreases. This manifests as increased internal resistance. At 0°C, a Li-ion battery may only be able to deliver 80% to 90% of its rated capacity, and its ability to accept charge (regenerative braking) is severely curtailed to prevent lithium plating.

- **High Temperature Effects**: Operating above 35-40°C accelerates the chemical side reactions that drive aging. While capacity might initially increase slightly due to lower resistance, the long-term penalty is severe lifespan reduction.

Tugboat battery rooms are invariably climate-controlled (Liquid Cooling + HVAC) to maintain an optimal window of 20°C–25°C.

A conservative Temperature Correction Factor of 1.05 to 1.10 ($5 - 10\%$ oversizing) is applied. This ensures that the rated energy is available even if the ambient conditions in the battery room deviate from the ISO standard day.

e) Capacity Rating Factor and Peukert Efficiency (k_{c-rate})

Peukert's Law ($C_p = I^k t$) describes how the available capacity of a battery decreases as the discharge rate increases. For Lithium-ion batteries, the effect is minimal, with k values typically between 1.01 and 1.05.

- **Low Power Applications**: For hotel loads ($< 0.5C$), the factor is negligible (~ 1.0).

- **High Power Applications**: For propulsion boost ($> 2C$), a derating factor is prudent.

To calculate this factor, an initial C_{rate} can be made based on the required energy E_{load} and the service power $P_{service}$ using the following formula [21]:

$$C_{rate}[h^{-1}] = \frac{E_{load}[kWh]}{P_{service}[kWh]} \tag{4.4}$$

This provides an approximation of the C_{rate} to determine the value of the k_{c-rate} factor.

4.1.1 Battery Sizing

Based on the factors described previously, the calculations used to obtain the total installed energy $E_{\text{installed}}$ are presented:

1. **Total Load**: This corresponds to adding the energy from the load profile E_{prof} to the energy used for hotel loads E_{hotel}
$$E_{load} = E_{prof} + E_{hotel}$$

2. **C-rate**: Initial calculation of the C_{rate} to determine the associated factor k_{c-rate}
$$C_{rate} = \frac{E_{load}}{P_{service}} < 0 \Rightarrow k_{c-rate} = 1$$
$$C_{rate} = \frac{E_{load}}{P_{service}} \geq 0 \Rightarrow k_{c-rate} > 1$$

3. **System Efficiency**: To supply power to the mechanical shafts, the battery must generate
$$E_{batt\_out} = E_{load}/\eta_{sys}$$

4. **Max DOD**: Ensure that the battery is only used to the appropriate depth of discharge relative to its initial nominal capacity (beginning of life, BOL). To do this, calculate
$$E_{BOL} = E_{batt\_out}/DOD_{max}$$

5. **Aging Factor**: To ensure that the battery bank has a long service life, the aging factor must be taken into account
$$E_{EOL} = E_{BOL}/k_{aging}$$

6. **Apply Temperature/Rating Factors**: Consider the variation in temperature and battery discharge rate through
$$E_{installed} = E_{EOL} \cdot k_{temp} \cdot k_{c-rate}$$

The following equation is presented to obtain the total installed capacity based on the factors explained above.

$$E_{installed} = \frac{E_{load} \cdot k_{temp} \cdot k_{c-rate}}{DOD_{max} \cdot k_{aging} \cdot \eta_{sys}} \tag{4.5}$$

Based on equation (4.5), the factors used for sizing both battery banks proposed for hybridization are presented.

Table 4.1: Energy Factors

Energy Factors	
η_{sys}	0.88
DOD_{max}	0.8
k_{aging}	0.8
k_{temp}	1.05

Based on the information shown in Figures 3.19, 3.21, 3.20 and 3.22, it can be seen that one maneuver demands the most power from the battery bank in both scenarios. Thus, the total installed power for the two different battery bank options can be calculated. This can be seen in Table 4.2.

Table 4.2: Comparison of Propulsion Systems in Tugboats and Performance.

	BESS 15% P_{nom}	BESS 25% P_{nom}
E_{prof}	1,278[kWh]	3,102[kWh]
E_{hotel}	192[kWh]	192[kWh]
$\mathbf{E_{load}}$	**≈ 1,470[kWh]**	**≈ 3,300[kWh]**
$P_{service}$	600[kW]	1,000[kW]
$\mathbf{C_{rate}}$	≈ **0.4**[h^{-1}]	≈ **0.3**[h^{-1}]
k_{c-rate}	1[-]	1[-]
E_{bat_out}	1,670[kWh]	3,750[kWh]
E_{BOL}	2,088[kWh]	4,688[kWh]
E_{EOL}	2,610[kWh]	5,859[kWh]
$\mathbf{E_{installed}}$	**≈ 2,800[kWh]**	**≈ 6,300[kWh]**

4.2 Nominal Battery Voltage

The system voltage should be determined by the cost, desired generator and propulsion motor drive voltage, converter design, load requirements, cable and bus-bar rating, and the fault energy [30]. Generally, low-voltage dc (LVdc) systems are lower than 1 kV, with power system capacity in the range of 1–20MW, while medium-voltage dc (MVdc) systems are higher than 1 kV, with 20–100 MW.

- **750V DC**: This range is common in earlier hybrid designs, smaller tugboats, or retrofits where space for series-connected modules might be limited [73]. It balances safety requirements with the need for power density but becoming less favorable for tugs with MW-scale power requirements due to the heavy cabling required.
- **1000V DC**: This is the emerging standard for high-power vessels. Higher voltage reduces the current required for the same power ($P = VI$) and copper losses ($P_{loss} = I^2R$).

The battery bank is constructed of modules in series to achieve the voltage.

$$N_{series} = V_{bus\_req}/V_{cell\_nom} \tag{4.6}$$

For the battery bank in both options, a nominal voltage of $V_{bus\_req} = 750$[V] will be proposed.

4.3 Battery technologies applicable to tugboats

Although several battery technologies for different vessels were mentioned in section 1.5, in the field of hybrid and electric tugboats, three main lithium-ion chemical compositions dominate this sector [69]:

- **Nickel Manganese Cobalt (NMC)**,
- **Lithium Iron Phosphate (LFP)**,
- **Lithium Titanate Oxide (LTO)**.

4.3.1 Nickel Manganese Cobalt (NMC)

NMC technology ($LiNiMnCoO_2$) is preferred for electric vehicles and within the maritime industry as its life cycle is long while the energy density is satisfying [21].

NMC has historically been the chemistry of choice for first-generation hybrid vessels due to its high specific energy.

Main features on NMC include the following:

- **Energy Density**: NMC cells offer between 150 and 220 Wh/kg, which translates into lower densities after taking into account the BMS, cooling, and rack [68].
- **C-rate**: Standard marine NMC modules (e.g., Corvus Orca Energy) support high discharge rates, typically 3C continuous [72], ideal for the high power density demands of a tugboat.
- **Cycle Life**: NMC typically offers between 1000 and 1500 cycles [77], which may be sufficient for operating modes where the battery is used for peak shaving rather than performing full cycles every day.
- **Cost**: Cell prices range from US $120 – US $150/kWh [68]. The main drawback is thermal stability; NMC has a lower thermal runaway temperature, requiring robust active cooling systems that could increase its cost [82].

4.3.2 Lithium Iron Phosphate (LFP)

LFP ($LiFePO_4$) is rapidly gaining market share in the maritime sector, driven by safety and longevity.

- **Energy Density**: The cell density is 90-160 Wh/kg [77]. Tugboats require considerable ballast to achieve the draft necessary for propeller immersion. LFP batteries could replace lead or iron ballast, serving a dual function as a "**stability weight**" [68], representing around 30% of the total ballast, which could reach approximately 55[m^3] [71].
- **Safety**: The simple chemistry of LFP batteries (no nickel or cobalt) makes them safe and robust. These batteries are used in applications where safety and long life are more important than compact size [83].
- **Cycle Life**: The LFP offers between 2,000 and more than 5,000 cycles, even with a high discharge depth [77]. Some sources suggest that it lasts between 2 and 3 times longer than the NMC [68], making it ideal for applications that require longer life and reliability.
- **Cost**: The LFP has an estimated cell cost of between US $80 – US $100/kWh, making it 30% cheaper than the NMC option in 2026 [68].

4.3.3 Lithium Titanate Oxide (LTO)

Lithium titanate batteries ($Li_4Ti_5O_{12}$) are the only ones with a unique anode surface made from lithium and titanium oxides. These batteries for electric boats, also known as LTO batteries, operate in extreme weather conditions, offering greater safety [82].

- **Performance**: One of the main advantages of LTO is its high C-rate capacity (i.e., high maximum power) [90]. It offers exceptional service life (more than 10,000 cycles) [77].
- **Drawbacks**: LTO has the lowest energy density (60–120 Wh/kg) and the highest cost [77].
- **Cycle Life**: LTO batteries can operate safely for 25,000 cycles, achieving a service life of more than 20 years [90].

4.4 Battery Bank Design

Taking into account the battery technologies used in marine environments, the total energy required by the operating profile, and the voltage of the battery bank, the steps to follow to obtain the total number of cells needed to find the right battery bank are shown below [74].

To this end, the selected cells shall be considered to have the following input parameters, as provided by the manufacturer. This is shown in Table 4.3.

Table 4.3: Technical input information.

Input technical information	
V_{bc}	Battery cell voltage [V]
E_{bc}	Battery cell energy [Wh]
C_{bc}	Battery cell capacity [Ah]
C-rate$_{bcc}$	Continuous C-rate of the battery cell (discharge) [h^{-1}]
C-rate$_{bpc}$	Peak C-rate of the battery cell [h^{-1}]
m_{bc}	Battery cell mass [kg]
vol_{bc}	Battery cell volume [m^3]

In addition, the calculated energy and the proposed nominal voltage for each bank to be designed shall be considered as inputs. Below is the information that will be used for the small battery bank and the large battery bank.

- Small Battery Bank (15% P_{nom}):
  - V_{bp} (Battery pack voltage): 750 [V].
  - E_{bp} (Battery pack total energy): 2,800[kWh].
- Large Battery Bank 25% P_{nom}):
  - V_{bp} (Battery pack voltage): 750 [V].
  - E_{bp} (Battery pack total energy): 6,300[kWh].

The battery bank will be considered to consist of several strings connected in parallel, each of which consists of several cells connected in series. Also, the volume of the battery pack will be calculated taking into account only the battery cells, without considering other factors such as electronic circuits, cooling circuit, battery casing, wiring, etc.

Next, we will present the basic calculations used to characterize the battery bank. These include:

- Essential electrical parameters.
- Electrical performance parameters.
- Mechanical parameters.

4.4.1 Calculation of essential electrical parameters

The number of battery cells connected in series $N_{sc}[-]$ in a string is calculated by dividing the nominal battery pack voltage $V_{bp}[V]$ to the voltage of each battery cell $V_{bc}[V]$. The number of strings must be an integer. Therefore, the result of the calculation is rounded to the higher integer.

$$N_{sc} = \frac{V_{bp}}{V_{bc}} \tag{4.7}$$

The energy content of a string $E_{bs}[Wh]$ is equal with the product between the number of battery cells connected in series $N_{sc}[-]$ and the energy of a battery cell $E_{bc}[Wh]$.

$$E_{bs} = N_{sc} \cdot E_{bc} \tag{4.8}$$

Total number of strings of the battery pack $N_{sb}[-]$ is calculated by dividing the battery pack total energy $E_{bp}[Wh]$ to the energy content of a string $E_{bs}[Wh]$. The number of strings must be an integer. Therefore, the result of the calculation is rounded to the higher integer.

$$N_{sb} = \frac{E_{bp}}{E_{bs}} \tag{4.9}$$

The battery pack capacity $C_{bp}[Ah]$ is calculated as the product between the number of strings $N_{sb}[-]$ and the capacity of the battery $C_{bc}[Ah]$.

$$C_{bp} = N_{sb} \cdot C_{bc} \tag{4.10}$$

Total number of cells of the battery pack $N_{bp}[-]$ is calculated as the product between the number of strings $N_{sb}[-]$ and the number of cells in a string $N_{sc}[-]$.

$$N_{bp} = N_{sb} \cdot N_{sc} \tag{4.11}$$

4.4.2 Calculation of electrical performance parameters

The string continuous current $I_{scc}[A]$ is the product between the continuous C-rate of the battery cell C-rate$_{bcc}[h^{-1}]$ and the battery cell capacity $C_{bc}[Ah]$.

$$I_{scc} = \text{C-rate}_{bcc} \cdot C_{bc} \tag{4.12}$$

The battery pack continuous current I_{bpc}[A] is the product between the string continuous current I_{scc}[A] and the number of strings of the battery pack $N_{sb}[-]$.

$$I_{bpc} = I_{scc} \cdot N_{sb} \tag{4.13}$$

The battery pack continuous power P_{bpc}[W] is the product between battery pack continuous current I_{bpc}[A] and the battery pack voltage V_{bp}[V].

$$P_{bpc} = I_{bpc} \cdot V_{bp} \tag{4.14}$$

The string peak current I_{spc}[A] is the product between the peak C-rate of the battery cell C-rate$_{bpc}$[h^{-1}] and the battery cell capacity C_{bc}[Ah].

$$I_{spc} = \text{C-rate}_{bpc} \cdot C_{bc} \tag{4.15}$$

The battery pack peak current I_{bpp}[A] is the product between the string peak current I_{spc}[A] and the number of strings of the battery pack $N_{sb}[-]$.

$$I_{bpp} = I_{spc} \cdot N_{sb} \tag{4.16}$$

The battery pack peak power P_{bpp}[W] is the product between battery pack peak current I_{bpp}[A] and the battery pack voltage V_{bp}[V].

$$P_{bpp} = I_{bpp} \cdot V_{bp} \tag{4.17}$$

4.4.3 Calculation of mechanical parameters

The battery pack mass (cells only) m_{bp}[kg] is the product between the total number of cells $N_{bp}[-]$ and the mass of each battery cell m_{bc}[kg].

$$m_{bp} = N_{bp} \cdot m_{bc} \tag{4.18}$$

The volume of the battery pack (cells only) vol$_{bp}$[m^3] is the product between the total number of cells $N_{bp}[-]$ and the volume of each battery cell vol$_{bc}$[m^3]. This volume is only used to estimate the final volume of the battery pack, since it does not take into account the auxiliary components/systems of the battery [74].

$$\text{vol}_{bp} = N_{bp} \cdot \text{vol}_{bc} \tag{4.19}$$

4.4.4 Calculated NMC battery banks

Three different models of NMC batteries and their technical specifications are presented in Table 4.4.

Table 4.4: Technical specification of NMC Batteries.

NMC Batteries			
	ABL-012010P	**ABL-012030P**	**E4050T1**
V_{bc}	12[V]	12[V]	2.84[V]
E_{bc}	120[Wh]	360[Wh]	264[Wh]
C_{bc}	10[Ah]	30[Ah]	128[Ah]
C-rate$_{bcc}$	1 [h^{-1}]	1 [h^{-1}]	1 [h^{-1}]
C-rate$_{bpc}$	3 [h^{-1}]	3 [h^{-1}]	3 [h^{-1}]
m_{bc}	0.8[kg]	2.1[kg]	0.48[kg]
vol$_{bc}$	$0.49 \cdot 10^{-3}$[m^3]	$1.13 \cdot 10^{-3}$[m^3]	$4.3 \cdot 10^{-3}$[m^3]

The technical specifications of each model, in addition to the input parameters of the battery bank, allow the necessary calculations to be made to characterize both small and large NMC battery banks.

a) NMC Small Battery Bank (Battery Bank 15% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the small NMC battery bank are presented below in Tables 4.5 and 4.6, respectively.

Table 4.5: Essential parameters of the small NMC battery bank.

Essential parameters, NMC battery bank 15% P_{nom}			
	ABL-012010P	**ABL-012030P**	**E4050T1**
N_{sc}	63	63	264
E_{bs}	7.56[kWh]	22.68[kWh]	113[kWh]
N_{sb}	371	124	25
C_{bp}	3,710[Ah]	3,720[Ah]	3,200[Ah]
N_{bp}	23,373	7,812	6,600

Table 4.6: Performance parameters of the small NMC battery bank.

Performance parameters, NMC battery bank 15% P_{nom}			
	ABL-012010P	**ABL-012030P**	**E4050T1**
I_{scc}	5[A]	30[A]	128[A]
I_{bpc}	1.86[kA]	3.72[kA]	3.2[kA]
P_{bpc}	1,391[kW]	2,790[kW]	2,400[kW]
I_{spc}	20[A]	60[A]	750[A]
I_{bpp}	7.42[kA]	7.44[kA]	18.75[kA]
P_{bpp}	5,565[kW]	5,580[kW]	14,062[kW]
m_{bp}	18,698[kg]	16,405[kg]	3,143[kg]
vol$_{bp}$	11.45[m^3]	8.80[m^3]	28.11[m^3]

b) NMC Large Battery Bank (Battery Bank 25% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the large NMC battery bank are presented below in Tables 4.7 and 4.8, respectively.

Table 4.7: Essential parameters of the large NMC battery bank.

Essential parameters, NMC battery bank 25% P_{nom}			
	ABL-012010P	**ABL-012030P**	**E4050T1**
N_{sc}	63	63	264
E_{bs}	7.56[kWh]	22.68[kWh]	113[kWh]
N_{sb}	834	278	56
C_{bp}	8,340[Ah]	8,340[Ah]	7,168[Ah]
N_{bp}	52,542	17,514	14,784

Table 4.8: Performance parameters of the large NMC battery bank.

Performance parameters, NMC battery bank 25% P_{nom}			
	ABL-012010P	**ABL-012030P**	**E4050T1**
I_{scc}	5[A]	30[A]	128[A]
I_{bpc}	4.17[kA]	8.34[kA]	7.17[kA]
P_{bpc}	3,128[kW]	6,255[kW]	5,376[kW]
I_{spc}	20[A]	60[A]	750[A]
I_{bpp}	16.68[kA]	16.68[A]	42[kA]
P_{bpp}	12,510[kW]	12,510[kW]	31,500[kW]
m_{bp}	40,034[kg]	36,779[kg]	7,096[kg]
vol_{bp}	25.75[m^3]	19.73[m^3]	28.11[m^3]

4.4.5 Calculated LFP battery banks

As in the previous case, Table 4.9 shows three different battery models, but now LFP, with their technical specifications.

Table 4.9: Technical specification of LFP Batteries.

LFP Batteries			
	URB1270	**PSL-SC-12500**	**U27-36XP**
V_{bc}	12.8[V]	12.8[V]	38.4[V]
E_{bc}	97[Wh]	640[Wh]	1,920[Wh]
C_{bc}	7.6[Ah]	50[Ah]	50[Ah]
C-rate$_{bcc}$	2 [h^{-1}]	1 [h^{-1}]	1 [h^{-1}]
C-rate$_{bpc}$	7 [h^{-1}]	3 [h^{-1}]	3 [h^{-1}]
m_{bc}	1[kg]	6.3[kg]	18.7[kg]
vol_{bc}	$0.9 \cdot 10^{-3}$[m^3]	$5.59 \cdot 10^{-3}$[m^3]	$11.84 \cdot 10^{-3}$[m^3]

These technical specifications for each model enable the necessary calculations to be made to characterize both small and large LFP battery banks.

a) LFP Small Battery Bank (Battery Bank 15% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the small LFP battery bank are presented below in Tables 4.10 and 4.11, respectively.

Table 4.10: Essential parameters of the small LFP battery bank.

Essential parameters, LFP battery bank 15% P_{nom}			
	URB1270	**PSL-SC-12500**	**U27-36XP**
N_{sc}	59	59	20
E_{bs}	5.72[kWh]	37.76[kWh]	38.40[kWh]
N_{sb}	490	75	73
C_{bp}	3,724[Ah]	3,750[Ah]	3,650[Ah]
N_{bp}	28,910	4,425	1,460

Table 4.11: Performance parameters of the small LFP battery bank.

Performance parameters, LFP battery bank 15% P_{nom}			
	URB1270	**PSL-SC-12500**	**U27-36XP**
I_{scc}	15[A]	50[A]	50[A]
I_{bpc}	7.35[kA]	3.75[kA]	3.65[kA]
P_{bpc}	5,513[kW]	2,813[kW]	2,738[kW]
I_{spc}	55[A]	140[A]	150[A]
I_{bpp}	26.95[kA]	10.5[kA]	10.95[kA]
P_{bpp}	20,212[kW]	7,875[kW]	8,213[kW]
m_{bp}	28,910[kg]	27,877[kg]	27,302[kg]
vol_{bp}	26.27[m^3]	24.73[m^3]	17.28[m^3]

b) LFP Large Battery Bank (Battery Bank 25% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the large LFP battery bank are presented below in Tables 4.12 and 4.13, respectively.

Table 4.12: Essential parameters of the large LFP battery bank.

Essential parameters, LFP battery bank 25% P_{nom}			
	URB1270	**PSL-SC-12500**	**U27-36XP**
N_{sc}	59	59	20
E_{bs}	5.72[kWh]	37.76[kWh]	38.4[kWh]
N_{sb}	1,101	167	165
C_{bp}	8,368[Ah]	8,350[Ah]	8,250[Ah]
N_{bp}	64,969	9,853	3,300

Table 4.13: Performance parameters of the large LFP battery bank.

Performance parameters, LFP battery bank 25% P_{nom}			
	URB1270	**PSL-SC-12500**	**U27-36XP**
I_{scc}	15[A]	50[A]	50[A]
I_{bpc}	16.52[kA]	8.35[kA]	8.25[kA]
P_{bpc}	12,386[kW]	6,263[kW]	6,188[kW]
I_{spc}	55[A]	140[A]	150[A]
I_{bpp}	60.56[kA]	23.38[kA]	24.75[kA]
P_{bpp}	45,416[kW]	17,535[kW]	18,563[kW]
m_{bp}	64,960[kg]	62,074[kg]	61,710[kg]
vol_{bp}	59.05[m^3]	55.10[m^3]	39.08[m^3]

4.4.6 Calculated LTO battery banks

Finally, the alternatives found for LTO are analyzed using Table 4.14, which shows the technical specifications.

Table 4.14: Technical specification of LTO Batteries.

LTO Batteries			
	SLB12400L151	**SLB08115L140**	**ELB 1500mAh LTO**
V_{bc}	2.4[V]	2.4[V]	2.3[V]
E_{bc}	0.36[Wh]	0.03[Wh]	3.6[Wh]
C_{bc}	0.15[Ah]	0.014[Ah]	1.5[Ah]
C-rate$_{bcc}$	1 [h^{-1}]	0.5 [h^{-1}]	1 [h^{-1}]
C-rate$_{bpc}$	20 [h^{-1}]	20 [h^{-1}]	8 [h^{-1}]
m_{bc}	0.09[kg]	0.0012[kg]	0.046[kg]
vol_{bc}	$0.02 \cdot 10^{-3}$[m^3]	$0.002 \cdot 10^{-3}$[m^3]	$0.02 \cdot 10^{-3}$[m^3]

a) LTO Small Battery Bank (Battery Bank 15% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the small LTO battery bank are presented below in Tables 4.15, 4.16, and **??**, respectively.

Table 4.15: Essential parameters of the small LTO battery bank.

Essential parameters, LTO battery bank 15% P_{nom}			
	SLB12400L151	**SLB08115L140**	**ELB 1500mAh LTO**
N_{sc}	313	313	327
E_{bs}	113[Wh]	10.52[Wh]	1.18[kWh]
N_{sb}	24,850	266,241	2,379
C_{bp}	3,727[Ah]	3,727[Ah]	3,568[Ah]
N_{bp}	7,778,050	83,333,433	777,933

Table 4.16: Performance parameters of the small LTO battery bank.

Performance parameters, LTO battery bank 15% P_{nom}			
	SLB12400L151	**SLB08115L140**	**ELB 1500mAh LTO**
I_{scc}	150[mA]	7[mA]	1.5[A]
I_{bpc}	3.73[kA]	1.86[kA]	3.57[kA]
P_{bpc}	2,795[kW]	1,398[kW]	2,676[kW]
I_{spc}	3[A]	280[mA]	12[A]
I_{bpp}	74.55[kA]	74.55[kA]	28.55[kA]
P_{bpp}	55,913[kW]	55,910[kW]	21,411[kW]
m_{bp}	700,025[kg]	100,000[kg]	35,785[kg]
vol_{bp}	152.72[m^3]	192.68[m^3]	13.15[m^3]

b) LTO Large Battery Bank (Battery Bank 25% P_{nom})

The essential electrical characteristics, performance parameters, and physical characteristics of the large LTO battery bank are presented below in Tables 4.17 and 4.18, respectively.

Table 4.17: Essential parameters of the large LTO battery bank.

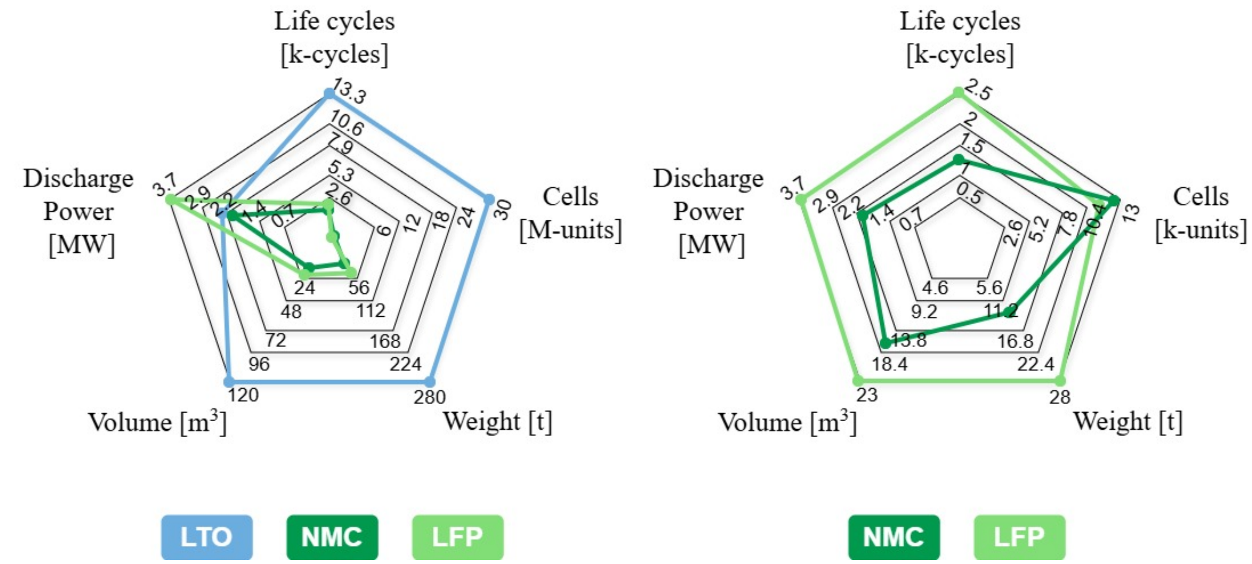
Essential parameters, LTO battery bank 25% P_{nom}			
	SLB12400L151	**SLB08115L140**	**ELB 1500mAh LTO**
N_{sc}	313	313	327
E_{bs}	113[Wh]	10.52[Wh]	1.18[kWh]
N_{sb}	55,911	599,042	5,352
C_{bp}	8,386[Ah]	8,387[Ah]	8,029[Ah]
N_{bp}	17,500,143	187,500,146	1,750,104

Table 4.18: Performance parameters of the large LTO battery bank.

Performance parameters, LTO battery bank 25% P_{nom}			
	SLB12400L151	**SLB08115L140**	**ELB 1500mAh LTO**
I_{scc}	150[mA]	7[mA]	1.5[A]
I_{bpc}	8.38[kA]	4.19[kA]	8.03[kA]
P_{bpc}	6,290[kW]	3,145[kW]	6,021[kW]
I_{spc}	3[A]	280[mA]	12[A]
I_{bpp}	167.73[kA]	167.73[kA]	64.22[kA]
P_{bpp}	125,800[kW]	125,800[kW]	48,168[kW]
m_{bp}	1,575,013[kg]	225,000[kg]	80,505[kg]
vol_{bp}	343.61[m^3]	433.54[m^3]	29.59[m^3]

4.4.7 Comparison between technologies

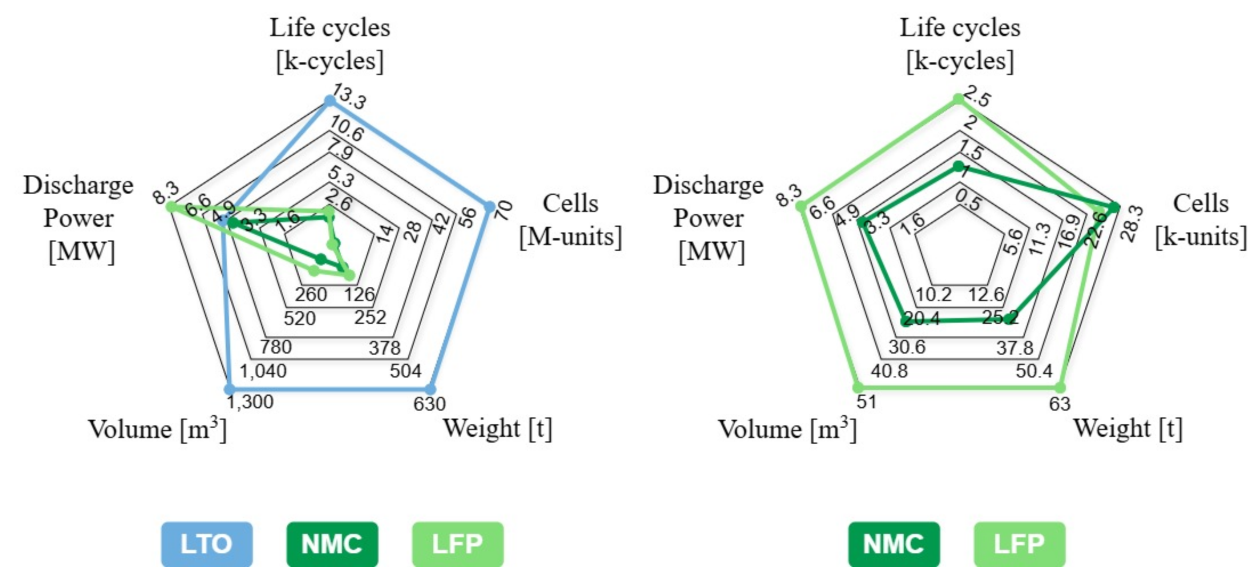
Based on the technical information provided by battery suppliers and on the calculations performed, the values obtained in each calculation for each technology are averaged to obtain average parameters for comparing technologies.



(a) Radar chart showing a technical comparison of small battery bank designs using the three technologies (average).

(b) Radar chart showing a technical comparison of small battery pack designs using NMC and LFP technologies (average).

Figure 4.1: Comparison of technologies for the technical analysis of small battery bank design.



(a) Radar chart showing a technical comparison of large battery bank designs using the three technologies (average).

(b) Radar chart showing a technical comparison of small battery pack designs using NMC and LFP technologies (average).

Figure 4.2: Comparison of technologies for the technical analysis of large battery bank design.

For comparison purposes, based on the options chosen for each technology, an average value is obtained for the technical parameters of interest to be observed, in order to provide a general analysis of the performance of each technology for the design of large and small banks.

What can be observed in Figures 4.1a and 4.2a is that LTO technology significantly outperforms most technical parameters of interest. This is because small cells were chosen for the battery banks, which can pose greater technical challenges in their implementation.

Thus, a comparison is made only between NMC and LFP technologies through Figures 4.1b and 4.1b. In general, if technology performance is paramount, both in terms of service life and discharge power, LFP is the appropriate choice. But if there are physical restrictions within the vessel, NMC is the most suitable technology.

4.4.8 Mechanical design considerations

The mechanical design is focused on the calculation of the total weight and volume of the battery bank [9].

As mentioned in section 2.1.3, **gross tonnage (GT)** is a measure of a ship's total internal volume, while **net tonnage (NT)** is a ship's gross tonnage minus the space occupied by crew quarters, machinery, navigation equipment, engine rooms, and fuel storage. Therefore, net tonnage represents the space available for passenger, accommodation and cargo storage.

Additionally, **deadweight tonnage (DWT)** measures how much weight a vessel can safely carry, including the combined weight of cargo, fuel, fresh water, ballast water, provisions, passengers, and crew.

Thus, the mechanical parameters of the vessel are presented below in Table 4.19.

Table 4.19: Mechanical parameters of the Chilean tugboat

Deadweight [kg]	**Gross Tonnage [m^3]**	**Net Tonage [m^3]**
220,000	761.81	228.54

It is necessary to take into account the volume and weight of the main generator presented in section 3.5.1, in order to consider the available space and the weight allowed for the installation of the batteries. These values are presented in table 4.20.

Table 4.20: Mechanical parameters of MTU 16V 4000.

Weight [kg]	**Volume [m^3]**
9,050	10.65

In this way, Table 4.21 shows the available space and the permitted weight for the designed battery bank.

Table 4.21: Permitted dimensions inside the vessel.

Allowed weight [kg]	**Available space [m^3]**
210,950	217.88

In this way, battery banks of different technologies that could be mechanically enabled for incorporation into the vessel are identified. Options that account for less than 10% of the vessel's capacity are highlighted in green (Table 4.21), while technologies that, given their technical characteristics, do not qualify as solutions are highlighted in red. This is shown in Table 4.22 and 4.23.

Table 4.22: Physical aspects of technologies (small battery banks).

Battery Bank 15% P_{nom}		
	m_{bp}	vol_{bp}
ABL-012010P	18,698[kg]	11.45[m^3]
ABL-012030P	16,405[kg]	8.80[m^3]
E4050T1	3,143[kg]	28.11[m^3]
URB1270	28,910[kg]	26.27[m^3]
PSL-SC-12500	27,877[kg]	24.73[m^3]
U27-36XP	27,302[kg]	17.28[m^3]
SLB12400L151	700,025[kg]	152.72[m^3]
SLB08115L140	100,000[kg]	192.68[m^3]
ELB 1500mAh LTO	35,785[kg]	13.15[m^3]

Table 4.23: Physical aspects of technologies (large battery banks).

Battery Bank 25% P_{nom}		
	m_{bp}	vol_{bp}
ABL-012010P	40,034[kg]	25.75[m^3]
ABL-012030P	36,779[kg]	19.73[m^3]
E4050T1	7,096[kg]	28.11[m^3]
URB1270	64,960[kg]	59.05[m^3]
PSL-SC-12500	62,074[kg]	55.10[m^3]
U27-36XP	61,710[kg]	39.08[m^3]
SLB12400L151	1,575,013[kg]	343.61[m^3]
SLB08115L140	225,000[kg]	433.54[m^3]
ELB 1500mAh LTO	80,505[kg]	29.59[m^3]

4.5 Economic Analysis

To analyze the economic viability of a hybrid tugboat, it is necessary to first understand the higher initial capital expenditures (CAPEX) and operating expenses (OPEX).

In addition, two additional factors related to emissions taxes and government subsidies are taken into account to analyze how this affects economic indicators and how sensitive they are:

- A CO_2 emissions tax of \$50/t_{CO_2}.
- A government subsidy of 10% of CAPEX.

4.5.1 Capital Expenditure (CAPEX)

One of the major problems facing the maritime transport sector with regard to the advancement of hybrid propulsion is how to achieve a fully satisfactory implementation of a cost-effective hybrid propulsion plant [84]. The main challenge is CAPEX, i.e., keeping the capital expenditure of hybrid equipment in line with that of conventional propulsion plants.

The cost of marine battery systems is significantly higher than automotive packs due to the ”marine tax”—the cost of ruggedization, certification, and low-volume manufacturing.

BESS is a mature technology and there are already hybrid tugboats that have been deployed [75]. A BESS contains Li-ion batteries, a battery management system, ventilation/cooling, and a dedicated fire extinguishing system. The BESS must be installed in a dedicated compartment with ventilation. However, Li-ion batteries have a very low energy density, and large battery installations will entail space and weight costs on the vessel
Fire safety is also a concern, as battery fires can be difficult to extinguish. The lifetime of the BESS is also limited by the number of charge/discharge cycles. The typical BESS lifetime is 8,000 to 12,000 cycles.

It is important to note that this is the estimated cost for the first tugboat [75]. This distinction is important because the first hybrid tugboat is considered a novel design, which requires additional engineering and design work hours.Therefore, 3.5% of the gross cost will be considered, which is a similar engineering cost in proportion to that seen in other projects [75].

The following are the main elements that are considered for the hybridization and retrofitting of a conventional tugboat.

a) Battery Bank NMC

Their life cycles are around 1,000–2,500 cycles. In a heavy-duty tug application with frequent peaks, NMC packs may degrade faster, necessitating a costly replacement mid-lifecycle (Year 7–10), which severely impacts the NPV [76].

- **System Price (2024/2025)** [81]: US$600 – US$1000 per kWh.

The higher cost is due to the use of expensive raw materials such as cobalt and the more complex manufacturing processes required to achieve the high energy density and performance characteristics [81]. For example, in electric vehicles where space and weight are critical factors, NMC batteries are often preferred despite the higher cost, as they can provide longer driving ranges without adding excessive bulk to the vehicle.

b) Battery Bank LFP

While LFP has a lower energy density (90 to 160 Wh/kg) than NMC, it offers a superior cycle life, typically achieving 2,000 to 6,000+ cycles. In many maritime applications, such as cargo vessels or mid-speed ferries, the slightly higher weight of LFP is an acceptable trade-off for the increased safety and the ability to avoid replacement for 10 years or more [77].

- **System Price (2024/2025)** [81]: US$300 – US$600 per kWh.

The lower cost can be attributed to the use of more abundant and less expensive raw materials, as well as a simpler manufacturing process [81]. Although they have a slightly lower energy density compared to NMC batteries, they are a popular choice for applications where safety and long-term reliability are of utmost importance, such as in home energy storage systems and some electric vehicles where cost is a significant consideration.

c) Battery Bank LTO

LTO systems can withstand 10,000 to 15,000+ cycles and are capable of ultra-fast charging, often reaching full capacity in 10 to 30 minutes [77].

- **System Price (2024/2025)** [81]: US$800 – US$1,200 per kWh.

The higher cost is mainly due to the production process of lithium titanate, which is more complex and requires specialized equipment. Additionally, the raw materials used in LTO batteries can be costly [81]. Despite their high cost, LTO batteries are finding applications in areas where rapid charging and high power output are crucial, such as in some electric buses and high-performance electric vehicles.

d) Power Electronics

- **DC/DC Converters**: The DC/DC converters convert and regulate the power supplied from the lithium ion batteries to the voltage requirement of the DC bus [78].
  - **System Price** [79] [80]: US$100 – US$400 per kW.
- **Inverters**: Bidirectional AFE that regulates the voltage from a battery-linked DC/DC converter to an AC propulsion bus, with marine-grade and liquid-cooled drives [86].
  - **System Price**: US$250 – US$450 per kW.
- **Variable Frequency Drives**: AC drives function as the primary interface between the electrical distribution system—whether it be a traditional AC grid or a modern common DC bus—and the three-phase motors that provide mechanical thrust [87].
  - **System Price**: US$120 – US$250 per kW.

e) Main genset and Electric motors

- **Main Genset**: The price ranges for diesel generators with a power output of 2000 kW (2 MW) vary significantly based on the manufacturer (Tier 1 vs. budget brands), application (Marine vs. Industrial), and condition (New vs. Used).
  - **System Price**: US$75 – US$750 per kW.
- **Electric Motors:** The price for a 1,000 kW (1 MW) squirrel cage induction motor suitable for hybrid tugboat propulsion varies significantly depending on the manufacturer (ABB, Siemens (SIMOTICS M), WEG, Danfoss Editron), cooling method, and classification status.
  - **System Price**: US$90 – US$250 per kW.

f) Integration and Safety systems

- **Integration/cabling**: It is considered to be around 20% of the cost of equipment.
  - **System Price**: US$100, 000 – US$175, 0000.

- **Fire/Safety Systems:** Battery rooms must be A-60 fire-rated, gas-tight, and equipped with explosion relief vents and specialized ventilation to handle potential off-gassing, adding structural steel and HVAC costs [88].
  - **System Price**: US$50,000 – US$80,000.

g) Total Cost

Tables 4.24 and 4.25 show the total cost obtained for the different elements identified for both the small battery bank and the large battery bank, using an average of the highest and lowest costs for each element.

Table 4.24: BESS implementation costs, without taxes or subsides (small bank).

Battery Bank 15% P_{nom}	
NMC Battery Bank	US $2,240,000
LFP Battery Bank	US $1,260,000
LTO Battery Bank	US $2,800,000
DC/DC Converters	US $150,000
Inverter and Drives	US $580,000
Main Genset	US $825,000
Electric Motors	US $340,000
Integration/Cabling/Equipment	US $137,500
Fire/Safety Systems	US $65,000

Table 4.25: BESS implementation costs, without taxes or subsides (small bank).

Battery Bank 25% P_{nom}	
NMC Battery Bank	US $5,040,000
LFP Battery Bank	US $2,835,000
LTO Battery Bank	US $6,300,000
DC/DC Converters	US $250,000
Inverter and Drives	US $720,000
Main Genset	US $825,000
Electric Motors	US $340,000
Integration/Cabling/Equipment	US $137,500
Fire/Safety Systems	US $65,000

Using the cost information provided in the tables, the cost associated with engineering is calculated, considered to be 3.5% of the total costs. This can be seen in tables 4.26 and 4.27.

Table 4.26: Engineering cost associated with the small battery bank.

Battery Bank 15% P_{nom}	
Engineering (NMC)	US $151,812
Engineering (LFP)	US $117,512
Engineering (LTO)	US $171,412

Table 4.27: Engineering cost associated with the large battery bank.

Battery Bank 25% P_{nom}	
Engineering (NMC)	US $258,212
Engineering (LFP)	US $181,037
Engineering (LTO)	US $302,312

Engineering costs will be added to the costs obtained for hybrid propulsion system equipment, thus obtaining the CAPEX for each battery bank. This can be seen in Tables 4.28 and 4.29.

Table 4.28: Calculated CAPEX, without taxes or subsides (small bank).

Battery Bank 15% P_{nom}	
$CAPEX_{NMC}$	US $4,489,313
$CAPEX_{LFP}$	US $3,239,988
$CAPEX_{LTO}$	US $4,726,088

Table 4.29: Calculated CAPEX, without taxes or subsides (large bank).

Battery Bank 25% P_{nom}	
$CAPEX_{NMC}$	US $7,635,713
$CAPEX_{LFP}$	US $5,353,538
$CAPEX_{LTO}$	US $8,939,813

This additional capital expenditure must be recouped through reduced operating costs resulting from the use of electricity instead of diesel fuel and reduced maintenance [85]. The local price of electricity and the local price of fuel are very important for the business case. This improves substantially when authorities provide cheap green electricity at the port to achieve their zero-emission targets.

4.5.2 Operational Expenditure (OPEX)

OPEX savings are derived primarily from fuel reduction during ”loitering” (low load) and maintenance intervals extended by reduced engine runtime [84].

Thus, the following variables are considered in operating expenses.

a) Fuel Consumption Savings

A conventional tugboat consumes approximately several thousand liters of fuel just idling during its daily workday. A hybrid tugboat in electric mode consumes 0 liters in that mode. The batteries allow the load on the main engines to be balanced, thereby improving the specific fuel consumption (BSFC).

b) Maintenance Engine Reduction

Hybrid tugs can reduce main engine operating hours by up to 60% per year by using batteries for transit and standby mode. This extends the overhaul intervals for propulsion engines, reducing annualized maintenance costs. However, these savings in propulsion engine maintenance are partially offset by the need to budget for battery replacement, so an "amortization fund" must be established in the financial model to cover battery pack replacement [85]. For NMC/LFP, replacement will be considered after 15 years, while for LTO, a useful life of 30 years will be considered, exceeding the project period, so no replacement cost will be amortized for this technology.

c) Total Costs

To calculate operating costs, a fuel cost of \$0.8/L will be used, and the annual consumption for both one maneuver and two maneuvers will be averaged. These consumption figures are obtained from section 3.7.3.

Table 4.30: Operating costs, excluding taxes and subsidies (small bank).

Battery Bank 15% P_{nom}		
	Conventional Tug	**Hybrid Tug**
Total fuel cost	US \$1,715,646	US \$974,404
Electricity	-	US \$30,000
Engine Maintenance	US \$40,000	US \$20,000
NMC Battery Replacement	-	US \$149,400
LFP Battery Replacement	-	US \$84,000
LTO Battery Replacement	-	-

Table 4.31: Operating costs, excluding taxes and subsidies (large bank).

Battery Bank 25% P_{nom}		
	Conventional Tug	**Hybrid Tug**
Total fuel cost	US \$1,715,646	US \$870,014
Electricity	-	US \$30,000
Engine Maintenance	US \$40,000	US \$20,000
NMC Battery Replacement	-	US \$336,000
LFP Battery Replacement	-	US \$189,000
LTO Battery Replacement	-	-

With the operating costs, we proceed to calculate the OPEX for each technology, which can be seen in Tables 4.32 and 4.33.

Table 4.32: Calculated OPEX (annual), without taxes or subsides (small bank).

Battery Bank 15% P_{nom}	
$OPEX_{NMC}$	US \$1,173,737
$OPEX_{LFP}$	US \$1,108,404
$OPEX_{LTO}$	US \$1,024,404

Table 4.33: Calculated OPEX (annual), without taxes or subsides (large bank).

Battery Bank 25% P_{nom}	
$OPEX_{NMC}$	US $\$1,256,014$
$OPEX_{LFP}$	US $\$1,109,014$
$OPEX_{LTO}$	US $\$920,014$

Finally, the annual benefit can be obtained by calculating the difference between the OPEX of the conventional tugboat ($OPEX_{tug}$) and the OPEX of each technology. These results are shown in Tables 4.34 and 4.35.

Table 4.34: Calculated profit (annual), without taxes or subsides (small bank).

Battery Bank 15% P_{nom}	
$OPEX_{TUG}$	US $\$1,755,646$
Profit (NMC)	US $\$581,909$
Profit (LFP)	US $\$647,242$
Profit (LTO)	US $\$731,242$

Table 4.35: Calculated profit (annual), without taxes or subsides (large bank).

Battery Bank 25% P_{nom}	
$OPEX_{TUG}$	US $\$1,755,646$
Profit (NMC)	US $\$499,632$
Profit (LFP)	US $\$646,632$
Profit (LTO)	US $\$835,632$

The economic benefit provides insight into economic sensitivity indicators, which are essential for analyzing the life of a project.

4.5.3 Financial Performance

Based on the information described above, the viability of the project can be determined using economic indicators such as net present value (NPV), internal rate of return (IRR), and finally return on investment (ROI).

a) Net Present Value (NPV)

The following equation will be used to calculate the NPV.

$$\mathrm{NPV} = \sum_{t=1}^{\mathrm{LFS}} \frac{\mathrm{Prf}}{(1+\mathrm{ds})^t} - \mathrm{Inv}. \tag{4.20}$$

Where:

- **LFS**: Lifespan of the project (years).
- **Prf**: Annual profit.

- **ds**: Discount rate.
- **Inv**: Initial Investment.

For all scenarios, a **discount rate** of 8% and a useful life of 10 and 20 years for the hybrid tugboat will be proposed. In this way, the positive NPV will demonstrate that the project will generate a return on investment.

In addition, it is proposed that the batteries used will have a lifespan of 3,000 to 5,000 cycles during the lifetime of this project.

This provides the net present values for each technology and for each battery bank requested.

Table 4.36: Calculated NPV, without taxes or subsides (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	$-\$584,658$	$\$491,522$	$\$1,223,953$
NPV_{LFP}	$\$1,103,059$	$\$2,300,067$	$\$3,114,730$
NPV_{LTO}	$\$180,606$	$\$1,532,963$	$\$2,453,354$

Table 4.37: Calculated NPV, without taxes or subsides (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-$\$4,283,141$	$-\$3,359,123$	-$\$2,730,252$
NPV_{LFP}	-$\$1,014,584$	$\$181,295$	$\$995,191$
NPV_{LTO}	-$\$2,766,444$	-$\$1,064,972$	$\$93,414$

Tables 4.36 and 4.37 show that for a small battery bank, the project is economically viable for all technologies over a short period of time. This contrasts with the large battery bank, where only LFP technology is viable, NMC is viable over a period of 15 years, and LTO is not viable in any of the proposed periods.

The case is also analyzed in terms of how it is affected by emissions taxes and government grants, effects that can be applied jointly or separately. This information is provided in Appendix A.1, Appendix A.2, and Appendix A.3.

b) Internal Rate of Return (IRR)

IRR is an indicator of a project's profitability that is used to decide whether to accept or reject an investment project [89]. To do this, the IRR is compared to a minimum rate or discount rate. If the project's rate of return—expressed by the IRR—exceeds the discount rate, it is accepted; otherwise, it is rejected. The IRR can be calculated using the following equation:

$$0 = \sum_{t=1}^{\mathrm{LFS}} \frac{\mathrm{Prf}}{(1+\mathrm{IRR})^t} - \mathrm{Inv}. \tag{4.21}$$

Where:

- **LFS**: Lifespan of the project (years).

- **Prf**: Annual profit.
- **Inv**: Initial Investment.
- **IRR**: Internal rate of return.

The results obtained for this economic variable for each design are shown below.

Table 4.38: Calculated IRR, without taxes or subsides (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	5.02%	9.75%	11.49%
IRR_{LFP}	15.07%	18.39%	19.40%
IRR_{LTO}	8.84%	13.00%	14.43%

Table 4.39: Calculated IRR, without taxes or subsides (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	−7.05%	−0.23%	2.71%
IRR_{LFP}	3.59%	8.55%	10.41%
IRR_{LTO}	−1.21%	4.56%	6.87%

For the cases presented in Tables 4.38 and 4.39, it can be seen that all small battery banks manage to exceed the discount rate imposed by the investment, even though LTO technology slightly exceeds the discount rate over a 10-year period. In the case of large battery banks, only LFP manages to be profitable within the 10-year period, and extending the period allows for the viability of NMC and LFP projects, which is not the case with LTO technology, even when considering a 20-year period.

The case is also analyzed in terms of how it is affected by emissions taxes and government grants, effects that can be applied jointly or separately. This information is provided in Appendix A.1, Appendix A.2, and Appendix A.3.

c) Return on Investment (ROI)/ Payback Period

Return on investment (ROI) is a measure of profitability. It indicates how much profit or loss a company makes from an investment in relation to its cost. Measuring return on investment helps identify which projects truly drive profitability, optimize resources, and strengthen the company's competitiveness.

The formula is as follows:

$$ROI = \frac{G - Inv}{Inv} \cdot 100 \tag{4.22}$$

- **G**: Annual gain from investment.
- **Inv**: Initial Investment.

To calculate the period in which the investment is recouped and profits are generated P_{ROI}, the following formula is used:

$$P_{ROI} = \frac{Inv}{G} \tag{4.23}$$

This provides the periods for each technology in which the return on investment begins to be positive. This can be seen in Tables X and Y.

Table 4.40: Calculated period for positive ROI, without taxes or subsides (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 7 years and 9 months
$ROI_{LFP} > 0$	≈ 5 years
$ROI_{LTO} > 0$	≈ 6 years and 6 months

Table 4.41: Calculated period for positive ROI, without taxes or subsides (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 14 years and 9 months
$ROI_{LFP} > 0$	≈ 8 years
$ROI_{LTO} > 0$	≈ 10 years and 4 months

Based on the ROI financial indicator, it can be seen that LFP technology is the one that allows profits to be obtained in a shorter period of time, both for small battery banks and large battery banks. An increase in fuel prices allows for higher annual profits when comparing conventional systems with hybrid systems, which in turn also shortens the payback period. Conversely, high onshore electricity rates can prolong the payback period.

The case is also analyzed in terms of how it is affected by emissions taxes and government grants, effects that can be applied jointly or separately. This information is provided in Appendix A.1, Appendix A.2, and Appendix A.3.

4.6 Chapter Summary

This chapter has the focus on the preliminary design, technical evaluation, and economic viability of an Energy Storage System (ESS) for a hybrid tugboat. The chapter outlines the methodology for sizing battery banks and compares three primary lithium-ion technologies: NMC, LFP, and LTO.

1. **Sizing Methodology and Requirements**: The design process begins by calculating the required energy capacity based on the tugboat's operational load profile (E_{load}), which includes both propulsion and "hotel" (auxiliary) loads. To determine the final installed capacity ($E_{installed}$), it was applied several correction factors.

2. **Technology Comparison**: It was compared Nickel Manganese Cobalt (NMC), Lithium Iron Phosphate (LFP), and Lithium Titanate Oxide (LTO), , based on market options. The comparison of alternatives was carried out in a particularly technical manner.

3. **Economic and Financial Results**: The economic analysis considers the Capital Expenditure (CAPEX) of retrofitting a tugboat and the Operational Expenditure (OPEX) savings from reduced fuel consumption.

The conclusion of the design analysis is that LFP technology is the most viable solution for hybrid tugboats, as it balances lower initial cost with a long service life, enabling a faster return on investment compared to NMC or LTO.

Chapter 5

Simulation results

This chapter presents the simulation of the battery bank following selection based on technical and economic criteria. The proposed model for the cell is presented and its electrical performance is analyzed, with the aim of analyzing the electrical performance of the battery bank.

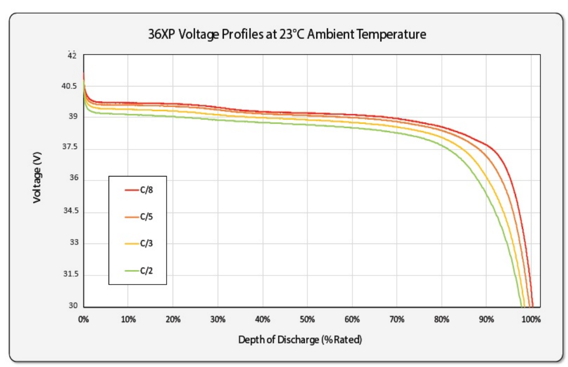
5.1 Cell Battery

The battery is chosen based on electrical performance, service life, and return on investment. Based on the comparison table, the best option is chosen.

In this case, based on the characteristics and economics, LFP technology is chosen to model both the small and large battery banks.

The selection will be made based on the best of the three alternatives presented for LFP technology.

(a) U27 36XP Valence LFP battery cell.



(b) Voltage profile, U27 36XP Valence LFP battery cell.

Figure 5.1: LFP battery selected for modeling. [91].

5.1.1 Battery mathematical model

One way to model the battery cell is to obtain a mathematical model that allows us to simulate the electrical behavior of the cell.

To do this, we will use Shepherd's order 0 mathematical model [92] [93], which models the battery cell as a Thevenin equivalent, parameterizing the controlled voltage source, as shown in the following equation:

$$V_{bat}(t) = E_0 - K\frac{Q_0}{Q_0 - Q(t)}Q(t) + A \cdot e^{-BQ(t)} \tag{5.1}$$

$$Q(t) = \int_0^t i_{bat}(t)dt \tag{5.2}$$

Based on equation 5.1, its parameters are described in Table 5.1 and its methods of obtaining it in Table 5.2, respectively, understanding that these parameters are obtained from technical information derived from the cell's discharge profile, as shown in Figure 5.2.

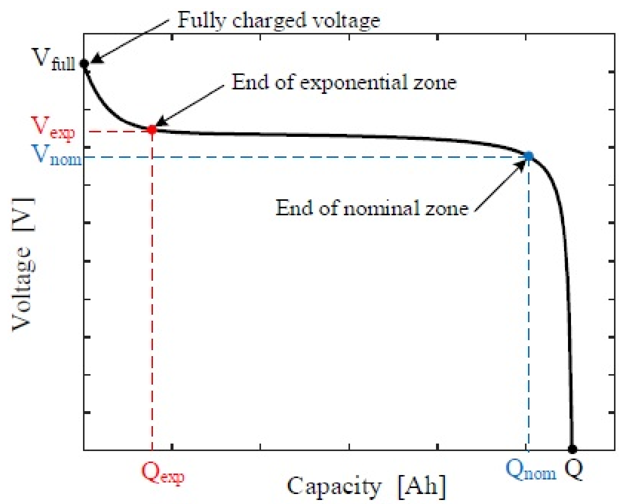


Figure 5.2: Discharge curve description [92].

Table 5.1: Technical input information for Shephard equation.

Equation fundamental parameters	
E_0	Open circuit voltage [V]
K	Polarization voltage [V]
A	Exponential zone amplitude [V]
B	Exponential zone inverse time constant [1/Ah]
Q_0	Nominal battery capacity [Ah]

$$\begin{aligned}
E_0 &= V_{full} + K + R \cdot I_{nom} - A \\
A &= V_{full} - V_{exp} \\
B &= \frac{3}{Q_{exp}} \\
K &= \frac{(V_{full} - V_{nom} + A(e^{-B \cdot Q_0} - 1)) \cdot \Delta Q}{Q_{full}} \\
\Delta Q &= Q_{full} - Q_0
\end{aligned} \tag{5.3}$$

Table 5.2: Technical description of sub-parameters.

Technical sub-parameters description	
V_{full}	Fully charged voltage [V]
V_{exp}	Voltage at end of exponential zone [V]
Q_{full}	Total charge delivered [Ah]
Q_{exp}	Charge depleted at end of exponential zone [Ah]
V_{nom}	Voltage at end of nominal zone [V]
Q_0	Charge depleted at end of nominal zone [Ah]
I_{nom}	Nominal Current [A]

5.1.2 Model simulation

The objective of the parameters shown in Table 5.1 is to modify the different areas of the curve provided by the mathematical model so that it matches the curve provided by the manufacturer. In this way, the parameters obtained and the manually adjusted parameters are shown.

Table 5.3: Parameters of the classic and adjusted Shepherd model for a battery cell.

Parameters	Classic Shepherd parameter value	Adjusted Sheperd parameter value
A	-[V]	1.44[V]
B	-[Ah^{-1}]	3[Ah^{-1}]
K	-[V]	0.0051[V]
E_0	-[V]	39.25[V]

Based on the adjusted parameters of the mathematically obtained curve, the comparison shown in Figure 5.3 allows to determine that the model obtained has a discharge profile similar to the discharge profile indicated by the manufacturer.

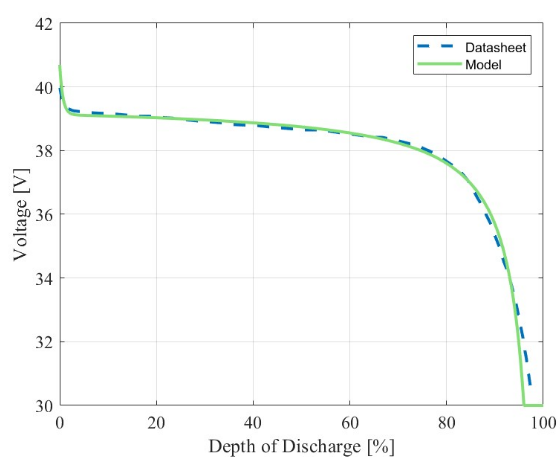


Figure 5.3: discharge curve of the cell battery.

5.2 Battery Bank

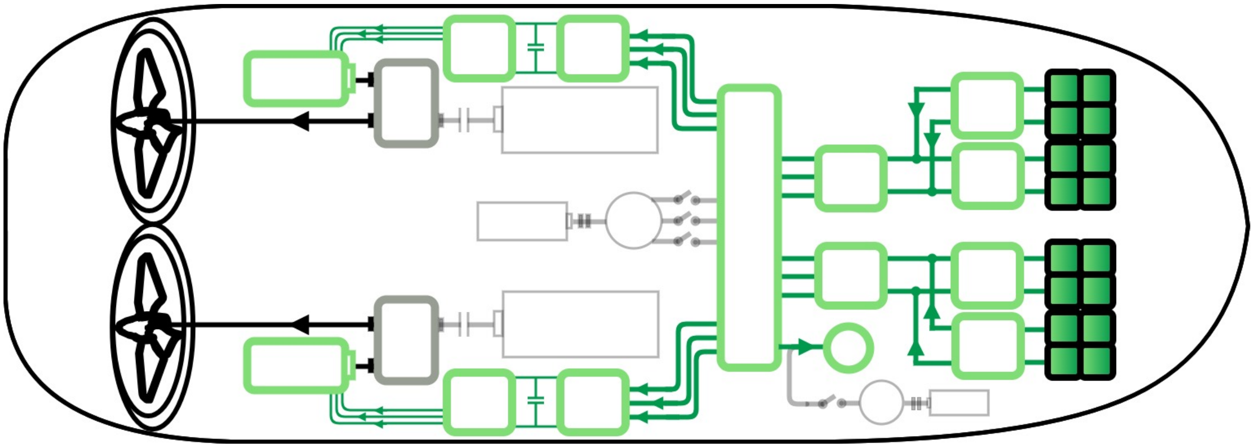


Figure 5.4: Proposed layout for battery bank implementation.

To simulate the battery bank, it is necessary to know its distribution within the tugboat, as shown in Figure 5.4.

This figure proposes dividing the battery bank into four equal packs in order to reduce the nominal power of the DC/DC converters that feed the proposed AC bus.

In this scenario, each battery pack is connected to a DC/DC converter, which connects to a 1,000 V DC bus and then to a three-phase inverter that delivers power to the 400 V, 50 Hz AC bus.

Based on the parameters obtained in Table 5.3 for the cell, they are scaled to obtain the discharge voltage profile of the battery bank based on the number of cells in series n_s and the number of strings in parallel n_p. This exercise is performed for both the small and large battery banks.

5.2.1 Small Battery Bank

The parameters in Table 5.4 allow the mathematical model for the small battery bank to be scaled. The discharge voltage profile of the bank is shown in Figure 5.5.

Table 5.4: Parameters small LFP bank.

Small battery bank parameters.	
n_s	20
n_p	73
E_0	784.9811
K	0.0012
A	28.8
B	0.0411

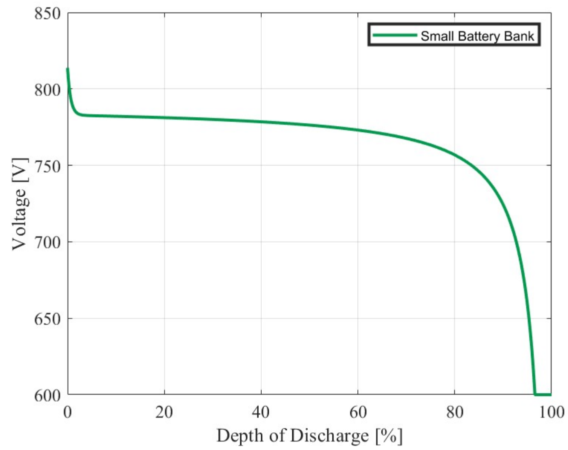


Figure 5.5: Voltage profile of the small battery bank.

The small battery bank model is simulated and subjected to the operating profiles proposed in Figures 3.19 and 3.21, which graph the battery bank's operation. The results of the bank's operation are shown in Figures 5.6 and 5.7.

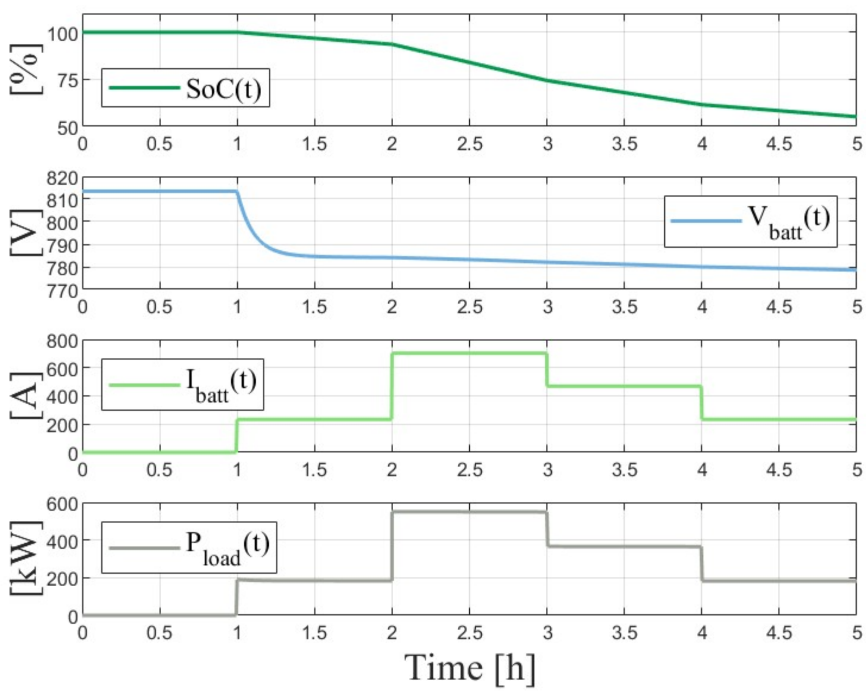


Figure 5.6: Operation and electrical performance of the small battery bank for one maneuver.

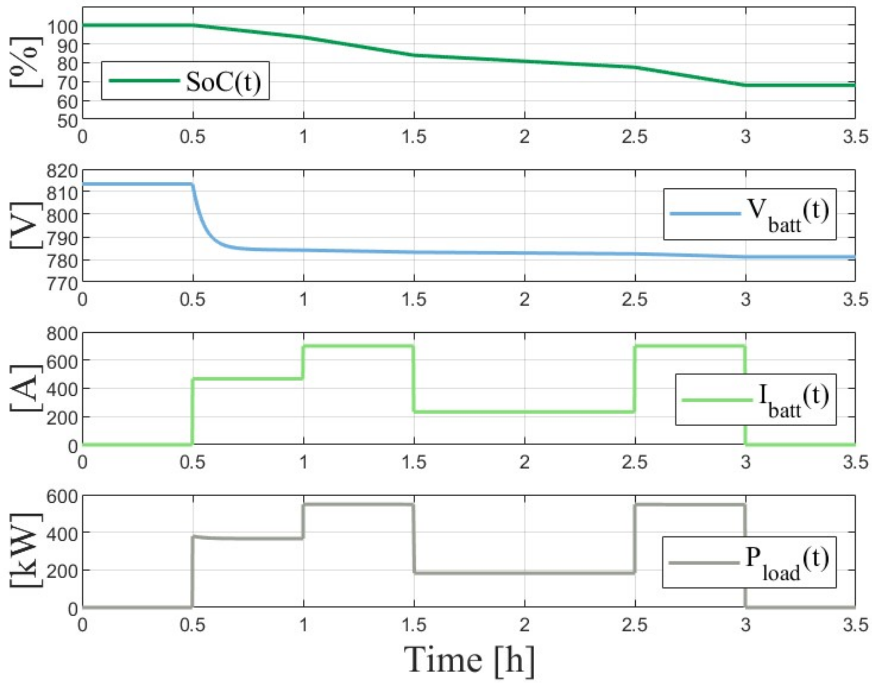


Figure 5.7: Operation and electrical performance of the small battery bank for two maneuvers.

Figures 5.6 and 5.7 show how the battery charge status does not exceed 50% discharge, which allows us to verify through simulation that the bank's dimensions, taking into account various considerations, were adequate for the load profiles indicated.

This allows for flexibility in charging the battery bank in port after service maneuvers and delivering power to hotel loads.

5.2.2 Large Battery Bank

As in the previous case, the battery cell parameters are rescaled, but now considering the large battery bank, which translates into a greater number of strings connected in parallel n_p. This information can be seen in Table 5.5, and the discharge voltage profile of this bank is shown in Figure 5.8.

Table 5.5: Parameters large LFP battery bank.

Large battery bank parameters.	
n_s	20
n_p	165
E_0	784.9811
K	0.0012
A	28.8
B	0.0182

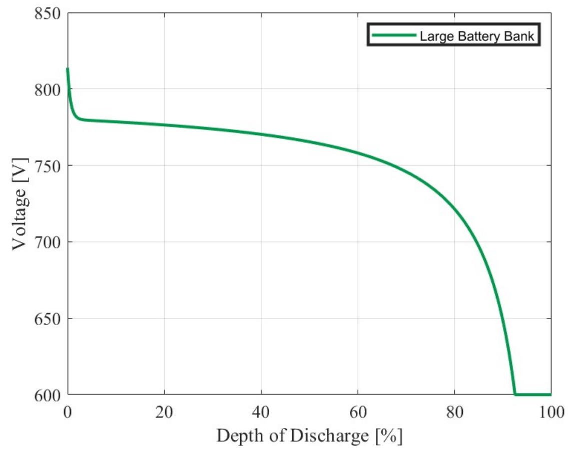


Figure 5.8: Voltage profile of the small battery bank.

Based on the discharge profile, the battery bank is subjected to electrical performance simulations, which allow us to determine how the battery bank will behave for the operating profiles proposed in Figures 3.20 and 3.22. The results are shown below.

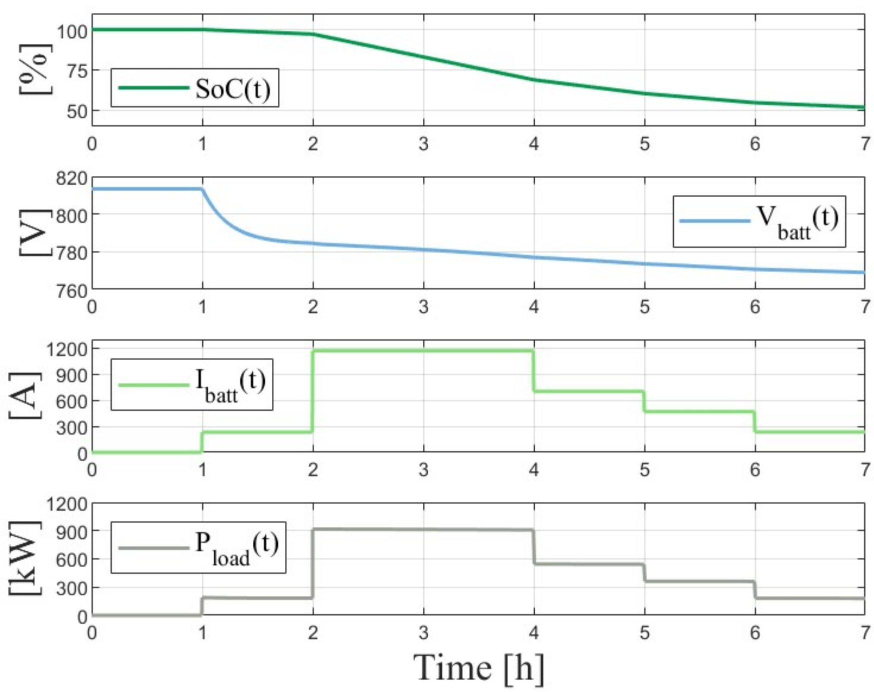


Figure 5.9: Operation and electrical performance of the large battery bank for one maneuver.

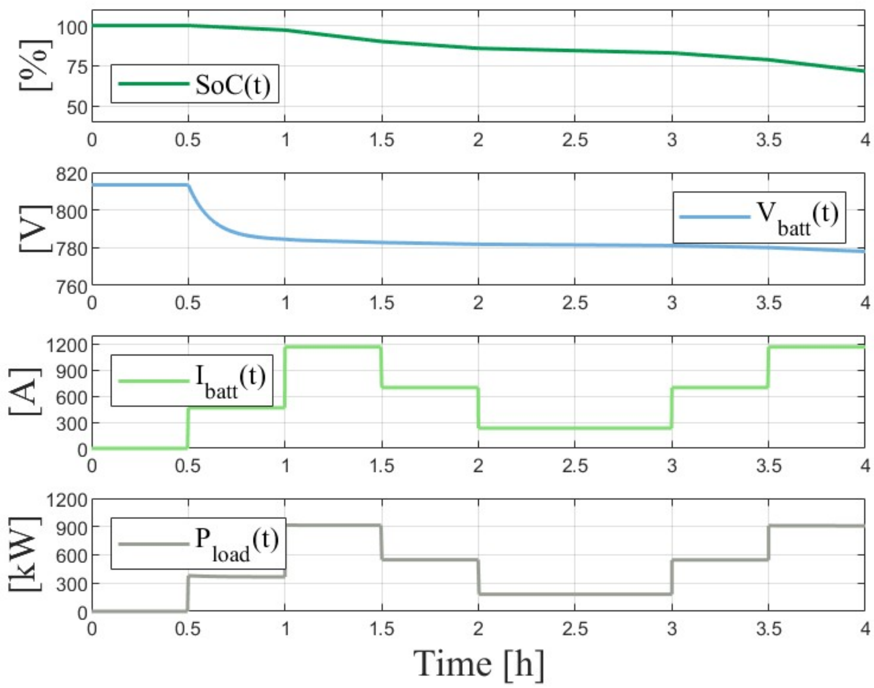


Figure 5.10: Operation and electrical performance of the large battery bank for two maneuvers.

The electrical performance of the large battery bank shows that, for the maneuvers studied, it is possible to deliver this power without exceeding 50% discharge of the battery bank. It can even be seen that for two maneuvers, the state of charge of the bank reaches around 75%, which allows flexibility for charging in port and could increase the useful life of the bank.

5.3 Bus DC

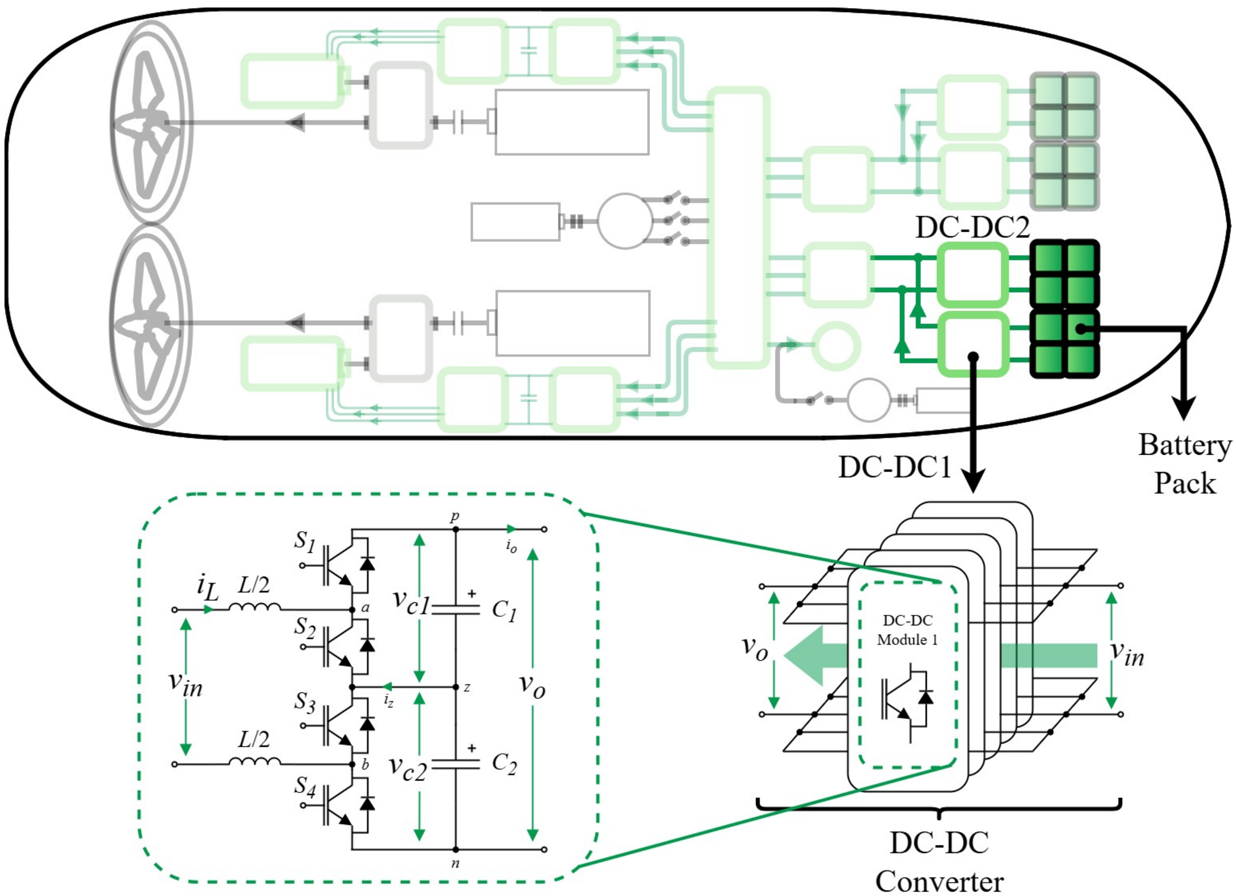


Figure 5.11: Power converter implementation for Bus DC regulation.

The performance of the battery bank for regulating a 1,000[V] DC bus is analyzed below. To do this, the operation of several power converters connected in parallel to the battery pack is analyzed, as well as their electrical performance in regulation.

The proposed implementation for the power converters is shown in Figure 5.11. The proposal to use this power converter is related to the fact that, having more than two levels, there is less electrical stress on the internal components of this converter [99], which could extend the useful life of this converter and even the useful life of the hybrid vessel. The maximum stable operating power is achieved for both the small and large battery banks, as shown in Table 4.2. The performance of an individual power module will be analyzed, as well as the performance of the set of power modules that enable the DC-DC converter connected to the battery pack to be implemented.

The power module chosen is a TLBBC, which is explained in Appendix A.4.

5.3.1 Small Battery Bank

In the case of the small battery bank, each battery pack will have 18 strings in parallel. If the AC bus is designed to deliver a service power of 600[kW], then each pack will deliver a power of

150[kW]. This means that each DC-DC converter must be capable of operating at that power and, therefore, the power modules must operate at a minimum of 30[kW].

Next, we analyze the operation of a power module inserted in the DC-DC converter and then the characteristics of each DC-DC converter.

a) TLBBC

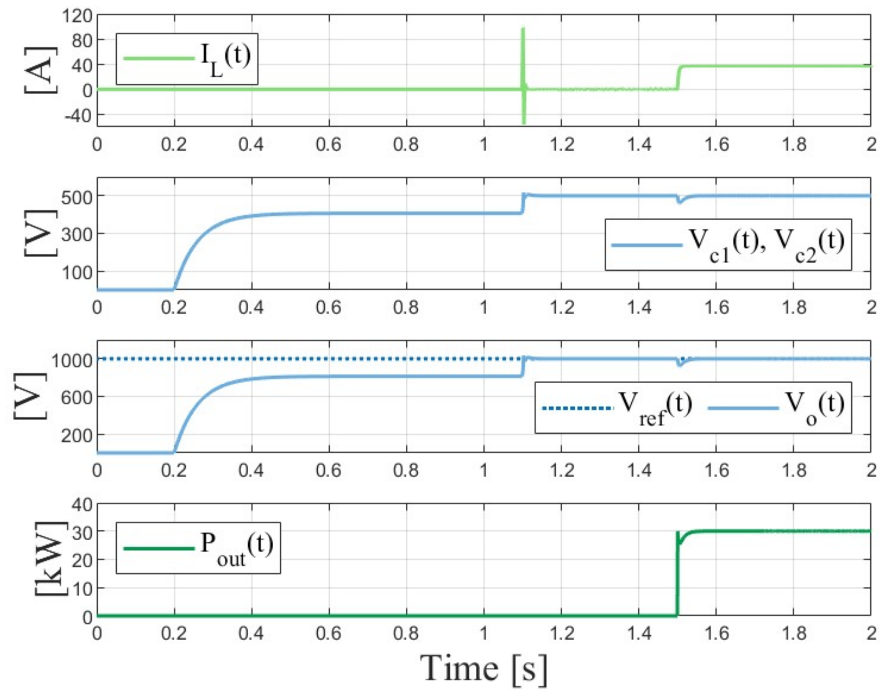


Figure 5.12: Electrical performance of power module under nominal operation.

Figure 5.12 shows the input current to the power module, the voltage across each capacitor (whether balanced or not), the module output voltage, and finally the operating power achieved by the module. Based on the voltage graphs, it can be seen that the module is initially precharged in order to reduce the current peak observed when the voltage reference is delivered to the module [95]. In addition, the difference in capacitor voltages is kept to a minimum, resulting in a stable output voltage, as required by the reference. Finally, it can be seen that when faced with a load step, the module manages to reach 30[kW], maintaining the output voltage at 1,000[V] and requiring a current of approximately 40[A] from the battery pack.

b) Bus DC

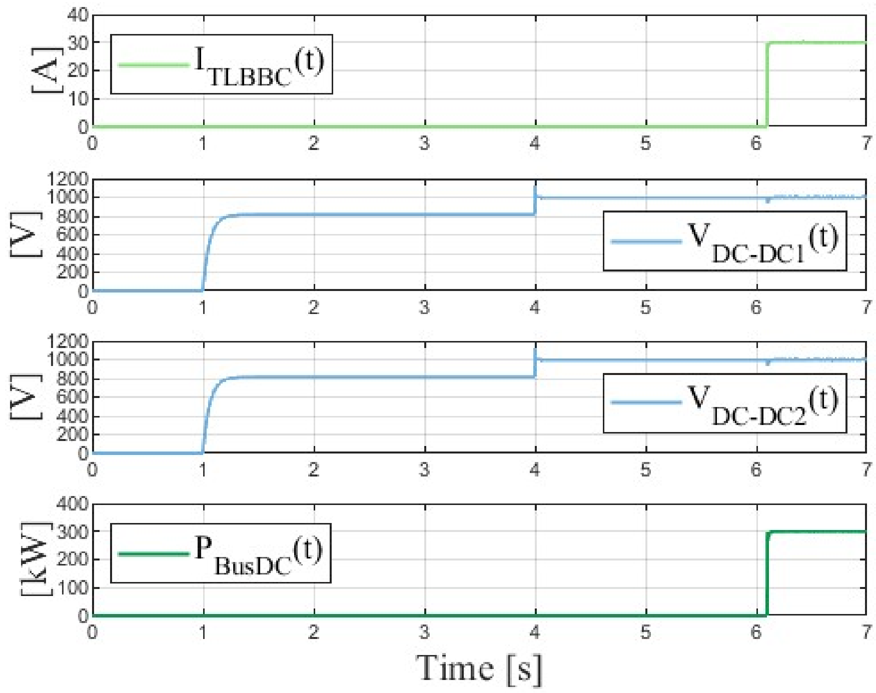


Figure 5.13: Electrical performance of power DC-DC converter under nominal operation.

The DC bus regulation for the small battery bank is shown below. To do this, Figure 5.13 analyzes the output power of each DC-DC converter power module, the output voltage of DC-DC1, the output voltage of DC-DC2, and finally the operation at rated power.

To perform these simulations, a master-slave control was implemented, in which one power module provided the references to the other nine modules in order to achieve coordination [94]. Once the power modules were precharged, the 1000[V] reference was delivered, followed by the necessary load step for the DC-DC converters to operate at rated power. In this way, it can be seen that no converter loses the 1000[V] voltage reference and also that each module operates at 30[kW], thus obtaining 300[kW] with 2 battery packs, which, with proper coordination, could reach 600[kW] with the 4 battery packs.

5.3.2 Large Battery Bank

In the case of the large battery bank, each battery pack will have 41 strings in parallel. If the AC bus is designed to supply 1000 [kW] of service power to the propulsion system, each pack will supply 250 [kW] of power. This means that each DC-DC converter must be capable of operating at that power level and, therefore, the internal power modules must operate at a minimum of 50 [kW].

a) TLBBC

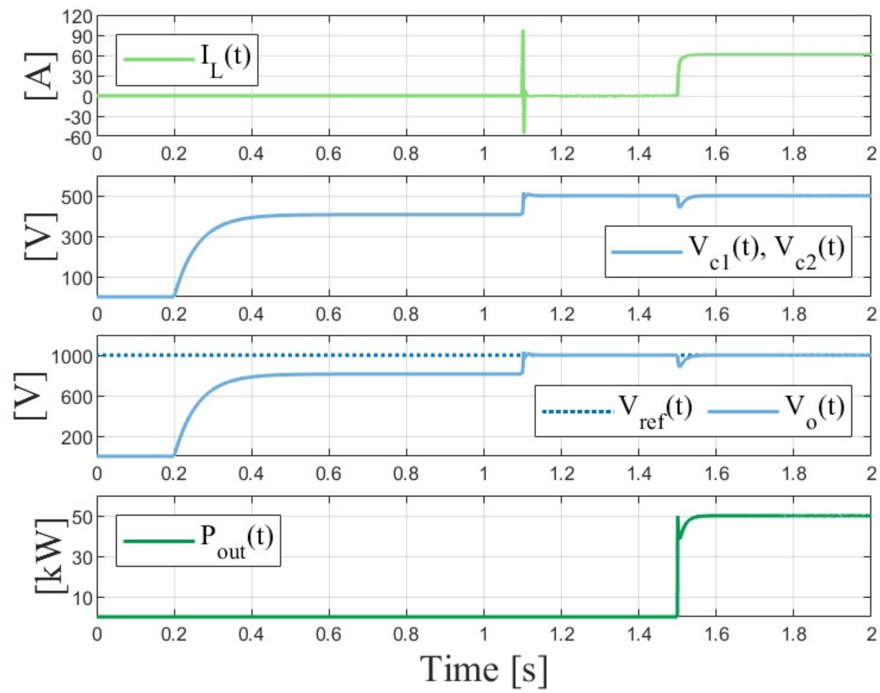


Figure 5.14: Electrical performance of power module under nominal operation.

Figure 5.14, as in the previous case, shows the input current to the power module, the voltage across each capacitor (whether balanced or not), the output voltage of the module, and finally, the operating power achieved by the module. Although a similar analysis can be performed, it can be seen that the power module is capable of regulating the output voltage at 1,000[V], reaching 50 [kW], requiring a current of approximately 60 [A] from the battery pack.

b) Bus DC

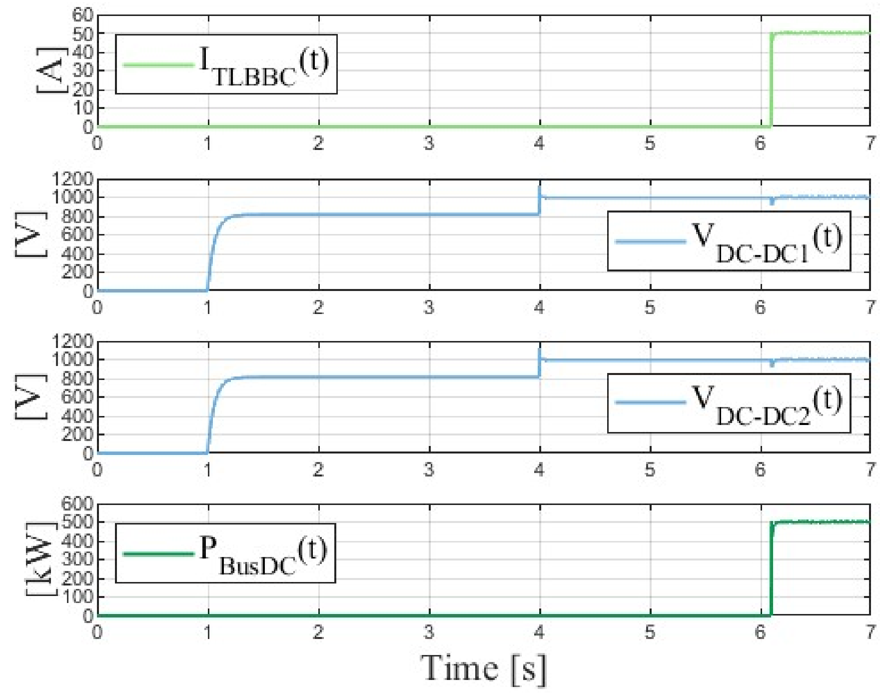


Figure 5.15: Electrical performance of power DC-DC converter under nominal operation.

Figure 5.15, as previously done, analyzes the output power of each power module of the DC-DC converter, the output voltage of DC-DC1, the output voltage of DC-DC2, and finally, the operation at rated power.

It can be seen that no converter loses the 1000 [V] voltage reference and also that each module operates at 50 [kW], thus obtaining 500 [kW] with 2 battery packs, which, with proper coordination, could reach 1000 [kW] with the 4 battery packs, maintaining the same control strategy as for the small battery bank [94] [95].

5.4 Chapter Summary

The simulations confirm that both the small and large banks are suitable for the load profiles studied, since the load state does not exceed 50% discharge in the maneuvers analyzed.

By maintaining a high residual charge level (around 75% after two maneuvers on the large bank), the system offers flexibility to power hotel loads and allows for charging in port without compromising the ship's availability.

The use of three-level converters (TLBBC) allows for stable regulation of the $1,000[V]$ bus, even in the event of sudden load steps, ensuring safe operation and reducing electrical stress on internal components, potentially extending their useful life.

It has been demonstrated that, through proper coordination of battery packs, it is possible to achieve operating powers of up to $1,000[kW]$ (in the case of the large bank), meeting the demands of hybrid propulsion systems.

Chapter 6

Conclusion

The development of a hybrid propulsion system through retrofitting involves multiple variables, with the technical and economic analysis of the battery bank as the central focus. The characterization of the load profile is the starting point, allowing the inefficiencies of the original system to be quantified and a battery bank to be sized according to the ship's operational requirements.
In addition to the technical analysis of the battery bank, it is crucial to integrate regulatory frameworks and physical limits (weight and volume), which are determining factors in the architecture of a tugboat. At the same time, profitability analysis—considering the optimal level of hybridization according to the asset's useful life—is vital to ensure return on investment (ROI) and attract capital interested in maritime decarbonization.

Once the technical and economic feasibility has been validated, mathematical modeling and subsequent analysis of the electrical performance of the battery bank and power converters are carried out. This comprehensive approach not only ensures that the system meets the demands of the propulsion system, but also establishes a robust methodology with multiple criteria, seeking to ensure that hybridization processes are successful and aligned with global emissions policies. In this way, this work seeks to lay the foundations for the creation of a methodology that allows the modeling of battery banks for hybrid tugboats, considering multiple variables in the design process.

Chapter 7

Future Work

Based on the methodology presented in this paper, it is proposed as future work:

- Model the other internal elements of the hybrid powertrain to ultimately obtain a simulated model of the propulsion system that allows different load profiles to be tested.
- Based on the information provided in this paper, related to technical and economic aspects, the aim is to work on an optimization function that incorporates information from all these areas and allows us to identify the best battery alternatives, optimizing variables such as project cost, number of cells, available space, etc.
- The complexity of the battery bank model can be increased by adding dynamics that provide a better understanding of performance when a more detailed load profile is obtained, not only in terms of hours but also in terms of minutes or seconds.
- Implement PTO mode and analyze the feasibility of replacing diesel engines with ones with lower nominal power, allowing the use of batteries for towing tasks.
- Consideration of port charger implementation, which could also impact the economic analysis of hybridization.
- Through sensorization, monitor the health status of the different elements within the hybrid power system, especially the battery bank, in order to generate predictive maintenance routines and also have an operating history.

Appendix A

Appendix

A.1 CO_2 tax emissions

In countries such as Finland, there is a so-called carbon tax, which seeks to mitigate the effects of climate change by discouraging the use of fossil fuels [96].

Based on this, a carbon tax of $50/ton$_{CO_2}$ will be assumed, and the impact this has on economic sensitivity variables will be determined, since this tax mainly affects the OPEX related to each battery.

A.1.1 Net Present Value (NPV)

Table A.1: Calculated NPV, with carbon taxes, without subsides (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	$241, 940	$1, 545, 943	$2, 433, 426
NPV_{LFP}	$1, 929, 657	$3, 354, 487	$4, 324, 203
NPV_{LTO}	$1, 007, 204	$2, 587, 384	$3, 662, 827

Table A.2: Calculated NPV, with carbon taxes and without subsides (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-$3, 340, 207	−$2, 156, 302	-$1, 350, 557
NPV_{LFP}	-$71, 650	$1, 384, 116	$2, 374, 886
NPV_{LTO}	-$1, 792, 895	$176, 901	$1, 517, 511

A.1.2 Internal Rate of Return (IRR)

Table A.3: Calculated IRR, with carbon taxes and without subsides (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	9.18%	13.29%	14.69%
IRR_{LFP}	19.91%	22.67%	23.43%
IRR_{LTO}	12.52%	16.17%	17.34%

Table A.4: Calculated IRR, with carbon tax and without subsides (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	−3.08%	3.01%	5.52%
IRR_{LFP}	7.70%	12.03%	13.54%
IRR_{LTO}	1.63%	6.91%	8.95%

A.1.3 Return on Investment (ROI)/ Payback Period

Table A.5: Calculated period for positive ROI, with carbon taxes and without subsides (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 6 years and 5 months
$ROI_{LFP} > 0$	≈ 4 years and 3 months
$ROI_{LTO} > 0$	≈ 5 years and 7 months

Table A.6: Calculated period for positive ROI, witho carbon taxes and without subsides (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 11 years and 7 months
$ROI_{LFP} > 0$	≈ 6 years and 7 months
$ROI_{LTO} > 0$	≈ 8 years and 11 months

A.2 Government grants

With the aim of contributing to carbon neutrality in the global maritime transport sector, there are government programs that promote the use of clean technologies that accelerate decarbonization, such as the Ministry of Oceans and Fisheries of the Republic of Korea (MOF), which announced its ”First National Plan for the Development and Popularization of Eco-friendly Ships (2021-2030),“ known as the ”2030-K Eco-friendly Ship Promotion Strategy” [97].

This strategy explores advanced zero-emission technologies that can reduce greenhouse gas (GHG) emissions in areas such as ship design, future fuels, renewable energy, and equipment, among others. In this way, the Ministry of Finance is facilitating the conversion of 15% of Korean-flagged ships (528 out of a total of 3,542), for both the public and private sectors.

This will be considered a 10% government subsidy, which will directly affect the CAPEX of each design and the economic sensitivity variables.

A.2.1 Net Present Value (NPV)

Table A.7: Calculated NPV, without taxes and with government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-$150,908	$925,272	$1,657,703
NPV_{LFP}	$1,438,809	$2,635,817	$3,405,480
NPV_{LTO}	$670,356	$2,022,713	$2,943,104

Table A.8: Calculated NPV, without taxes and with government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-$3,545,391	−$2,261,373	-$1,992,502
NPV_{LFP}	-$497,334	$698,545	$1,512,441
NPV_{LTO}	-$1,902,694	-$201,222	$956,771

A.2.2 Internal Rate of Return (IRR)

Table A.9: Calculated IRR, without taxes and with Government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	7.17%	11.57%	13.13%
IRR_{LFP}	18.04%	21.01%	21.86%
IRR_{LTO}	11.39%	15.16%	16.44%

Table A.10: Calculated IRR, without taxes and with Government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	−5.47%	1.06%	3.82%
IRR_{LFP}	5.66%	10.30%	11.98%
IRR_{LTO}	0.63%	6.08%	8.21%

A.2.3 Return on Investment (ROI)/ Payback Period

Table A.11: Calculated period for positive ROI, without carbon taxes and with government grants (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	$\approx$ 7 years
$ROI_{LFP} > 0$	$\approx$ 4 years and 6 months
$ROI_{LTO} > 0$	$\approx$ 5 years and 10 months

Table A.12: Calculated period for positive ROI, without taxes and with Government grants (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	$\approx$ 14 years and 9 months
$ROI_{LFP} > 0$	$\approx$ 8 years
$ROI_{LTO} > 0$	$\approx$ 10 years and 4 months

A.3 Taxes and Grants

The following section presents the effect of applying taxes and government subsidies. The values are as follows:

- Carbon Tax: $\$50/t_{CO_2}$
- A government subsidy of 10% on the cost of materials purchased for the development of the hybrid propulsion system.

A.3.1 Net Present Value (NPV)

Table A.13: Calculated NPV, with taxes and government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	$675,690	$1,979,693	$2,867,167
NPV_{LFP}	$2,265,407	$3,690,237	$4,659,953
NPV_{LTO}	$1,496,954	$3,077,134	$4,152,577

Table A.14: Calculated NPV, with taxes and government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
NPV_{NMC}	-$2,602,457	−$1,418,552	-$612,807
NPV_{LFP}	-$445,600	$1,901,366	$2,892,126
NPV_{LTO}	-$929,145	$1,040,651	$2,381,261

A.3.2 Internal Rate of Return (IRR)

Table A.15: Calculated IRR, with taxes and government grants (small bank).

Battery Bank 15% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	17.57%	15.34%	16.58%
IRR_{LFP}	23.25%	25.67%	26.28%
IRR_{LTO}	15.32%	18.61%	19.61%

Table A.16: Calculated IRR, with taxes and government grants (large bank).

Battery Bank 25% P_{nom}			
Lifespan	10 years	15 years	20 years
IRR_{NMC}	−1.34%	4.45%	6.78%
IRR_{LFP}	10.00%	13.99%	15.34%
IRR_{LTO}	3.60%	8.56%	10.42%

A.3.3 Return on Investment (ROI)/ Payback Period

Table A.17: Calculated period for positive ROI, with taxes and government grants (small bank).

Battery Bank 15% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 5 years and 10 months
$ROI_{LFP} > 0$	≈ 3 years and 10 months
$ROI_{LTO} > 0$	≈ 5 years

Table A.18: Calculated period for positive ROI, with taxes and government grants (large bank).

Battery Bank 25% P_{nom}	
	Period
$ROI_{NMC} > 0$	≈ 11 years and 6 months
$ROI_{LFP} > 0$	≈ 6 years and 7 months
$ROI_{LTO} > 0$	≈ 8 years and 11 months

A.4 Three Level Buck Boost Converter - TLBBC

The three-level buck-boost DC-DC converter (TLBBC) is a good alternative, as it has lower stress on the semiconductors compared to two-level topologies, and uses fewer switches compared to other three-level topologies.

Furthermore, one of its advantages is that this converter is capable of operating in a manner similar to an interleaved mode, reducing the requirements for passive filters, which represents an interesting possibility for bipolar DC power grids [98].

Thus, the modeling of this converter and how to obtain the transfer functions for the subsequent design of controllers are shown below.

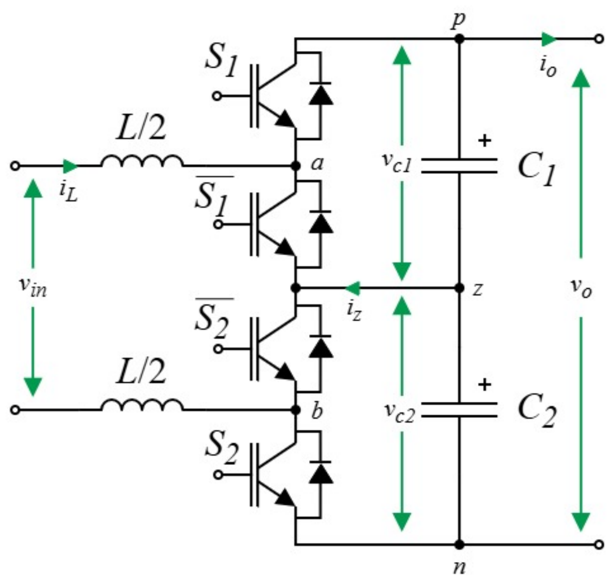


Figure A.1: Three Level Buck Boost Converter (TLBBC).

A.4.1 Modelling

For modeling purposes, we will first assume that the capacitors are identical.

$$C_1 = C_2 = C \tag{A.1}$$

Based on this, the states of the converter are analyzed based on the switching on and off of the power switches, with the aim of finding a pattern between the state variables and the states of the power switches. This is shown in the figures A.2, A.3, A.4 and A.5.

State E0

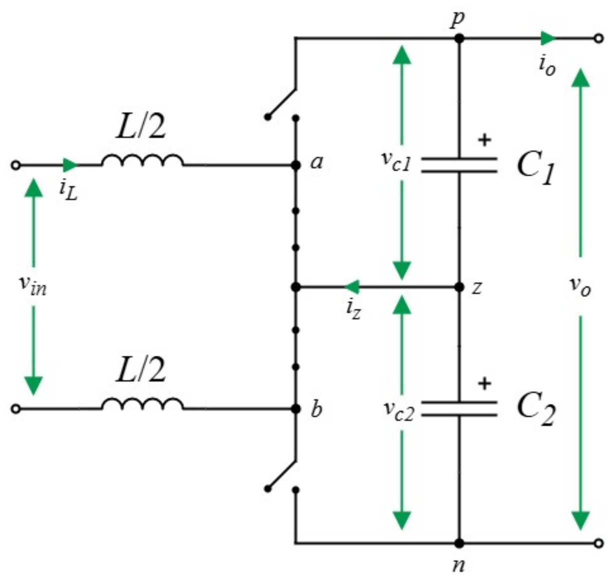


Figure A.2: State 0 (S1:Off, S2:Off).

$$
\begin{aligned}
v_{in}(t) - v_{L/2}(t) - v_{L/2}(t) &= 0 \\
v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\
\Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t)
\end{aligned}
\tag{A.2}
$$

$$
\begin{aligned}
i_{c1}(t) &= -i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= -i_o(t)
\end{aligned}
\tag{A.3}
$$

$$
\begin{aligned}
i_{c2}(t) &= -i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= -i_o(t)
\end{aligned}
\tag{A.4}
$$

State E1

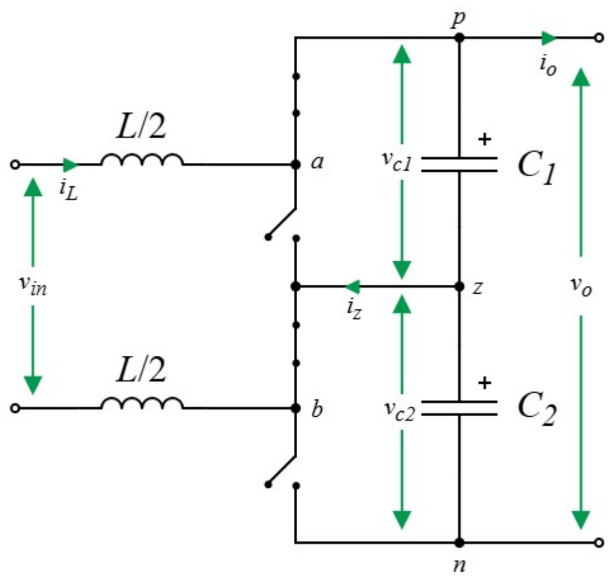


Figure A.3: State 1 (S1:On, S2:Off).

$$\begin{aligned} v_{in}(t) - v_{L/2}(t) - v_{c1}(t) - v_{L/2}(t) &= 0 \\ v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - v_{c1}(t) - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\ \Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) - v_{c1}(t) \end{aligned} \tag{A.5}$$

$$\begin{aligned} i_{c1}(t) &= i_L(t) - i_o(t) \\ \Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= i_L(t) - i_o(t) \end{aligned} \tag{A.6}$$

$$\begin{aligned} i_{c1}(t) &= i_L(t) + i_{c2}(t) \\ i_L(t) - i_o(t) &= i_L(t) + i_{c2}(t) \\ -i_o(t) &= i_{c2}(t) \\ \Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= -i_o(t) \end{aligned} \tag{A.7}$$

State E2

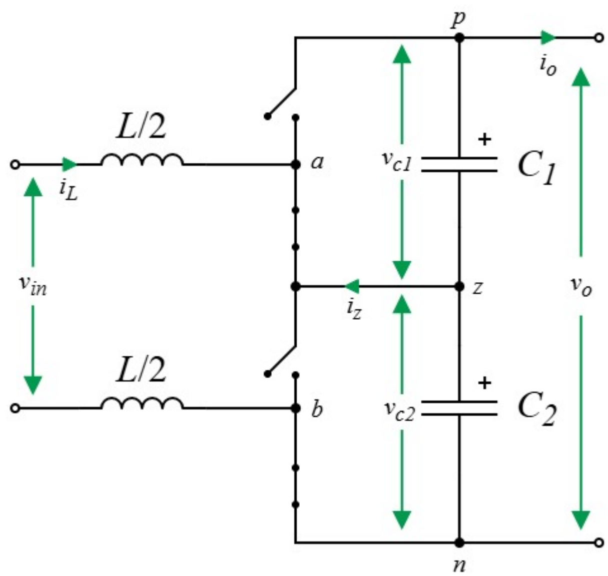


Figure A.4: State 2 (S1:Off, S2:On).

$$\begin{aligned} v_{in}(t) - v_{L/2}(t) - v_{c2}(t) - v_{L/2}(t) &= 0 \\ v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - v_{c2}(t) - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\ \Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) - v_{c2}(t) \end{aligned} \tag{A.8}$$

$$\begin{aligned} i_{c1}(t) &= -i_o(t) \\ \Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= -i_o(t) \end{aligned} \tag{A.9}$$

$$\begin{aligned} i_{c2}(t) &= i_L(t) - i_o(t) \\ \Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= i_L(t) - i_o(t) \end{aligned} \tag{A.10}$$

State E3

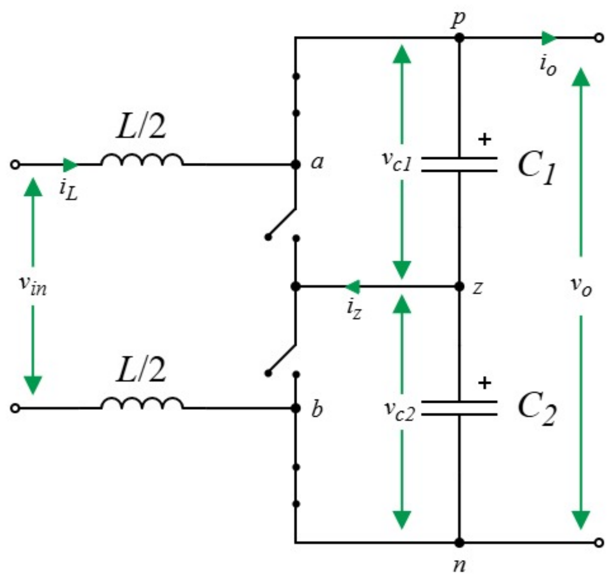


Figure A.5: State 3 (S1:On, S2:On).

$$
\begin{aligned}
v_{in}(t) - v_{L/2}(t) - v_{c1}(t) - v_{c2}(t) - v_{L/2}(t) &= 0 \\
v_{in}(t) - (L/2) \cdot \frac{di_L(t)}{dt} - v_{c1}(t) - v_{c2}(t) - (L/2) \cdot \frac{di_L(t)}{dt} &= 0 \\
\Rightarrow L \cdot \frac{di_L(t)}{dt} &= v_{in}(t) - v_{c1}(t) - v_{c2}(t)
\end{aligned} \tag{A.11}
$$

$$
\begin{aligned}
i_{c1}(t) &= i_L(t) - i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c1}(t)}{dt} &= i_L(t) - i_o(t)
\end{aligned} \tag{A.12}
$$

$$
\begin{aligned}
i_{c2}(t) &= i_L(t) - i_o(t) \\
\Rightarrow C \cdot \frac{dv_{c2}(t)}{dt} &= i_L(t) - i_o(t)
\end{aligned} \tag{A.13}
$$

General differential Model

Based on the states identified above, the different combinations are analyzed and the pattern of switches with the variables of interest is identified. This is shown in equation A.14.

$$\frac{di_L(t)}{dt} = \frac{v_{in}(t) - v_{c1}(t) \cdot \mathbf{S_1}[t] - v_{c2}(t) \cdot \mathbf{S_2}[t]}{L} \tag{A.14}$$
$$\frac{dv_{c1}(t)}{dt} = \frac{i_L(t) \cdot \mathbf{S_1}[t] - i_o(t)}{C} \tag{A.15}$$
$$\frac{dv_{c2}(t)}{dt} = \frac{i_L(t) \cdot \mathbf{S_2}[t] - i_o(t)}{C} \tag{A.16}$$
$$\tag{A.17}$$

where

$$\mathbf{S_1}, \mathbf{S_2} \in \{0, 1\} \tag{A.18}$$

The equations described in are nonlinear, given that the states of the switches consider discrete states, so it is necessary to perform a conversion related to the duty cycle of the switches.

An average model consider:

$$\begin{aligned} d_i &= \frac{t_i^{ON}}{t_i^{ON} + t_i^{OFF}} \\ S_1[t] &\Rightarrow d_1(t) \\ S_2[t] &\Rightarrow d_2(t) \\ d_1, d_2 &\in (0, 1) \end{aligned} \tag{A.19}$$

If the change of variables is performed:

$$\begin{aligned} u_i(t) &= \frac{d_1(t) + d_2(t)}{2} \\ u_v(t) &= \frac{d_1(t) - d_2(t)}{2} \\ \Rightarrow u_i(t) + u_v(t) &= d_1(t) \\ \Rightarrow u_i(t) - u_v(t) &= d_2(t) \end{aligned} \tag{A.20}$$

In this way:

$$\frac{di_L(t)}{dt} = \frac{v_{in}(t) - \mathbf{d_1}(t) \cdot v_{c1}(t) - \mathbf{d_2}(t) \cdot v_{c2}(t)}{L} \tag{A.21}$$
$$\frac{dv_{c1}(t)}{dt} = \frac{\mathbf{d_1}(t) \cdot i_L(t) - i_o(t)}{C} \tag{A.22}$$
$$\frac{dv_{c2}(t)}{dt} = \frac{\mathbf{d_2}(t) \cdot i_L(t) - i_o(t)}{C} \tag{A.23}$$

This allows the differential equations to contain only continuous variables, thus obtaining an average model of the system.

A.5 Linearization

The state variables to be linearized are defined.

$$
\begin{aligned}
x(t) &= [i_L(t) \quad v_{c1}(t) \quad v_{c2}(t)]^T \\
u(t) &= [d_1(t) \quad d_2(t)]^T \\
y(t) &= [i_L(t) \quad (v_{c1}(t) - v_{c2}(t)) \quad v_o(t)]^T
\end{aligned} \tag{A.24}
$$

where $x(t)$ are the state variables, $u(t)$ are the inputs of the system and $y(t)$ are the output of the system.

To perform linearization, a signal corresponding to its equilibrium point plus a small variation will be taken into account.

$$
\begin{aligned}
x(t) &= \Delta x(t) + x_Q \\
u(t) &= \Delta u(t) + u_Q
\end{aligned} \tag{A.25}
$$

The purpose is to represent the state variables, inputs, and outputs using the following matrix equation:

$$\Delta \dot{x}(t) = A\Delta x(t) + B\Delta u(t) + E\Delta u(t) \tag{A.26}$$

$$\Delta y(t) = C\Delta x(t) + D\Delta u(t) \tag{A.27}$$

In this way, and applying a low-order Taylor series to linearize, the linearized differential equations are found.

$$\frac{d\Delta i_L(t)}{dt} = \frac{\Delta v_{in}(t) - \Delta v_{c1}(t)\cdot d_{1Q} - \Delta v_{c2}(t)\cdot d_{2Q} - \Delta d_1(t)\cdot v_{c1Q} - \Delta d_2(t)\cdot v_{c2Q}}{L} \tag{A.28}$$

$$\frac{d\Delta v_{c1}(t)}{dt} = \frac{\Delta i_L(t)\cdot d_{1Q} + \Delta d_1(t)\cdot i_{LQ} - \Delta i_o(t)}{C} \tag{A.29}$$

$$\frac{d\Delta v_{c2}(t)}{dt} = \frac{\Delta i_L(t)\cdot d_{2Q} + \Delta d_2(t)\cdot i_{LQ} - \Delta i_o(t)}{C} \tag{A.30}$$

In this way, the matrices of interest can be found in state variables within the linearized differential equations.

$$
\begin{aligned}
\frac{d}{dt}\begin{bmatrix}\Delta i_L(t)\\ \Delta v_{c1}(t)\\ \Delta v_{c2}(t)\end{bmatrix} &= \begin{bmatrix}0 & -d_{1Q}/L & -d_{2Q}/L\\ d_{1Q}/C & 0 & 0\\ d_{2Q}/C & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta i_L(t)\\ \Delta v_{c1}(t)\\ \Delta v_{c2}(t)\end{bmatrix} + \begin{bmatrix}-v_{c1Q}/L & -v_{c2Q}/L\\ i_{LQ}/C & 0\\ 0 & i_{LQ}/C\end{bmatrix}\begin{bmatrix}\Delta d_1(t)\\ \Delta d_2(t)\end{bmatrix} \\
&\quad + \begin{bmatrix}1/L & 0\\ 0 & -1/C\\ 0 & -1/C\end{bmatrix}\begin{bmatrix}\Delta v_{in}(t)\\ \Delta i_o(t)\end{bmatrix} \\
\begin{bmatrix}\Delta i_L(t)\\ \Delta v_{c1}(t) - \Delta v_{c2}(t)\\ \Delta v_o(t)\end{bmatrix} &= \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & -1\\ 0 & 1 & 1\end{bmatrix}\begin{bmatrix}\Delta i_L(t)\\ \Delta v_{c1}(t)\\ \Delta v_{c2}(t)\end{bmatrix} + \begin{bmatrix}0 & 0\\ 0 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}\Delta d_1(t)\\ \Delta d_2(t)\end{bmatrix}
\end{aligned} \tag{A.31}
$$

A.5.1 Transfer Functions

Given a state model defined by the matrices (A,B,C,D), the transfer function can be expressed as follows

$$H(s) = C(sI - A)^{-1}B + D$$

Whose transfer function allows the following expression as:

$$Y(s) = H(s)U(s) \tag{A.32}$$

A transfer function will be found for each element related to the inputs available. Thus, matrix C can take different values, depending on the variables desired as output, considering D = 0.

$$y(t) = \Delta i_L(t) \longrightarrow C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$y(t) = (\Delta v_{c1}(t) - \Delta v_{c2}(t)) \longrightarrow C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$y(t) = \Delta v_0(t) \longrightarrow C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

In this way, the following transfer functions can be obtained:

$$Hi_L d_2 = \frac{-s \cdot Cv_{cQ} - I_{LQ} d_Q}{s^2 \cdot LC + 2d_Q^2} \tag{A.33}$$

$$Hi_L d_2 = \frac{-s \cdot Cv_{cQ} - I_{LQ} d_Q}{s^2 \cdot LC + 2d_Q^2} \tag{A.34}$$

$$H\tilde{v}_c d_1 = \frac{i_{LQ}}{s \cdot C} \tag{A.35}$$

$$H\tilde{v}_c d_2 = -\frac{i_{LQ}}{s \cdot C} \tag{A.36}$$

$$Hv_o d_1 = \frac{-(s \cdot I_{LQ} L + 2d_Q v_{cQ})}{s^2 \cdot LC + 2d_Q^2} \tag{A.37}$$

$$Hv_o d_2 = \frac{-(s \cdot I_{LQ} L + 2d_Q v_{cQ})}{s^2 \cdot LC + 2d_Q^2} \tag{A.38}$$

$$\tag{A.39}$$

With these transfer functions, current and voltage controllers can be designed to control the input current to the converter and the voltage balance at the output, as is shown in figure A.6.

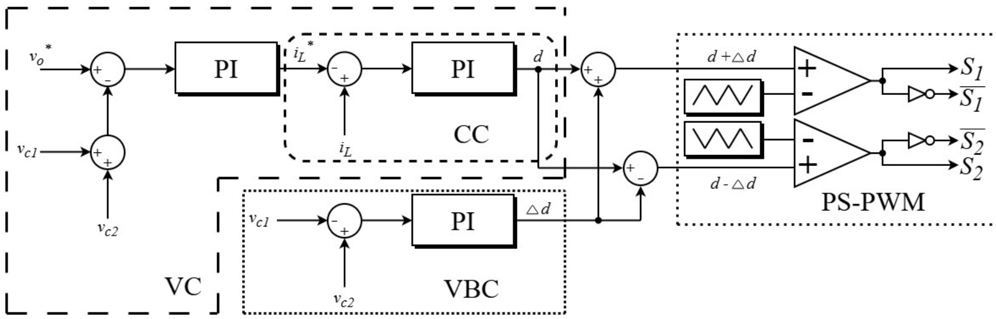


Figure A.6: Proposed control strategy.

Bibliography

[1] "45 Latest Greenhouse Gas & Climate Change Statistics", TheRoundup.org [Online]: theroundup.org

[2] "How to reduce shipping emissions? Best way to cut emissions & save money in 2024", Container xChange [Online]: container-xchange.com

[3] M. B. Othman, N. P. Reddy, P. Ghimire, M. K. Zadeh, A. Anvari-Moghaddam and J. M. Guerrero, "A Hybrid Power System Laboratory: Testing Electric and Hybrid Propulsion," in IEEE Electrification Magazine, vol. 7, no. 4, pp. 89-97, Dec. 2019.

[4] DNV, "Energy Transition Outlook 2025 - Maritime Forecast to 2025, A deep dive into shipping's decarbonization journey", 2025

[5] Völker, Thorsten. "Hybrid Propulsion Concepts on Ships." 2013.

[6] N. Bennabi, J. F. Charpentier, H. Menana, J. Y. Billard and P. Genet, "Hybrid propulsion systems for small ships: Context and challenges," *XXII International Conference on Electrical Machines (ICEM)*, Lausanne, Switzerland, 2016, pp. 2948-2954, doi: 10.1109/ICELMACH.2016.7732943.

[7] C. A. Reusser, H. A. Young, J. R. Perez Osses, M. A. Perez and O. J. Simmonds, "Power Electronics and Drives: Applications to Modern Ship Propulsion Systems," in *IEEE Industrial Electronics Magazine*, vol. 14, no. 4, pp. 106-122, Dec. 2020, doi: 10.1109/MIE.2020.3002493.

[8] B. Kwasieckyj, "Hybrid Propulsion Systems: Efficiency analysis and design methodology of hybrid propulsion systems", Master Thesis conducted at MAN Diesel & Turbo SE, Augsburg, 2013.

[9] L. Solis-Zamora, J. Perez and M. A. Perez, "Design and Sizing of an Energy Storage System for a Hybrid Tugboat," 2025 IEEE 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG), Antalya, Turkiye, 2025, pp. 1-6.

[10] Lloyd Register, DNV, "ASSESSMENT OF IMO MANDATED ENERGY EFFICIENCY MEASURES FOR INTERNATIONAL SHIPPING" in Project Final Report, MEPC 63/INF.2, Annex, Page 1, 2011.

[11] SAAM, "Harbour Towage". [Online]: saamtowage.com

[12] Wärtsilä, "Encyclopedia of Marine And Energy Technology". [Online]: https://www.wartsila.com/encyclopedia/term/steam-turbine

[13] Maritime Education,"Marine Steam Turbines". https://maritimeducation.com/marine-steam-turbines/

[14] Science Direct,"Gas Turbine Engines". https://www.sciencedirect.com/topics/physics-and-astronomy/gas-turbine-engines

[15] S. Grigoryev,"Gas turbines in marine propulsion",Bachelor's thesis 2025, LUT University.

[16] "The Ultimate Guide to Tug Boats: Types, Functions, and Applications", Marine Insight [Online]: marineinsight.com

[17] D. Griffiths, "Marine Medium Speed Diesel Engines", Institute of Marine Engineers, London 1999.

[18] C. Wik, "Reducing Medium-Speed Engine Emissions", ournal of Marine Engineering and Technology, A17, 8, 2010.

[19] P. H. Bauer and E. Pérez-Bernabeu, "The effect of the operating point choice on fuel economy in series hybrids," 2013 World Electric Vehicle Symposium and Exhibition (EVS27), Barcelona, Spain, 2013, pp. 1-4, doi: 10.1109/EVS.2013.6914858.

[20] DNV GL AS Maritime, "Electrical Energy Storage for Ships", EMSA European MAritime Safety Agency, 2019-0217, Rev. 04, 2020.

[21] MAN Energy Solutions, "Batteries on Board ocean-going Vessels", 2019

[22] Chin, C.S.; Tan, Y.-J.; Kumar, M.V. "Study of Hybrid Propulsion Systems for Lower Emissions and Fuel Saving on Merchant Ship during Voyage". J. Mar. Sci. Eng. 2022, 10, 393.

[23] Z. Fan, C. Liang, Y. Dong, G. Chang, W. Pan and S. Shao, "Control of the Diesel-Electric Hybrid Propulsion in a Practical Cruise Vessel," 2021 IEEE International Conference on Power, Intelligent Computing and Systems (ICPICS), Shenyang, China, 2021, pp. 311-316, doi: 10.1109/ICPICS52425.2021.9524194.

[24] Kolodziejski, M.; Michalska-Pozoga, I. "Battery Energy Storage Systems in Ships' Hybrid/Electric Propulsion Systems". Energies 2023, 16, 1122.

[25] R. Barrera-Cardenas, O. Mo and G. Guidi, "Optimal Sizing of Battery Energy Storage Systems for Hybrid Marine Power Systems," 2019 IEEE Electric Ship Technologies Symposium (ESTS), Washington, DC, USA, 2019, pp. 293-302.

[26] Lucà Trombetta, G.; Leonardi, S.G.; Aloisio, D.; Andaloro, L.; Sergi, F. "Lithium-Ion Batteries on Board: A Review on Their Integration for Enabling the Energy Transition in Shipping Industry". Energies 2024, 17, 1019. https://doi.org/10.3390/en17051019

[27] DNV Maritime Forecast to 2050—Energy Transition Outlook 2022.

[28] T. Hofman, M. Naaborg and E. Sciberras, "System-Level Design Optimization of a Hybrid Tug," 2017 IEEE Vehicle Power and Propulsion Conference (VPPC), Belfort, France, 2017, pp. 1-6, doi: 10.1109/VPPC.2017.8331009.

[29] B. A. Kumar, R. Selvaraj, K. Desingu, T. R. Chelliah and R. S. Upadhyayula, "A Coordinated Control Strategy for a Diesel-Electric Tugboat System for Improved Fuel Economy," in IEEE Transactions on Industry Applications, vol. 56, no. 5, pp. 5439-5451, Sept.-Oct. 2020, doi: 10.1109/TIA.2020.3010779.

[30] L. Xu et al., "A Review of DC Shipboard Microgrids—Part I: Power Architectures, Energy Storage, and Power Converters," in IEEE Transactions on Power Electronics, vol. 37, no. 5, pp. 5155-5172, May 2022.

[31] INEAL, "Descubre las top 5 Funciones Principales del Remolcador de Puerto". [Online]: ineal.com.mx

[32] ESC Group, "Comprender el Tirón de los Bolardos: Un Factor Clave en las Maniobras de los Barcos". [Online]: www.escglobalgroup.com

[33] A. Bruce, A. Bermejo, "Remolque en la Mar", Trabajo de Fin de Grado en Náutica y Transporte Marítimo, Universidad de La Laguna, sept. 2020

[34] Murimar, "Remolcadores, elemento clave en la logística marítima", [Online]: murimar.com

[35] Ingeniero Marino, "Remolcador",[Online]: https://ingenieromarino.com

[36] Algeciras al minuto.es,"Los remolcadores del puerto de Algeciras denuncian la excesiva carga de trabajo",[Online]: https://www.algecirasalminuto.es

[37] Robert Allan,"Rastar Series",[Online]: https://ral.ca

[38] Vessel Finder,"SAR BALDER",[Online]: https://www.vesselfinder.com

[39] Prácticos de Puerto,"Remolque Portuario",[Online]: https://www.practicosdepuerto.es

[40] Salvamento Marítimo, "Remolcadores de Salvamento", [Online]: https://www.salvamentomaritimo.es

[41] Cotecmar, "Remolcado Offshore", [Online]: https://www.cotecmar.com

[42] Portal Portuario, "SAAM recibe dos nuevos remolcadores para reforzar su flota en Chile, Mayo 2016", [Online]: https://portalportuario.cl

[43] M. Jiménez Fernández, Dr. J. González Almeida, "Remolcadores: Diseño y Desempeño," Grado en Náutica y Transporte Marítimo, Escuela Politécnica Superior de Ingeniería Sección Náutica, Máquinas y Radioelectrónica Naval, Universidad de La Laguna, Marzo de 2025.

[44] J. Farina Orihuela, A. Poleo Mora, "Maniobras con Remolcadores " Grado en Náutica y Transporte Marítimo, Escuela Politécnica Superior de Ingeniería Sección Náutica, Máquinas y Radioelectrónica Naval, Universidad de La Laguna, Junio de 2015.

[45] wle online, "Measuring Length Overall", [Online]: https://wleonline.com

[46] HZ-Containers, "Glosario: Beam-Ancho del Barco", [Online]: https://hz-containers.com

[47] Discover Boating, "What Is a Boat Beam and Why Is It Important?", [Online]: https://www.discoverboating.com

[48] Marine Rubber Fender, "Difference between Gross Tonnage, Net Tonnage, and Deadweight Tonnage". [Online]: https://www.chinarubberfender.com

[49] Britannica, "Cars and Other Vehicles: Tonnage", [Online]: https://www.britannica.com

[50] G. Çakıroğlu, B. Sener, A. Balin, “Applying a fuzzy-ahp for the selection of a suitable tugboat based on propulsion system type”. Brodogradnja. 69. 1-13. 10.21278/brod69401, 2018.

[51] Shiphandling for Professionals, “Types of Tugs”, [Online]: https://www.shiphandlingpro.com

[52] Yacht Buyer, “Horsepower vs Brake Horsepower (hp vs bhp): What’s the Difference?”, [Online]: https://www.yachtbuyer.com

[53] S. -Y. Kim, S. Choe, S. Ko and S. -K. Sul, ”A Naval Integrated Power System with a Battery Energy Storage System: Fuel efficiency, reliability, and quality of power,” in IEEE Electrification Magazine, vol. 3, no. 2, pp. 22-33, June 2015.

[54] F. Derman, L. Kallryd, “Development of a flexible control for a simulation model of a hybrid tug propulsion system”, Master’s thesis in Naval Architecture and Ocean Engineering, Department of Shipping and Marine Technology, Chalmers University of Technology, Gothenburg, Sweden 2016.

[55] G. Wortmann, Q. Zhang, “ELectric Innovative Commuter Aircraft (ELICA) - Hybrid Electric Propulsion Architecture Report”, Rolls-Royce Electrical Deutschland, March 2020.

[56] Volvo Penta, ‘Serial or parallel hybrid marine propulsion?”, [Online]: https://www.volvopenta.com

[57] S. Qazi, P. Venugopal and G. Rietveld, “Comprehensive Study of a Hybrid Electric Ship for Optimum Onboard Fuel Consumption,” 2022 IEEE 16th International Conference on Compatibility, Power Electronics, and Power Engineering (CPE-POWERENG), Birmingham, United Kingdom, 2022, pp. 1-6.

[58] Nautech, “The renaissance of the direct current”, [Online]: https://www.nautechnews.it

[59] R.D. Geertsma, R.R. Negenborn, K. Visser, J.J. Hopman, “Design and control of hybrid power and propulsion systems for smart ships: A review of developments”, Applied Energy, Volume 194, 2017, Pages 30-54, ISSN 0306-2619,

[60] ABB, “The step forward Onboard DC Grid”, Norway, 2014.

[61] Wärtsilä, “DC or not DC, that is the question”. [Online]: https://www.wartsila.com

[62] C. Zhang and B. -z. JIA, “The Research of Power Allocation in Diesel-electric Hybrid Propulsion System,” 2019 Chinese Automation Congress (CAC), Hangzhou, China, 2019, pp. 3664-3668.

[63] L. W. Y. Chua, T. Tjahjowidodo, G. G. L. Seet and R. Chan, “Implementation of Optimization-Based Power Management for All-Electric Hybrid Vessels,” in IEEE Access, vol. 6, pp. 74339-74354, 2018.

[64] V. Jayaram, M. Yusuf Khan, J. Wayne Miller, W. A Welch, K. Johnson, D. R Cocker, “Evaluating Emission Benefits of a Hybrid Tug Boat”, Final Report, College of Engineering-Center for Environmental Research and Technology, University of California, October 2010.

[65] DAMEN,“DAMEN ASD TUG 2810”, datasheet, 2014.

[66] ”IEEE Guide for the Selection and Sizing of Batteries for Uninterruptible Power Systems,” in IEEE Std 1184-1994 , vol., no., pp.1-36, 6 June 1995.

[67] IEngineering, “Battery Energy Storage System Design”.[Online]: https://ieng.tech

[68] UFine Blog, “LFP vs. NMC Battery: What is the Difference?”. [Online]: https://www.ufinebattery.com

[69] Maritime Battery Forum, “Which battery for your ship? A simplified guide to help you decide”.

[70] Battery University, “BU-808c: Coulombic and Energy Efficiency with the Battery”. [Online]: https://www.batteryuniversity.com

[71] Cotecmar, “Remolcador Offshore - Remolcador de apoyo logístico multipropósito”.

[72] Corvus Energy,“Technical Specifications, Corvus Orca ESS”, February 2025.

[73] Valence, “Hybrid Tugboats Case Study”. [Online]: https://www.lithionbattery.com

[74] Valence, “Hybrid Tugboats Case Study”. [Online]: https://x-engineer.org

[75] B. Lee,“The Hydrogen Zero-Emission Tugboat Project”, in Gas Research and Development Program, Energy Reserach and Development (500), California Energy Commision, May 2024.

[76] Anker Solix, “LFP vs NMC Batteries: Safety, Lifespan, and Value in 2025”. [Online]: https://www.ankersolix.com

[77] EV Lithium, “NMC vs LFP vs LTO Batteries: Complete Comparison Guide”. [Online]: https://www.evlithium.com

[78] Professional Mariner,“Foss’s second hybrid tugboat employs new, more-powerful lithium polymer batteries”.[Online]: https://professionalmariner.com

[79] M. Akbarzadeh, J. D. Smet and J. Stuyts, “Cost Assessment of Battery Hybrid Energy Storage System for Full-Electric Marine Applications,” 2022 22nd International Scientific Conference on Electric Power Engineering (EPE), Kouty nad Desnou, Czech Republic, 2022, pp. 1-6.

[80] Test Equipment Depot, “Preen ADG-PLUS-1000-100” [Online]: https://www.testequipmentdepot.com

[81] Ritar Power, “The Price of 50 kWh Lithium Ion Batteries: A Comprehensive Analysis” [Online]: https://www.ritarpower.com

[82] Sunreef Yachts,“High-Performance Electric Boat Batteries for Smooth, Silent Sailing” [Online]: https://sunreef-yachts.com

[83] Bonnen, “LFP Battery vs NMC Battery”, [Online]: https://www.bonnenbatteries.com

[84] MAN Energy Solutions, “Hybrid power workboats, a holisticapproach for better CAPEX”, 2018.

[85] Wärtsilä, “Hybrid and electric tug viability: the future’s bright”, Part of Tug Technology Webinar Week, September 2020.

[86] Wärtsilä, “Hybrid marine system – Wärtsilä HY”,[Online]: https://www.wartsila.com

[87] Radica, G.; Vidović, T.; Šimunović, J.; Jurić, Z. “Overview of Hybrid Marine Energy System Configurations and System Component Modeling Approaches”. Energies 2025, 18, 1189.

[88] Sjofartsdirektoratet, “Consultation: Draft Regulations on ships using battery systems with lithium-ion cells with a total capacity of 20 kWh or more”, August 2025.

[89] Ministerio de Hacienda, “Tasa Interna de Retorno (TIR)”. [Online]: https://old.hacienda.cl

[90] Echandia, “What batteries are used in Sparky the tugboat?”[Online]: https://echandia.se

[91] Valence, “Group 27, 36 Volt 50 Ah Lithium Ion Battery Module”.

[92] S. Moussa and M. J. Ben Ghorbal, “Shepherd Battery Model Parametrization for Battery Emulation in EV Charging Application,” 2022 IEEE International Conference on Electrical Sciences and Technologies in Maghreb (CISTEM), Tunis, Tunisia, 2022, pp. 1-6.

[93] M. Ahmed, “Modeling Lithium-ion Battery Chargers in PLECS”, Plexim Inc, 2016.

[94] Z. Zhang, Z. Zhang, S. Xie and C. Yang, “A control strategy for paralleled bi-directional DC-DC converters used in energy storage systems,” 2016 IEEE Energy Conversion Congress and Exposition (ECCE), Milwaukee, WI, USA, 2016, pp. 1-6.

[95] B. Munari, A. Schneer, “How to design a precharge circuit for hybrid and eectric vehicle applications”,Technical white paper, Sensata Technologies.

[96] United Nations,“Carbon Tax (CO_2 Tax)”[Online]: https://sdgs.un.org

[97] Ministry of Oceans and Fisheries, “2030 Greenship-K Promotion Strategy to Dominate the Global Green Ship Market”[Online]: https://www.mof.go.kr

[98] V. Monteiro, T. J. C. Sousa, D. Pedrosa, S. Coelho and J. L. Afonso, “A Three-Level dc-dc Converter for Bipolar dc Power Grids: Analysis and Experimental Validation,” IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society, Singapore, 2020, pp. 3761-3766.

[99] R. González, C. A. Rojas and L. Callegaro, “Three-level DC-DC GaN-based Converter with Active Thermal Control for Powertrain applications in Electric Vehicles,” 2021 22nd IEEE International Conference on Industrial Technology (ICIT), Valencia, Spain, 2021, pp. 502-507.